

Supporting Information:

xTB-Based High-Throughput Screening of TADF Emitters: 747-Molecule Benchmark

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Stokes Shifts and Estimation of Relaxed Singlet-Triplet Excitation Energies

Direct geometry optimization of the S_1 state is not straightforward with the current xTB implementation used. Therefore, an estimation for the relaxed singlet-triplet gap, $\Delta E_{\text{ST,relaxed}}$, was made. The geometric energy relaxation for the T_1 state is $\Delta E_{\text{relax}}(T_1) = \Delta E_v(S_0 \rightarrow T_1)_{\text{xTB}} - \Delta E_r(S_0 \rightarrow T_1)_{\text{xTB}}$. The Stokes shift for fluorescence ($S_1 \rightarrow S_0$) is approximated by assuming a similar geometric relaxation energy for S_1 as for T_1 , i.e., $S_{\text{shift}} \approx \Delta E_{\text{relax}}(T_1)$. This is a strong approximation. Then, the relaxed S_1 energy is $E_r(S_1) \approx E_v(S_1)_{\text{sTDA/sTD-DFT}} - S_{\text{shift}}$. And $\Delta E_{\text{ST,relaxed}} \approx E_r(S_1) - E_r(T_1)_{\text{xTB}}$.

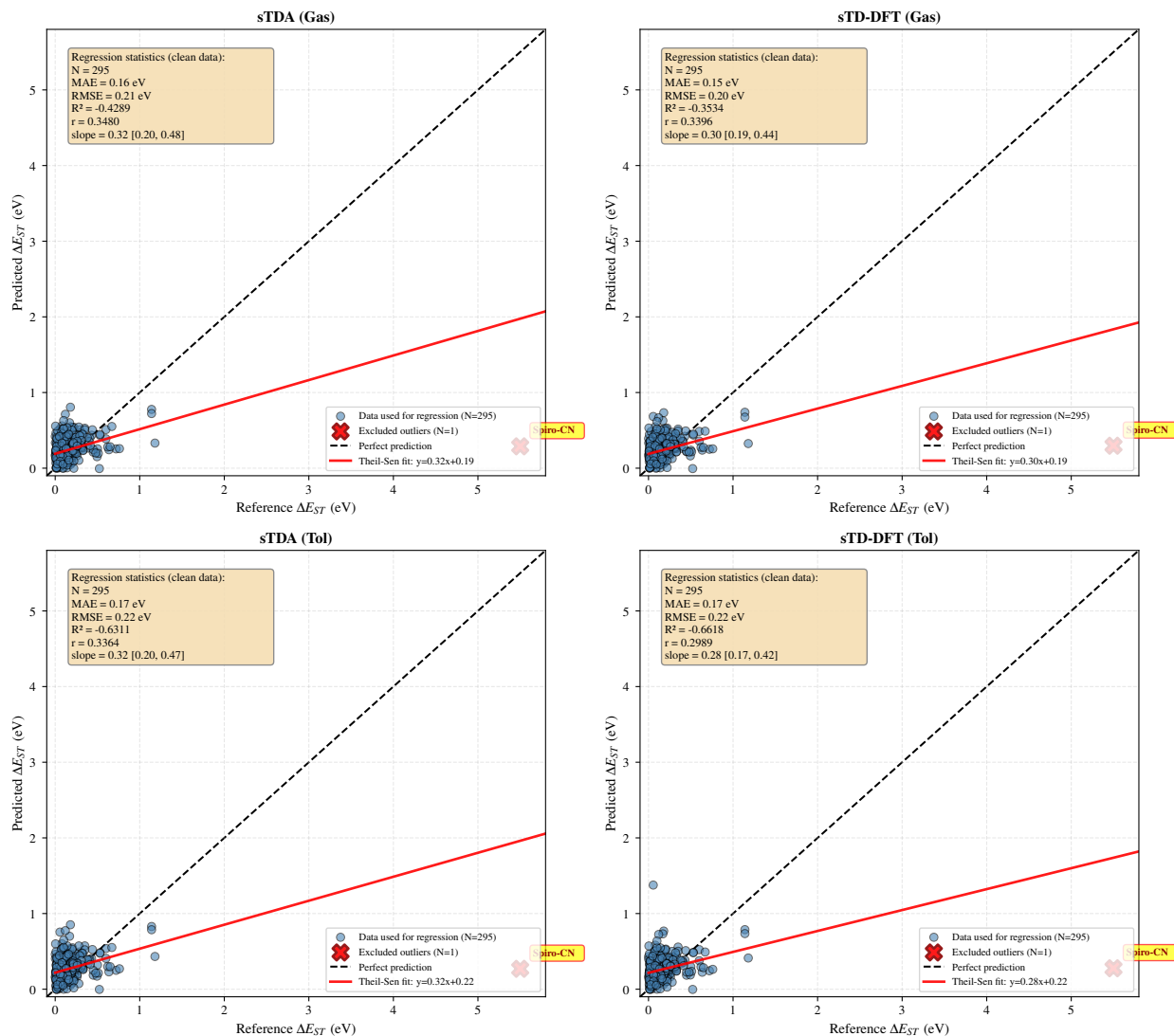


Figure S1: Correlation between predicted and reference singlet-triplet energy gaps (ΔE_{ST}) for TADF molecules. One physically unrealistic outlier (Spiro-CN, reference $\Delta E_{ST} = 5.50$ eV, shown as red X) was excluded from regression analysis. Statistics show Theil-Sen robust regression with 95% bootstrap confidence intervals for slope. Despite moderate scatter, the correlations indicate that semi-empirical methods capture qualitative trends in ΔE_{ST} .

Dataset

Table S1: Molecular architecture of the studied TADF emitters. This table provides the SMILES representation of each molecule.

Molecule Id	SMILES	Molecule Id	SMILES
4CzTPN	<chem>N#Cc1c(-n2c3cccc3c3cccc32)c(-n2c3cccc3c3cccc32)c(C#N)c(-n2c3cccc3c3cccc32)c1-n1c2cccc2c2cccc21</chem>	SPBP-DPAC	<chem>O=C1c2cccc2C2(c3cc(N4c5cccc5C(c5cccc5)c5cccc54)ccc31)c1cccc1-c1cccc12</chem>
DMAC-DPS	<chem>Cc1cc(C)c(B(c2ccc(-c3nc(C1)nc(-c4cccc4)n3)cc2)c2c(C)cc(C)cc2C)c(C)c1</chem>	PX-TRZ	<chem>Cc1ccc(N(c2ccc(C)cc2)c2ccc2C2(c3cccc3N(c3ccc(C)cc3)c3ccc(C)cc3)CCCC2)cc1</chem>
PSPCz	<chem>C1=CC=C(C=C1)S(=O)(=O)C2=CC=CC(=C2)N3C4=CC=CC=C4C5=CC=CC=C53</chem>	C2MPI	<chem>CCCCCCCCN1C(=O)c2c(-c3ccc(-n4c5cccc5c5cccc54)cc3)sc(-c3ccc(-n4c5cccc5c5cccc54)cc3)c2C1=O</chem>
Px2BP	<chem>C1=CC=C2C(=C1)N(C3=CC=CC=C3O2)C4=CC=C(C=C4)C(=O)C5=CC=C(C=C5)N6C7=CC=CC=C7O</chem>	p-PCzTPD	<chem>CCCCCCCCN1C(=O)c2c(-c3ccc(-n4c5cccc5c5cccc54)cc3)c2C1=O</chem>
CzS2	<chem>N2C4=CC=C(C=C4)S(=O)(=O)C5=CC=C(C=C5)N6C7=CC=CC=C7C8=CC=CC=C86</chem>	m-PCzTPD	<chem>c(-n4c5cccc5c5cccc54)c3)sc(-c3cccc(-n4c5cccc5c5cccc54)c3)c2C1=O</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
2TCz-DPS	<chem>CC(C)(C)C1=CC2=C(C=C1)N(C3=C2C=C(C=C3)C(C)(C)C)C4=CC=C(C=C4)S(=O)(=O)C5=CC=C(C=C5)N6C7=C(C=C(C=C7)C(C)(C)C)C8=C6C=CC(=C8)C(C)(C)C</chem>	o-PCzTPD	<chem>CCCCCCCCN1C(=O)c2c(-c3cccc3-n3c4cccc4c4cccc43)s c(-c3cccc3-n3c4cccc4c4cccc43)c2C1=O</chem>
TDBA-DI	<chem>CC(C)(C)c1ccc2c(c1)B1c3ccc(C(C)(C)C)ccc3Oc3cc(-n4c5cccc5c5c6c(c7cccc7n6-c6cccc6)c6c(c7cccc7n6-c6cccc6)c54)cc(c31)O2</chem>	CPBzFC	<chem>c1ccc2c(c1)c1cccc1n2-c1ccc(-c2ccc(-c3ccc4c(c3)Cc3cc(-n5c6cccc6c6cccc65)c3-4)c3nnsc23)cc1c1cncc(-c2ccc([Si](c3ccc(-c4cccnc4)cc3)(c3ccc(-n4c5cccc5c5cccc54)cc3)c3cc c(-n4c5cccc5c5cccc54)cc3)cc2)c1</chem>
OPDPO	<chem>O=CC1=CCC(c2cccc2N2c3cccc3Sc3cccc32)(P(=O)(c2cccc2)c2cccc2)C=C1</chem>	SiCz2Py2	<chem>c1cncc(-c2ccc([Si](c3ccc(-c4cccnc4)cc3)(c3ccc(-c4ccccnc4)cc3)c3ccc(-n4c5cccc5c5cccc54)cc3)cc2)c1</chem>
2CzTPEPCz	<chem>C(=C(c1ccc(-n2c3cccc3c3cccc32)cc1)c1ccc(-n2c3cccc3c3cccc32)cc1)c1ccc(-c2ccc(-n3c4cccc4c4cccc43)cc2)cc1</chem>	SiCz1Py3	<chem>c1cncc(-c2ccc([Si](c3ccc(-c4ccccnc4)cc3)c3ccc(-n4c5cccc5c5cccc54)cc3)cc2)c1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
TBRb	<chem>CC(C)(C)c1ccc(-c2c3ccc(C(C)(C)C)cc3c(-c3ccccc3)c3c(-c4ccc(C(C)(C)C)cc4)c4cc</chem>	SiCz3Py1	<chem>c1cncc(-c2ccc([Si](c3ccc(-n4c5ccccc5c5ccccc54)cc3)(c3ccc(-n4c5ccccc5c5ccccc54)cc3)c3ccc(-n4c5ccccc5c5ccccc54)cc3)cc2)c1</chem>
TBPe	<chem>CC(C)(C)c1cc2cc(C(C)(C)C)cc3c4cc(C(C)(C)C)cc5cc(C(C)(C)C)cc(c(c1)c23)c54</chem>	POBBPE	<chem>CC(C)(C)c1ccc(0c2ccc(C(C)(C)C)cc2P(=O)(c2ccccc2)c2ccccc2)cc1</chem>
2CzPN	<chem>N#Cc1cc(-c2cccc3c2[nH]c2cccc23)c(-c2cccc3c2[nH]c2cccc23)cc1C#N</chem>	CzppT	<chem>c1ccc(-c2c(-c3ccccc3)c(-c3ccc(-c4ccccc5c4[nH]c4cccc45)cc3)c(-c3ccccc3)c(-c3ccccc3)c2-c2ccc(-c3cccc4c3[nH]c3ccccc34)cc2)cc1</chem>
3CzFTFP	<chem>Fc1nc(F)c(F)c(C2(c3ccc4c(c3)c3ccccc3n4-c3ccccc3)c3ccccc3-c3ccccc32)c1F</chem>	SBDBQ-PXZ	<chem>50c5ccccc54)cc3nc2-c2cccc2)cc1</chem>
DDPhCz-DCPP	<chem>N#Cc1nc2c3ccc(-n4c5ccc(-c6ccccc6)cc5c5cc(-c6ccccc6)ccc54)cc3c3cc(-n4c5ccc(-c6ccccc6)ccc54)ccc3c2nc1C#N</chem>	DBQ-3PXZ	<chem>c1ccc2c(c1)0c1ccccc1N2c1ccc(-c2nc3ccc(N4c5ccccc50c5ccccc54)cc3nc2-c2ccc(N3c4ccccc40c4ccccc43)cc2)cc1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
ACRXTN	<chem>CC1(C)c2cccc2N(c2ccc3c(=O)c4cccc4oc3c2)c2cccc21</chem>	P4CzCN-BCz	<chem>C=Cc1ccc(C0c2ccc(C(C)(C)c3ccc(0)cc3)cc2)cc1</chem>
TXO-TPA	<chem>O=C1c2cccc2S(=O)(=O)c2cc</chem>	NAP-1,5-DPA	<chem>C0c1ccc(N(c2ccc(OC)cc2)c2cccc3c(N(c4ccc(OC)cc4)c4cc(OC)cc4)cccc23)cc1</chem>
PCTXO	<chem>N#Cc1c(-c2ccc(N(c3cccc3)0=C1c2cccc2S(=O)(=O)c2cc</chem>	3DPA3CN	<chem>c3cccc3)cc2)c(C#N)c(-c2cc(N(c3cccc3)c3cccc3)cc2)c(C#N)c1-c1ccc(N(c2cccc2)c2cccc2)cc1</chem>
IP-PPI	<chem>c1ccc(-n2c(-c3ccc(-c4ccc(-c5cn6cccc6n5)cc4)cc3)nc3c4cccc4c4cccc4c32)cc1</chem>	TPA-AN-NA	<chem>C0c1ccc2c(c1)c1cccc1n2-c1cc(-n2c3cccc3c3cc(OC)cc32)c(C#N)c(-n2c3cccc3c3cc(OC)ccc32)c1</chem>
DMDHNP-DPS	<chem>CC1(C)c2cnccc2Nc2ccncc21.O=S(=O)(c1cccc1)c1cccc1</chem>	OCzBN	<chem>c1ccc(-c2nnc(-c3ccc(-c4cc</chem>
4NTAZ-PPI	<chem>c(-c5nc6c7cccc7c7cccc7c6n5-c5cccc5)cc4)cc3)n2-c2cccc3cccc23)cc1</chem>	BBCz	<chem>c1ccc2c(c1)[nH]c1cc3[nH]c4cccc4c3cc12</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
TAZ-PPI	<chem>c1ccc(-c2nnc(-c3ccc(-c4cc</chem>	IQ-Se	<chem>C0c1c(C#N)c(C)cc2c([Se][S</chem>
	<chem>c(-c5nc6c7cccc7c7cccc7c</chem>		<chem>e]c3c(C)nc(N4CCCCC4)c4c(0</chem>
	<chem>6n5-c5cccc5)cc4)cc3)n2-c</chem>		<chem>C)c(C#N)c(C)cc34)c(C)nc(N</chem>
	<chem>2cccc2)cc1</chem>		<chem>3CCCCC3)c12</chem>
PTZ-mPYR	<chem>Cc1cc(N2c3cccc3Sc3cccc3</chem>	dTolmp	<chem>Cc1cccc1P(CP(CP(c1cccc1</chem>
	<chem>2)ccc1-c1cc(-c2ccc(N3c4cc</chem>		<chem>C)c1cccc1C)c1cccc1)c1cc</chem>
	<chem>ccc4Sc4cccc43)cc2C)ncn1</chem>		<chem>ccc1C</chem>
	<chem>N#Cc1ccc(-n2c3cccc3c3cc(</chem>		<chem>N#Cc1cccc(-c2nc(-c3cccc([</chem>
pCNBCzoCF3	<chem>-c4ccc5c(c4)c4cccc4n5-c4</chem>	SiTCNCz	<chem>Si](c4cccc4)(c4cccc4)c4</chem>
	<chem>ccc(C#N)cc4C(F)(F)F)ccc32</chem>		<chem>cccc4)c3)nc(-n3c4cccc4c</chem>
	<chem>)c(C(F)(F)F)c1</chem>		<chem>4cccc43)n2)c1</chem>
	<chem>CC1(C)OB(c2ccc(C(=C(c3ccc</chem>		<chem>Cc1cc(P(c2cccc2)c2cccc2</chem>
PTZTPE-2	<chem>cc3)c3cccc3)c3cccc3)cc2</chem>	dpmb	<chem>)c(P(c2cccc2)c2cccc2)cc</chem>
	<chem>)OC1(C)C</chem>		<chem>1C</chem>
	<chem>N#Cc1c(-n2c3cccc3c3cccc</chem>		<chem>N#Cc1c(-n2c3cccc3c3cccc</chem>
4CzIPN	<chem>32)c(C#N)c(-n2c3cccc3c3c</chem>	3CzBN	<chem>32)cc(-n2c3cccc3c3cccc3</chem>
	<chem>cccc32)c(-n2c3cccc3c3ccc</chem>		<chem>2)cc1-n1c2cccc2c2cccc21</chem>
	<chem>cc32)c1-n1c2cccc2c2cccc</chem>		<chem>21</chem>
	<chem>21</chem>		<chem>OC1(C)c2cccc2N(c2ccc(-c3</chem>
DMAC-TRZ	<chem>nc(-c4cccc4)nc(-c4cccc4</chem>	TCB	<chem>cccc3c3cc(-c4ccc(N(c5cccc</chem>
	<chem>)n3)cc2)c2cccc21</chem>		<chem>c5)c5cccc5)cc4)ccc32)cc1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
TCzTrz	<chem>c1ccc(-c2nc(-c3ccccc3)nc(-c3cc(-n4c5ccccc5c5ccccc54)c(-n4c5ccccc5c5ccccc54)c3)n2)cc1</chem>	CPzPyC	<chem>c1ccc2c(c1)c1ccccc1n2-c1cncc(-c2cncc(-n3c4ccccc4cccc43)n2)c1</chem>
δ -2CbPN	<chem>N#Cc1cc(-n2c3ccccc3c3nccc32)c(-n2c3ccccc3c3nccc32)cc1C#N</chem>	CPzPzC	<chem>c1ccc2c(c1)c1ccccc1n2-c1cncc(-c2cncc(-n3c4ccccc4cccc43)n2)n1</chem>
CzDCbTrz	<chem>c1ccc(-c2nc(-c3ccccc3)nc(-c3cc(-n4c5ccccc5c5ncccc54)c(-n4c5ccccc5c5ncccc54)c3)n2)cc1</chem>	CPzPC	<chem>c1cc(-c2cncc(-n3c4ccccc4cccc43)n2)cc(-n2c3ccccc3c3ccccc32)c1</chem>
α -2CbPN	<chem>N#Cc1cc(-n2c3ccccc3c3cccn32)cc1C#N</chem>	CBIQD	<chem>O=C1c2cccc3cccc(c23)C(=O)N1c1nc2ccc(Cl)cc2s1</chem>
CN-QP	<chem>N#Cc1ccc2nc(-c3ccc(N4c5cccc50c5ccccc54)cc3)c(-c3ccc(N4c5ccccc50c5ccccc54)c3)nc2c1</chem>	PXZ-QL	<chem>c1ccc2c(c1)0c1ccccc1N2c1cc2ccccc2n1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
TFM-QP	<chem>FC(F)(F)c1ccc2nc(-c3ccc(N4c5cccc50c5cccc54)cc3)c(-c3ccc(N4c5cccc50c5cccc54)cc3)nc2c1</chem>	SBA	<chem>c1ccc2c(c1)Nc1cccc1C21c2cccc2Nc2cccc21</chem>
NDN	<chem>c1ccc2c(c1)c1cccn1n2-c1ccc2sc3ccc(-n4c5cccc5c5ccn4)cc3c2c1</chem>	TATP-BP	<chem>O=C(c1ccc(-n2c3cccc3c3c4c(c5cccc5n4-c4cccc4)c4c(c5cccc5n4-c4cccc4)c32)cc1)c1ccc(-n2c3cccc3c3c4c(c5cccc5n4-c4cccc4)c4c(c5cccc5n4-c4cccc4)c32)cc1</chem>
mCP	<chem>c1cc(-n2c3cccc3c3cccc32)cc(-n2c3cccc3c3cccc32)c1</chem>	Pd-1	<chem>CCCCN(CCCC)c1ccc(C=Nc2cc(C#N)c(C#N)cc2N=Cc2ccc(N(CCC)CCCC)cc2O)c(O)c1</chem>
ZDZ	<chem>c1ccc2c(c1)c1cccc1n2-c1ccc2sc3ccc(-n4c5cccc5c5ccc4)cc3c2c1</chem>	C3F7	<chem>Cc1cc(C)c(B(c2cccc2-n2c3cccc3c3cccc32)c2c(C)cc(C)cc2C)c(C)c1</chem>
ZDN	<chem>c1ccc2c(c1)c1cccc1n2-c1ccc2sc3ccc(-n4c5cccc5c5ccn4)cc3c2c1</chem>	BuCzCF3oB	<chem>Cc1cc(C)c(B(c2ccc(C(F)(F)F)cc2-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c2c(C)cc(C)cc2C)c(C)c1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
BTCz3CzTPN	<chem>N#Cc1c(-n2c3cccc3c3cccc32)c(-n2c3cccc3c3cccc32)c(C#N)c(-n2c3cccc3c3c4sc5cccc5c4ccc32)c1-n1c2ccccc2c2cccc21</chem>	phenazasiline	<chem>BrC1ccc2nc3c4ccc(Br)cc4c4cccc4c3nc2c1</chem>
2BTCz2CzTPN	<chem>N#Cc1c(-n2c3cccc3c3cccc32)c(-n2c3cccc3c3c4sc5ccc5c4ccc32)c(C#N)c(-n2c3cccc3c3cccc32)c1-n1c2ccccc2c2sc4cccc4c3ccc21</chem>	D-A-D	<chem>BrC1ccc2nc3c4ccc(Br)cc4c4cccc4c3nc2c1</chem>
APDC-DTPA	<chem>N#Cc1nc2c(nc1C#N)-c1ccc(-c3ccc(N(c4cccc4)c4cccc4)cc3)c3c(-c4ccc(N(c5cccc5)c5cccc5)cc4)ccc-2c13</chem>	pzpy	<chem>c1ccc(-c2nc3cccc3n2-c2ccc(-c3c4cccc4c(-c4ccc(-n5c(-c6cccc6)nc6cccc65)cc4)c4cccc34)cc2)cc1</chem>
BMIM	<chem>N#Cc1nc2c(nc1C#N)-c1ccc(-c3ccc(N(c4cccc4)c4cccc4)cc3)c3c(-c4ccc(N(c5cccc5)c5cccc5)cc4)ccc-2c13</chem>	m-PO-ABN	<chem>N#Cc1ccc(-c2c3cccc3c(-c3cccc(-c4nc5c6cccc6c6cccc6c5o4)c3)c3cccc23)cc1</chem>
ADO-TPA	<chem>O=C1C(=O)c2ccc(-c3ccc(N(c4cccc4)c4cccc4)cc3)c3ccc1c23</chem>	PO-ABN	<chem>N#Cc1ccc(-c2c3cccc3c(-c3cccc(-c4nc5c6cccc6c6cccc6c5o4)c3)c3cccc23)cc1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
DCzTrz	<chem>c1ccc(-c2nc(-c3ccccc3)nc(-c3cc(-n4c5ccccc5c5ccccc54)cc(-n4c5ccccc5c5ccccc54)c3)n2)cc1</chem>	p-PO-ABN	<chem>N#Cc1ccc(-c2c3ccccc3c(-c3ccc(-c4nc5c6ccccc6c6ccccc6c5o4)cc3)c3ccccc23)cc1</chem>
BDPCC	<chem>c1ccc(-c2ccc3[nH]c4c(-c5ccc6c([nH]c7ccccc76)c5-c5cc(-c6ccccc6)cc6c5[nH]c5cc(-c7ccccc7)cc56)cc(-c5ccccc5)cc4c3c2)cc1</chem>	teracridan	<chem>c1ccc2c(c1)Cc1c(cccc1-c1ccc3c(c1-c1cccc4c1Cc1ccccc1N4)Cc1ccccc1N3)N2</chem>
SpiroAC-TRZ	<chem>c1ccc(-c2nc(-c3ccccc3)nc(-c3ccc(N4c5ccccc5C5(c6cccc6-c6ccccc65)c5ccccc54)c3)n2)cc1</chem>	PTPC	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-c2ccc(-c3cc(-c4ccccc4)c(-c4ccc(-c5nc6c7ccccc7c7ccccc7c6n5-c5ccccc5)cc4)cc3-c3ccccc3)cc2)cc1</chem>
DBT-BZ-DMAC	<chem>CC1(C)c2ccccc2Nc2cccc(C(=O)c3ccccc3)c21.c1ccc2c(c1)sc1ccccc12</chem>	TPTI	<chem>O=c1[nH]c2ccsc2c2cc3c(=O)[nH]c4ccsc4c3cc12</chem>
DQP	<chem>c1ccc2nc3c(nc2c1)c1nc2cccc2nc1c1nc2ccccc2nc31</chem>	BDBTCz-Trz	<chem>c1ccc(-c2nc(-c3ccccc3)nc(-c3ccc(-n4c5ccc(-c6ccc7sc8ccccc8c7c6)cc5c5cc(-c6cc7sc8ccccc8c7c6)ccc54)cc3)n2)cc1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
TPA-DCPP	<chem>N#Cc1nc2c3ccc(-c4ccc(N(c5cccc5)c5cccc5)cc4)cc3c3cc(-c4ccc(N(c5cccc5)c5cccc5)cc4)ccc3c2nc1C#N</chem>	BDBFCz-Trz	<chem>c1ccc(-c2nc(-c3cccc3)nc(-c3ccc(-n4c5ccc(-c6ccc7oc8cccc8c7c6)cc5c5cc(-c6ccc7oc8cccc8c7c6)ccc54)cc3)n2)cc1Cc1cc(C2=NC3=NC(c4ccc(F)c(C)c4)=NC4=NC(c5ccc(F)c(C)c5)=NC(=N2)N34)ccc1F</chem>
2PXZ-OXD	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc(-c2nnc(-c3ccc(N4c5cccc5)cc3)oc2)cc1CC1(C)c2cccc2N(c2ccc(-c3nc(-c4ccc(N5c6cccc6C(C)(C)c6cccc65)cc4)nc(-c4ccc(N5c6cccc6C(C)(C)c6cccc65)cc4)n3)cc2)c2cccc21C0c1ccc(N(c2ccc(OC)cc2)c2ccc(-c3ccc4c(c3)C(=O)c3cc(-c5ccc(N(c6ccc(OC)cc6)c6ccc(OC)cc6)cc5)cc3C4=O)c2)cc1</chem>	HAP-3MF	<chem>CC1(C)c2cccc2-c2ccc(-c3ncc(-c4cc(-n5c6cccc6c6cccc65)cc4)cn3)cc21</chem>
BmPyPhB	<chem>N#Cc1ccc2c(c1)C1(c3cc(C#N)ccc3-2)c2cccc2N(c2cccc2)c2cccc21</chem>	FlCz	<chem>c1ccc2c(c1)-c1cccc1C21c2cccc2-c2ccc(-c3ncc(-c4cc(-n5c6cccc6c6cccc65)cc4)cn3)cc21</chem>
MPPA	<chem>N#Cc1ccc2c(c1)C1(c3cc(C#N)ccc3-2)c2cccc2N(c2cccc2)c2cccc21</chem>	SFCz	<chem>c1ccc(N(c2cccc2)c2cccc2-c2c3cccc3c(-c3ccc4ccc5c(-c6cccc6N(c6cccc6)c6ccc6)ccc6ccc3c4c65)c3cccc23)cc1</chem>
ACRFLCN	<chem>N#Cc1ccc2c(c1)C1(c3cc(C#N)ccc3-2)c2cccc2N(c2cccc2)c2cccc21</chem>	o-TPA-AP-TPA	

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
PTZ-DCPP	<chem>N#Cc1nc2c3ccc(N4c5cccc5Sc5cccc54)ccc3c2nc1C#N</chem>	p-TPA-AP-TPA	<chem>c1ccc(N(c2cccc2)c2ccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7ccc(N(c8cccc8)c8cccc8)cc7)ccc7ccc4c5c76)c4cccc34)cc2)cc1N#Cc1ccc(-c2ccc3c(c2)c2cc(-c4ccc(C#N)cc4)ccc2c2nc4cc(-c5ccc(N(c6cccc6)c6ccc6)cc5)c(-c5ccc(N(c6cccc6)c6cccc6)cc5)cc4nc32)</chem>
BBA-Pyr	<chem>c1ccc(-c2ccc(N(c3ccc(-c4cccc4)cc3)c3ccc(-c4ccc(-c5cc(-c6cccc6)nc(-c6cccc6)n5)cc4)cc3)cc2)cc1</chem>	TNPZ	<chem>cc1c1ccc(N(c2cccc2)c2cccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7cccc(N(c8cccc8)c8cccc8)c7)ccc7ccc4c5c76)c4cccc34)c2)cc1CC1(C)C(=O)c2cc(-c3ccc(N4c5cccc50c5cccc54)cc3)c(-c3ccc(N4c5cccc50c5cccc54)cc3)cc2C1=O</chem>
PhICz-Pyr	<chem>c1ccc(-c2ccc3c(c2)c2cc(-c4cccc4)cc4c5cc(-c6ccc(-c7cc(-c8cccc8)nc(-c8cccc8)n7)cc6)ccc5n3c24)cc1</chem>	m-TPA-AP-TPA	<chem>cc1c1ccc(N(c2cccc2)c2cccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7cccc(N(c8cccc8)c8cccc8)c7)ccc7ccc4c5c76)c4cccc34)c2)cc1CC1(C)C(=O)c2cc(-c3ccc(N4c5cccc50c5cccc54)cc3)c(-c3ccc(N4c5cccc50c5cccc54)cc3)cc2C1=O</chem>
PhCz-Pyr	<chem>c1ccc(-c2ccc3c(c2)c2cc(-c4cccc4)ccc2n3-c2ccc(-c3ccc(-c4cc(-c5cccc5)nc(-c5cccc5)n4)cc3)cc2)cc1</chem>	5,6PXZ-PIDO	

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
PDA-DP	<chem>CC1(C)c2cc(-c3ccccc3)ccc2N(c2ccc(S(=O)(=O)c3ccc(N4c5ccc(-c6ccccc6)cc5C(C)(C)cc2)cc2)c2ccc(-c3ccccc3)cc21O=P(c1ccccc1)(c1ccccc1)c1cccc2c1oc1c(P(=O)(c3ccccc3)c3ccccc3)cc(-c3ccc(C4(c5ccc(-n6c7ccccc7c7ccccc76)cc5)c5ccccc5-c5ccccc54)c3)cc12O=P(c1ccccc1)(c1ccccc1)c1cccc2c1oc1c(P(=O)(c3ccccc3)c3ccccc3)cc(-c3ccc(C4(c5ccc(-c6ccc(-n7c8ccccc8c8ccccc87)cc6)cc5)c5ccccc5-c5ccccc54)cc3)cc12O=P(c1ccccc1)(c1ccccc1)c1cccc2c1oc1ccc(-c3ccc(C4(c5ccc(-c6ccc(-n7c8ccccc8c8ccccc87)cc6)cc5)c5ccccc5-c5ccccc54)cc3)cc12</chem>	TRZ-3SO2	<chem>O=S(=O)(c1ccccc1)c1cccc(-c2nc(-c3ccccc(S(=O)(=O)c4ccccc4)c3)nc(-c3ccccc(S(=O)(=O)c4ccccc4)c3)n2)c1</chem>
9CzFDBFDPO	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1cc(C(=O)c2cncnc2)cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1</chem>	35tCzBPym	<chem>O=C(c1cncnc1)c1cc(-n2c3ccccc3c3ccccc32)cc(-n2c3ccccc3c3ccccc32)c1</chem>
9PhCzFDBFDPO	<chem>O=C(c1cncnc1)c1cc(-n2c3ccccc3c3ccccc32)ccc1-n1c2ccccc2c2ccccc21</chem>	35CzBPym	
9PhCzFDBFSPO		25CzBPym	

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Molecule Id	SMILES	Molecule Id	SMILES
9CzFDBFSPO	<chem>O=P(c1ccccc1)(c1ccccc1)c1cccc2c1oc1ccc(-c3ccc(C4(c5ccc(-n6c7ccccc7c7ccccc76)cc5)c5ccccc5-c5ccccc54)c3)cc12</chem>	25tCzBPym	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c(C(=O)c2cncnc2)c1</chem>
9ArFDBFxPO	<chem>O=P(c1ccccc1)(c1ccccc1)c1cccc2c1oc1c(P(=O)(c3ccccc3)c3ccccc3)cc(-c3ccc(C4(c5ccc(-n6c7ccccc7c7ccccc76)cc5)c5ccccc5-c5ccccc54)c3)cc12</chem>	BPBiPA	<chem>c1ccc(-c2nc3ccccc3n2-c2cc(-c3c4ccccc4c(-c4ccc(-n5c(-c6ccccc6)nc6ccccc65)cc4)c4ccccc34)cc2)cc1</chem>
12BTCzTPN	<chem>N#Cc1c(-n2c3ccccc3c3ccccc32)c(-n2c3ccccc3c3ccc4c5ccccc5sc4c32)c(C#N)c(-n2c3ccccc3c3ccccc32)c1-n1c2ccc2c2ccc3c4ccccc4sc3c21</chem>	BTZ-DMAC	<chem>CC1(C)c2ccccc2N(c2ccc(N3c4ccccc4C(C)(C)c4ccccc43)c3nsnc23)c2ccccc21</chem>
oDTBPZ-DPXZ	<chem>Cc1ccccc1-c1cc2nc3c4ccc(N5c6ccccc60c6ccccc65)cc4c4cc(N5c6ccccc60c6ccccc65)c4c3nc2cc1-c1ccccc1C</chem>	BTZ-DPA	<chem>c1ccc(N(c2ccccc2)c2ccc(N(c3ccccc3)c3ccccc3)c3nsnc23)cc1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
3DMAC-BP-CN	<chem>CC1(C)c2cccc2N(c2ccc3c(c2)c2cc(N4c5cccc5C(C)(C)c5cccc54)ccc2c2nc4cc(N5c6cccc6C(C)(C)c6cccc65)c(C#N)cc4nc32)c2cccc21</chem>	2F	<chem>c1ccc(N(c2cccc2)c2ccc3c(c2)c2cc(N(c4cccc4)c4cccc4)ccc2n3-c2ccc(-c3ccc4ncnc4c3)cc2)cc1</chem>
3DMAC-BP-Br	<chem>CC1(C)c2cccc2N(c2ccc3c(c2)c2cc(N4c5cccc5C(C)(C)c5cccc54)ccc2c2nc4cc(N5c6cccc6C(C)(C)c6cccc65)c(Br)cc4nc32)c2cccc21</chem>	PFBP-2a	<chem>CC(C)(C)c1ccc2c(c1)C(C)(C)c1cc(C(C)(C)C)ccc1N2c1c(F)c(F)c(-c2c(F)c(F)c(F)c(F)c2F)c(F)c1F</chem>
TPA-PZCN	<chem>N#Cc1ccc2c(c1)c1cc(C#N)cc1c1nc3cc(-c4ccc(N(c5cccc5)c5cccc5)cc4)c(-c4ccc(N(c5cccc5)c5cccc5)cc4)c3nc21</chem>	PFBP-2b	<chem>CC(C)(C)c1ccc2c(c1)C(C)(C)c1cc(C(C)(C)C)ccc1N2c1c(F)c(F)c(-c2c(F)c(F)c(N3c4ccc(C(C)(C)C)cc4C(C)(C)c4cc(C(C)(C)C)ccc43)c(F)c2F)c(F)c1F</chem>
1PXZ-BP	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc2c(c1)c1cccc1c1nc3cccc3nc21</chem>	PXZ-PPO	<chem>Cc1cc2nc(N3c4cccc40c4cccc43)ccc2c(=O)o1</chem>
PTZ-DBTO2	<chem>O=S1(=O)c2cccc2-c2cc(N3c4cccc4Sc4cccc43)ccc21</chem>	PXZ-BOO	<chem>Cc1nc2cc(N3c4cccc40c4cccc43)ccc2c(=O)o1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
BACQ	<chem>CC1(C) c2c(ccc3ccccc23)-n2 c3ccccc3c(=O) c3cccc1c32</chem>	PPZTPI	<chem>CC(C)(C) c1ccc(-n2c(-c3ccc (N4c5ccccc5N(c5ccccc5) c5c cccc54) cc3) nc3c4ccccc4c4c cccc4c32) cc1 c1ccc(N2c3ccccc3N(c3ccc(- c4nc5c6ccccc6c6ccccc6c5n4 -c4ccccc4) cc3) c3ccccc32) c c1</chem>
2Cz-DMAC-TXO	<chem>CC1(C) c2cc(-n3c4ccccc4c4c cccc43) ccc2N(c2ccc3sc4ccc cc4c(=O) c3c2) c2ccc(-n3c4c cccc4c4ccccc43) cc21 CC1(C) c2cc(-n3c4ccccc4c4c cccc43) ccc2N(c2ccc3c(c2)S (=O)(=O) c2ccccc2S3(=O)=O) c2ccc(-n3c4ccccc4c4ccccc4 3) cc21 CC1(C) c2cc(-n3c4ccccc4c4c cccc43) ccc2N(c2ccc3c(c2)- c2ccccc2S3(=O)=O) c2ccc(-n 3c4ccccc4c4ccccc43) cc21 CC1(C) c2ccccc2N(c2cc(-c3c c(-c4ccc(-c5ccccc5) cc4) nc (-c4ccccc4) n3) cc(N3c4cccc c4C(C)(C) c4ccccc43) c2) c2c cccc21</chem>	PPZPPI	<chem>CC1(C) c2ccccc2N(c2ccnc(N3 c4ccccc4C(C)(C) c4ccccc43) c2) c2ccccc21</chem>
2Cz-DMAC-TTR		Pra-2DMAC	
2Cz-DMAC-BTB		BAPCN	<chem>CC1(C) c2ccccc2N(c2ccccc2) c2ccc3cc(-c4ccc(C#N) cc4) c cc3c21</chem>
DMAC-BPP		BACN	<chem>CC1(C) c2ccccc2N(c2ccccc2) c2ccc3cc(C#N) ccc3c21</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
Me-MOC	<chem>COC1CCC2C(C1)C1CCCCC1N2-C1CCC(-C2C(C#N)C(C)NC(C)C2C#N)CC1</chem>	BDQC-2	<chem>c1ccc(-c2ccc(N(c3ccc(-c4cccc4)cc3)c3ccc4c5cccc5n(-c5ccc6nc(-c7cccc7)c(-c7cccc7)nc6c5)c4c3)cc2)cc1</chem>
Me-DOC	<chem>COC1CCC2C(C1)C1CC(OC)CCC1N2-C1CCC(-C2C(C#N)C(C)NC(C)C2C#N)CC1</chem>	NIDPA-1	<chem>c1ccc(N(c2cccc2)c2ccc3c(c2)cc2c4cccc(N(c5cccc5)c5cccc5)c4n4c5cccc5c3c24)cc1</chem>
3DCPO	<chem>O=P(c1cccc1)(c1ccc2c(c1)c1cccc1n2-c1cccc1)c1ccc2c(c1)B1c3cc(C(C)(C)C)ccc3Oc3cc(-n4c5cccc5c5c(-c6cccc6)c6c(c(-c7cccc7)c54)c4cccc4n6-c4cccc4)cc(c31)O2</chem>	NIDPA-2	<chem>c1ccc(N(c2cccc2)c2ccc3c(c2)c2cc(N(c4cccc4)c4cccc4)cc4c5c6cccc6ccc5n3c24)cc1</chem>
TDBA-TPDICz	<chem>(C(C)(C)C)ccc3Oc3cc(-n4c5cccc5c5c(-c6cccc6)c6c(c(-c7cccc7)c54)c4cccc4n6-c4cccc4)cc(c31)O2</chem>	SABIQ	<chem>c1ccc(N2c3cccc3C3(c4cccc4-c4nc5cc6cccc6cc5nc43)c3cccc32)cc1</chem>
TRZ-pIC	<chem>(C(C)(C)C)ccc3Oc3cc(-n4c5cccc5c5c(-c6cccc6)c6c(c(-c7cccc7)c54)c4cccc4n6-c4cccc4)cc(c31)O2</chem>	Perylene	<chem>c1cc2cccc3c4cccc5cccc(c(c1)c23)c54</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
SBF-PXZ	<chem>c1ccc2c(c1)Oc1cccc1N2c1c cc2c(c1)C1(c3cccc3-c3ccc cc31)c1cc(N3c4cccc40c4cc ccc43)ccc1-2 CCC1=C(CC)c2cc3[nH]c(cc4n</chem>	SAIP	<chem>c1ccc(N2c3cccc3C3(c4cccc c4-c4nccnc43)c3cccc32)cc 1</chem>
PtOEP	<chem>c(cc5[nH]c(cc1n2)c(CC)c5C C)C(CC)=C4CC)c(CC)c3CC.[P t+2]</chem>	FAC	<chem>c1ccc2c(c1)Nc1cccc1C21c2 cccc2-c2cccc21</chem>
PxCYP	<chem>N#Cc1ccc(-c2ccc(N3c4cccc 40c4cccc43)cc2)nc1-c1ccc (N2c3cccc30c3cccc32)cc1 CC1(C)c2cccc2N(c2ccc(-c3</chem>	phenylperylene	<chem>c1ccc(-c2ccc3cccc4c5cccc6 cccc(c2c34)c65)cc1</chem>
AcCYP	<chem>ccc(C#N)c(-c4ccc(N5c6cccc c6C(C)(C)c6cccc65)cc4)n3)cc2)c2cccc21 CC1(C)c2cccc2N2c3ccc4cc(</chem>	ImPy-2	<chem>C0c1ccc2cc(-c3nc(-c4cccc 4)c4ccccn34)ccc2c1</chem>
BCN	<chem>C#N)ccc4c3C(C)(C)c3cccc1c 32 O=S(=O)(c1cccc1)c1ccc(-c</chem>	ImPy-1	<chem>c1ccc(-c2nc(-c3ccc4cccc4 c3)n3cccc23)cc1</chem>
DPPSPF	<chem>2ccc3c(c2)C(c2cccc2)(c2c cccc2)c2cccc2-3)cc1</chem>	ImPy-3	<chem>c1ccc(-c2nc(-c3ccc4ccc5cc cc6ccc3c4c56)n3cccc23)cc 1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
Me2ACBO	CC(C)(c1ccc2c(c1)B1c3ccc(N4c5ccccc5C(C)(C)c5ccccc54)cc30c3ccc(c(c31)O2)c1ccc2c(c1)B1c3ccc(N4c5ccccc5C(C)(C)c5ccccc54)cc30c3ccc(c(c31)O2)CC1(C)c2ccccc2N(c2ccc3c(c3)O)c2cccc4c2B3c2cc(C3(c5ccc6c(c5)B5c7ccc(N8c9ccccc9C(C)(C)c9ccccc98)cc70c7ccc(c75)O6)c5ccccc5-c5ccc(cc53)ccc204)c2ccccc21O=S(=O)(c1ccccc1)c1ccc(-c2ccc3c(c2)C2(c4ccccc4-c4ccccc42)c2ccccc2-3)cc1O=C1c2ccccc2S(=O)(=O)c2ccc(-c3ccc([Si](c4ccccc4)(c4ccccc4)c4ccccc4)cc3)cc21FC(F)(F)c1ccc(-n2c(-c3ccc(-c4ccc(-n5c6ccccc6c6ccccc65)cc4)cc3)nc(-c3ccccc3)c2-c2ccccc2)cc1	BTaz-Ar-PXZ	c1ccc2c(c1)Oc1ccccc1N2c1ccc(-c2nc3ccccc3s2)cc1
F2AcBO	CC1(C)c2ccccc2-c2ccc(N(c3ccccc3-c3ccccc3)c3ccccc4c3-c3ccccc3C43c4ccccc4-c4ccccc43)cc21	FSF4A	
PSPSF	N#Cc1ccc(CCCc2ccc(N(c3ccccc3)c3ccccc3)cc2)cc1	TPA-L-BN	
TXO-P-Si	CC(C)c1ccc2occc(-c3ccc(N4c5ccccc5C(C)(C)c5cc6c(cc54)oc4ccccc46)cc3)c(=O)c2c1	IpCm-PhBzAc	
CzB-FMPIM	O=C1c2ccccc2C(c2ccccc2)(c2ccc(Br)cc2)c2ccccc21	DphAn-Br	

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Molecule Id	SMILES	Molecule Id	SMILES
CzB-FMPPI	<chem>FC(F)(F)c1ccc(-n2c(-c3ccc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cc3)nc3c4cccc4c4cccc4c32)cc1</chem>	5-DBFBI	<chem>c1ccc2c(c1)nc1n(-c3ccc(-c4ccc5oc6cccc6c5c4)cn3)c3cccc3n21</chem>
BACF	<chem>CC1(C)c2cccc2N(c2cccc2)c2ccc3cc(F)ccc3c21</chem>	4-DBFBI	<chem>c1ccc2c(c1)nc1n(-c3cc(-c4ccc5oc6cccc6c5c4)ccn3)c3cccc3n21</chem>
BACH	<chem>CC1(C)c2cccc2N(c2cccc2)c2ccc3cccc3c21</chem>	DMACMNPTO	<chem>Cc1cc(N2c3cccc3S(=O)(=O)c3cccc32)c(C)cc1N1c2cccc2C(C)(C)c2cccc21</chem>
PzTDBA	<chem>CC(C)(C)c1ccc2c(c1)B1c3cc(C(C)(C)C)ccc3Oc3cc(N4c5cccc5N(c5cc6c7c(c5)Oc5ccc(C(C)(C)C)cc5B7c5cc(C(C)(C)C)ccc5O6)c5cccc54)cc(c31)O2</chem>	1CzSO	<chem>Cc1cc(-n3c4ccc(C(C)(C)C)cc4c4cc(C(C)(C)C)ccc43)cc1S2(=O)=O</chem>
3TCPM	<chem>O=C(c1cccc1)c1ccc(-n2c3cccc3c3cc(-c4ccc(N(c5cccc5)c5cccc5)cc4)ccc32)cc1</chem>	DpAn-BzAc	<chem>CC1(C)c2cccc2N(c2ccc(C3(c4cccc4)c4cccc4C(=O)c4cccc43)cc2)c2c1ccc1c2oc2cccc21</chem>
TPA-PPO	<chem>c1ccc(-c2nc(-c3ccc(-c4ccc(N(c5cccc5)c5cccc5)cc4)cn3)oc2-c2cccc2)cc1</chem>	BzITz	<chem>N#Cc1cc(-c2nc(-c3cccc3)n(-c3cccc3)n2)ccc1-n1c2cccc2n2c3cccc3nc12</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
CzPh-PPO	<chem>c1ccc(-c2nc(-c3ccc(-c4ccc(-n5c6cccc6c6cccc65)cc4)cn3)oc2-c2cccc2)cc1</chem>	TPXZ-as-TAZ	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc(-c2nnc(-c3ccc(N4c5cccc50c5cccc54)cc3)c(-c3ccc(N4c5cccc50c5cccc54)cc3)n2)cc1</chem>
m-CF3PIBI	<chem>FC(F)(F)c1cccc(-n2c(-c3ccc(-c4ccc(-c5ccc(-c6nc7ccc7n6-c6cccc6)cc5)cc4)cc3)nc3c4cccc4c4cccc4c32)</chem>	TPA-CN-N4-2PY	<chem>CC(C)(C)c1ccc(N(c2ccc(-c3cc(N)c(N)cc3C#N)cc2)c2ccc(C(C)(C)C)cc2)cc1</chem>
DPTPEA	<chem>c1ccc(C(=C(c2cccc2)c2ccc(N(c3cccc3)c3cccc3)cc2)c2cccc2)cc1</chem>	TPA-CN-N4	<chem>c1ccc2nc3c4cccnc4c4ncccc4c3nc2c1</chem>
PICNAnCz	<chem>N#Cc1ccc(-n2c(-c3ccc(-c4c5cccc5c(Br)c5cccc45)cc3)nc3c4cccc4c4cccc4c32)c1</chem>	6,7-DCNQx-DICz	<chem>N#Cc1cc2nc(-c3cccc3)c(-c3ccc(-n4c5cccc5c5c6c(c7cccc7n6-c6cccc6)c6c(c7cccc7n6-c6cccc6)c54)cc3)n</chem>
DPA-DPQ	<chem>c1ccc(-c2cc3ncc(N(c4cccc4)c4cccc4)nc3cc2-c2cccc2)cc1</chem>	PBICT	<chem>c2cc1C#Nc1ccc(-c2nc(-n3c4cccc4c4ccc5c6cccc6n(-c6cccc6)c5c43)nc(-n3c4cccc4c4ccc5c6cccc6n(-c6cccc6)c5c43)n2)cc1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
PhQLPXZ	<chem>c1ccc(-c2cc(-c3ccccc3)c3cc(N4c5ccccc5Oc5ccccc54)cc</chem>	DCBNPE	<chem>O=[N+]([O-])c1ccc(/C(=C(/</chem>
	<chem>c3n2)cc1</chem>		<chem>c2ccc([N+](=O)[O-])cc2)n2</chem>
	<chem>c1ccc(-c2cc3nc(N(c4ccccc4</chem>		<chem>c3ccccc3c3ccccc32)n2c3ccc</chem>
2DPA-DPQ	<chem>)c4ccccc4)c(N(c4ccccc4)c4</chem>	TPA-BT-Q	<chem>cc3c3ccccc32)cc1</chem>
	<chem>ccccc4)nc3cc2-c2ccccc2)cc</chem>		<chem>Cc1ccc(N(c2ccc(C)cc2)c2cc</chem>
	<chem>1</chem>		<chem>c(C3(c4ccc(N(c5ccc(C)cc5)</chem>
	<chem>Fc1cc2nc(-c3ccccc3)c(-c3c</chem>		<chem>c5ccc(C)cc5)cc4)CCCC3)cc</chem>
FDQPXZ	<chem>cccc3)nc2cc1N1c2ccccc2Oc2</chem>	D5	<chem>2)cc1</chem>
	<chem>cccc21</chem>		<chem>c1cc(-c2ccc(-n3c4ccccc4c4</chem>
	<chem>c1ccc(-c2nc(-c3ccccc3)nc(</chem>		<chem>cccc43)cc2)cc(-c2ccc3c4c</chem>
	<chem>-c3ccc4c(c3)C3(c5ccccc5Oc</chem>		<chem>cccc4c4ccccc4c3c2)c1</chem>
DTRZSFX	<chem>5ccccc53)c3cc(-c5nc(-c6cc</chem>	D6	<chem>c1cc(-c2ccc3c4ccccc4c4ccc</chem>
	<chem>ccc6)nc(-c6ccccc6)n5)ccc3</chem>		<chem>cc4c3c2)cc(-n2c3ccccc3c3c</chem>
	<chem>-4)n2)cc1</chem>		<chem>cccc32)c1</chem>
E-o-DHMPA	<chem>OCc1ccccc1-c1c2ccccc2c(-c</chem>	PxPYM	<chem>c1ccc2c(c1)Oc1ccccc1N2c1c</chem>
	<chem>2ccccc2C0)c2ccccc12</chem>		<chem>cc(-c2cnc(-c3ccc(N4c5ccccc</chem>
			<chem>c5Oc5ccccc54)cc3)nc2)cc1</chem>
Z-o-DHMPA	<chem>OCc1ccccc1-c1c2ccccc2c(-c</chem>	D1-D6	<chem>c1cc(-c2ccc3c4ccccc4c4ccc</chem>
	<chem>2ccccc2C0)c2ccccc12</chem>		<chem>cc4c3c2)cc(-n2c3ccccc3c3c</chem>
			<chem>cccc32)c1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
p-DHMPA	<chem>O=Cc1ccc(-c2c3cccc3c(-c3ccc(CO)cc3)c3cccc23)cc1</chem>	DTPCZTZ	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-c2ccc(-c3nnc(-c4cccc4)n3-c3cccc3)cc2)cc1</chem>
STAC-BP-STAC	<chem>O=C(c1ccc(N2c3cccc3C3(c4cccc4Sc4cccc43)c3cccc32)cc1)c1ccc(N2c3cccc3C3(c4cccc4Sc4cccc43)c3cccc32)cc1</chem>	DHPZ-2BI	<chem>O=C(c1ccc(N2c3cccc3C3(c4cccc4Sc4cccc43)c3cccc32)cc1)c1ccc(N2c3cccc3C3(c4cccc4Sc4cccc43)c3cccc32)cc1</chem>
SXAC-BP-SXAC	<chem>O=C(c1ccc(N2c3cccc3C3(c4cccc4Sc4cccc43)c3cccc32)cc1)c1ccc(N2c3cccc3C3(c4cccc4Sc4cccc43)c3cccc32)cc1</chem>	DHPZ-2BN	<chem>N#Cc1ccc(N2c3cccc3N(c3ccc(C#N)cc3)c3cccc32)cc1</chem>
m-DHMPA	<chem>O=Cc1cccc(-c2c3cccc3c(-c3ccc(CO)c3)c3cccc23)c1</chem>	SAF-2NP	<chem>O=C(c1ccc(N2c3cccc3C3(c4cccc4-c4cccc43)c3cccc32)cc1)c1ccc2c(c1)-c1cccc1C21c2cccc2N(c2ccc3nc4c5cccnc5c5ncccc5c4nc3c2)c2cccc21</chem>
SFAC-BP-SFAC	<chem>O=C(c1ccc(N2c3cccc3C3(c4cccc4-c4cccc43)c3cccc32)cc1)c1ccc(N2c3cccc3C3(c4cccc4-c4cccc43)c3cccc32)cc1</chem>	BTPO	<chem>O=S1(=O)c2cccc2N2c3cccc3S(=O)(=O)c3cccc1c32</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
PBTPA	<chem>CCN1c2cccc2Sc2cc(-c3ccc(-c4ccc(N(c5cccc5)c5cccc5)cc4)c4nsnc34)ccc21</chem>	PyDCN-DMAC	<chem>CC1(C)c2cccc2N(c2ccc(-c3cc(-c4ccc(C#N)cc4)nc(-c4ccc(C#N)cc4)c3)cc2)c2cccc21</chem>
HDP	<chem>O=C1c2cccc2C(=O)c2nc3c(c c(0)c4cccc43)nc21 CC(C)(C)c1ccc(N(c2ccc(-c3cc4nc5c6ccc(C#N)cc6c6cc(C#N)ccc6c5nc4cc3-c3cccnc3)cc2)c2ccc(C(C)(C)C)cc2)cc1</chem>	PyDCN	<chem>N#Cc1ccc(-c2cc(-c3cccc3)cc(-c3ccc(C#N)cc3)n2)cc1</chem>
PY-TPA	<chem>CC(C)(C)c1ccc(N(c2ccc(-c3cc4nc5c6ccc(C#N)cc6c6cc(C#N)ccc6c5nc4cc3-c3cccnc3)cc2)c2ccc(C(C)(C)C)cc2)cc1</chem>	PXZ-OXD	<chem>c1ccc(-c2nnc(-c3ccc(N4c5cccc50c5cccc54)cc3)o2)cc1</chem>
CN-TPA	<chem>CC(C)(C)c1ccc(N(c2ccc(-c3cc4nc5c6ccc(C#N)cc6c6cc(C#N)ccc6c5nc4cc3C#N)cc2)c2ccc(C(C)(C)C)cc2)cc1</chem>	TCz-3PA-TCz	<chem>CC(C)(C)c1ccc(-c2ccc3c(c2)c2cc(-c4cnc(-c5ccc6c(c5)c5cc(-c7ccc(C(C)(C)C)cc7)ccc5n6-c5ccc(C(C)(C)C)cc5)cn4)ccc2n3-c2ccc(C(C)(C)C)cc2)cc1 Cc1c(-c2cccc2)cc(N2C(=O)c3cccc4c(N(c5ccc(-c6cccc6)cc5)c5ccc6c(c5)C(C)(C)C5cccc5-6)ccc(c34)C2=O)cc1-c1cccc1</chem>
DMIC-TRZ	<chem>CC1(C)CC(c2cccc(-c3nc(-c4cccc4)nc(-c4cccc4)n3)c2)C=c2[nH]c3cc4c(cc3c21)-c1cccc1C=4</chem>	NAI-BiFA	

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
NyDPAC	<chem>CC1(C)c2cccc2Nc2cccc(-c3ccc(-c4ccc5nc(-c6ccc(-c7cccc8c7C(C)(C)c7cccc7N8)c6)ccc5n4)cc3)c21</chem>	NAI-PhBiFA	<chem>Cc1c(-c2cccc2)cc(N2C(=O)c3cccc4c(-c5ccc(N(c6ccc(-c7cccc7)cc6)c6ccc7c(c6)C(C)(C)c6cccc6-7)cc5)ccc(c34)C2=O)cc1-c1cccc1CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(S(=O)(=O)c2ccc(-n3c4ccc(C(C)(C)C)cc4cc(C(C)(C)C)ccc43)cn2)nc1O=C(c1ccc(N2c3cccc3C3(c4cccc4-c4cccc43)c3cc4ccc4cc32)cc1)c1ccc(N2c3cccc3C3(c4cccc4-c4cccc43)c3cc4cccc4cc32)cc1</chem>
oCBP	<chem>c1ccc(-n2c3cccc3c3cccc3c2)c(-c2cccc2-n2c3cccc3c3cccc32)c1</chem>	pDTCz-2DPyS	
PABP	<chem>Cc1cc(-c2c3cccc3c(-c3cccc3)c3cccc23)ccc1-c1ccc(-c2c3cccc3c(-c3cccc3)c3cccc23)cc1C</chem>	2SPBAC-BP	
DHID-DPS	<chem>CCn1c2cccc2c2c1c1cccc1n2-c1ccc(S(=O)(=O)c2ccc(-n3c4cccc4c4c3cccc3n4CC)cc2)cc1</chem>	SPBAC-PH	

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
4FBTPI	<chem>CC(C)(C)c1ccc(-n2c(-c3ccc(-c4c(F)c(F)c(-c5ccc(-c6nc7c8cccc8c8cccc8c7n6-c6ccc(C(C)(C)C)cc6)cc5)c(F)c4F)cc3)nc3c4cccc4c4cccc4c32)cc1</chem>	DFPBI	<chem>CC1(C)c2cc(-c3nc(-c4cccc4)c(-c4cccc4)n3-c3cccc3)ccc2-c2ccc(N(c3cccc3)c3cccc3)cc21</chem>
BCzBMes	<chem>Cc1cc(C)c(B(c2c(C)cc(C)cc2C)n2c3cccc3c3cc(-c4ccc5c(c4)c4cccc4n5-c4cccc4)ccc32)c(C)c1</chem>	TDPA-TRZ	<chem>c1ccc(N(c2cccc2)c2ccc(-c3nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)n3)cc2)cc1</chem>
CCDMB	<chem>Cc1cc(C)c(B(c2ccc(-n3c4cccc4c4cc(-n5c6cccc6c6cccc65)ccc43)cc2)c2c(C)cc(C)cc2C)c(C)c1</chem>	DMAC-11-DPPZ	<chem>CC1(C)c2cccc2N(c2ccc3nc4c5cccc5c5cccc5c4nc3c2)c2cccc21</chem>
PCDMB	<chem>Cc1cc(C)c(B(c2ccc(-n3c4cccc4c4cc(N5c6cccc60c6cccc65)ccc43)cc2)c2c(C)cc(C)cc2C)c(C)c1</chem>	PXZ-11-DPPZ	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc2nc3c4cccc4c4cccc4c3nc2c1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
BTrztCz	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1cccc1-c1nc(-c2cccc2)nc(-c2ccc(-c3nc(-c4cccc4)nc(-c4cccc4-n4c5ccc(C(C)(C)C)cc5c5cc(C(C)(C)C)ccc54)n3)cc2)n1</chem>	PXZ-10-DPPZ	<chem>c1ccc2c(c1)Oc1cccc1N2c1cccc2nc3c4cccc4c4cccc4c3nc12</chem>
DHID-DBS	<chem>CCCCn1c2cccc2c2c1c1cccc1n2-c1ccc2c(c1)-c1cc(-n3c4cccc4c4c3c3cccc3n4CCCC)ccc1S2(=O)=O</chem>	CCC	<chem>c1ccc2c(c1)Cc1c-2ccc2c1-c1cccc1C2</chem>
CZ-DPS-DMAC	<chem>CC1(C)c2cccc2N(c2ccc(S(=O)(=O)c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)cc2)c2cccc21Cc1ccc(N(c2ccc(C)cc2)c2ccc2C2(c3cccc3N(c3ccc(C)cc3)c3ccc(C)cc3)CCCC2)cc1</chem>	CzmPPC	<chem>c1ccc(-c2cccc(-c3nc(-c4ccc(-c5cccc5)c4)nc(-c4ccc(-c5cccc5)c4)n3)c2)cc1</chem>
XT-T	<chem>1</chem>	C2FLA-1	<chem>CCCC1(CCC)c2cccc2-c2ccc(C#Cc3ccc4c(c3)C(CCC)(CCC)c3cc(C#N)ccc3-4)cc21</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
3DPTIA	<chem>CC1(C)c2cccc2N(c2nc(-c3cccc3)nc(-c3cccc3)n2)c2c3c4cccc4n(-c4cccc4)c3cc21</chem>	MDP3FL	<chem>Cc1cccc(N(c2cccc2)c2ccc3c(c2)C(C)(C)c2cc(-c4ccc5c(c4)C(C)(C)c4cc(-c6ccc7c(c6)C(C)(C)c6cc(N(c8cccc8)c8cccc(C)c8)ccc6-7)ccc4-5)ccc2-3)c1CCCC1(CCCC)c2cc(C#Cc3ccc4c(c3)C(CCC)(CCC)c3cc(C#N)ccc3-4)ccc2-c2ccc(C#Cc3ccc4c(c3)C(CCC)(CCC)c3cc(C#N)ccc3-4)cc21O=P(c1cccc1)(c1cccc1)c1cccc10c1ccc(-c2ccc(C3(c4ccc(-n5c6cccc6c6cccc65)cc4)c4cccc4-c4cccc43)cc2)cc1P(=O)(c1cccc1)c1ccc1</chem>
PyIAAnTPh	<chem>c1ccc(-c2cc(-c3cccc3)cc(-c3c4cccc4c(-c4ccc(-c5nc6c7cccc8ccc9cccc(c9c87)c6n5-c5cccc5)cc4)c4cccc34)c2)cc1</chem>	C3FLA-2	<chem>c1ccc(-c2ccc(-c3cccc3)cc(-c3c4cccc4c(-c4ccc(-c5nc6c7cccc8ccc9cccc(c9c87)c6n5-c5cccc5)cc4)c4cccc34)c2)cc1</chem>
2CzP-INN	<chem>N#Cc1c(-c2ccc(-n3c4cccc4c4cccc43)cc2)cncc1-c1ccc(-n2c3cccc3c3cccc32)cc1</chem>	9CzFDPEPO	<chem>c1ccc(-c2ccc(-c3cccc3)cc(-c3c4cccc4c4cccc43)cc2)cc1P(=O)(c1cccc1)c1ccc1</chem>
VBNO	<chem>C=Cc1ccc(COc2ccc3c(c2)c2cccc2n3-c2cccc(-n3c4cccc4c4cc(Oc5ccc(C=C)cc5)ccc43)c2C#N)cc1</chem>	9CzFDPEPO	<chem>c1ccc(-c2ccc(-c3cccc3)cc(-c3c4cccc4c4cccc43)cc2)cc1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
PIAnTPh	<chem>c1ccc(-c2cc(-c3ccccc3)cc(-c3c4ccccc4c(-c4ccc(-c5nc6c7ccccc7c7ccccc7c6n5-c5ccccc5)cc4)c4ccccc34)c2)cc</chem>	CTM	<chem>O=C(c1ccc(-n2c3ccccc3c3ccc32)cc1)c1ccc2sc3ccccc3c2c1</chem>
BCZ-DPS-AD	<chem>CC1(C)c2ccccc2N(c2cc(N3c4ccccc4C(C)(C)c4ccccc43)cc(S(=O)(=O)c3ccc4c(c3)c3ccc3n4-c3ccccc3)c2)c2cccc</chem>	tCTM	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(C(=O)c2ccc3sc4ccccc4c3c2)cc1</chem>
2BTPA-INN	<chem>c21N#Cc1c(-c2ccc(N(c3ccc(-c4ccccc4)cc3)c3ccc(-c4ccccc4)cc3)cc2)cncc1-c1ccc(N(c2ccc(-c3ccccc3)cc2)c2ccc(-c3ccccc3)cc2)cc1</chem>	PSBC	<chem>CC(C)(C)c1ccc(-n2c(-c3ccc(-c4ccc(S(=O)(=O)c5ccc(-c6ccc(-n7c8ccccc8c8ccccc87)cc6)cc5)cc4)cc3)nc3c4ccc4c4ccccc4c32)cc1CC1(C)c2ccccc2N(c2ccc(C(=O)c3ccccc3(C(=O)c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)n3)cc2)c2ccccc21CC1(C)c2ccccc2N(c2ccc(C(=O)c3ccccc3(C(=O)c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)nc3)cc2)c2ccccc21</chem>
SpiroAc	<chem>c1ccc2c(c1)Nc1ccccc1C21c2ccccc2-c2ccccc21</chem>	26DAcBPy	
BuPCzBCO	<chem>Cc1ccc2c3cc(Br)c(C)cc3n(-c3ccc(C(C)(C)C)cc3)c2c1</chem>	25DAcBPy	

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
AQ-S-CZ	<chem>O=C1c2ccccc2C(=O)c2cc(Sc3ccccc(-n4c5ccccc5c5ccccc54)cc3)ccc21</chem>	26DPXZBPy	<chem>O=C(c1ccc(N2c3ccccc3Oc3ccccc32)cc1)c1cccc(C(=O)c2cc(N3c4ccccc4Oc4ccccc43)cc2)n1</chem>
DBP-3TPA	<chem>c1ccc(N(c2ccccc2)c2ccc(-c3ccc4nc5c6ccc(-c7ccc(N(c8ccccc8)c8ccccc8)cc7)cc6c6cc(-c7ccc(N(c8ccccc8)c8ccccc8)cc7)ccc6c5nc4c3)cc2)cc1</chem>	BTITrz	<chem>N#Cc1cc(-c2nc(-c3ccccc3)nc(-c3ccccc3)n2)ccc1-n1c2ccccc2c2c3ccccc3sc21</chem>
DCzIPN	<chem>N#Cc1cc(C#N)c(-n2c3ccccc3c3ccccc32)cc1-n1c2ccccc2c2ccccc21</chem>	mTPA-PPI	<chem>c1ccc(N(c2ccccc2)c2ccc(-c3ccccc(-c4nc5c6ccccc6c6ccccc6c5n4-c4ccccc4)c3)cc2)cc1</chem>
DMeCzIPN	<chem>Cc1cccc2c3ccccc3n(-c3cc(-n4c5ccccc5c5cccc(C)c54)c(C#N)cc3C#N)c12</chem>	PPI	<chem>c1ccc(-c2nc3c4ccccc4c4ccccc4c3n2-c2ccccc2)cc1</chem>
SAFDPA	<chem>c1ccc(N(c2ccccc2)c2ccc3c(c2)-c2ccccc2C32c3ccccc3N(c3ccccc3)c3ccccc32)cc1</chem>	CzTPA-m-Trz	<chem>c1ccc(-c2ccc(N(c3ccc(-c4cc5c(c4)c4ccccc4n5-c4ccccc4)cc3)c3ccccc(-c4nc(-c5ccccc5)n4)c3)cc2)cc1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
t-DABNA	<chem>CC(C)(C)c1ccc(N2c3ccc(C(C)(C)C)cc3B3c4cc(C(C)(C)C)ccc4N(c4ccc(C(C)(C)C)cc4)c4cccc2c43)cc1</chem>	TPA-2PPITPA	<chem>c1ccc(N(c2ccc(-c3ccc(-c4nc5c6cccc6c6cccc6c5n4-c4ccc(N(c5cccc5)c5cccc5)c4)cc3)cc2)c2ccc(-c3ccc(-c4nc5c6cccc6c6cccc6c5n4-c4ccc(N(c5cccc5)c5cccc5)cc4)cc3)cc2)cc1C(=C/c1nc2c3cccc3c3cccc3c2n1-c1ccc(N(c2cccc2)c2cccc2)cc1)\c1ccc(-c2ccc(N(c3cccc3)c3ccc(-c4ccc(/C=C/c5nc6c7cccc7c7cccc7c6n5-c5ccc(N(c6cccc6)c6cccc6)cc5)cc4)cc3)cc2)cc1CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1c(-n2c3cccc3c3cccc32)cc(-n2c3cccc3c3cccc32)c(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1C#N</chem>
3-AcBPtCN	<chem>CC1(C)c2cccc2N(c2ccc3c(c2)c2cc(C#N)ccc2c2nc4cc(C#N)c(C#N)cc4nc32)c2cccc21</chem>	TPA-2SPPITPA	
3-AcBPdCN	<chem>CC1(C)c2cccc2N(c2ccc3c(c2)c2cccc2c2nc4cc(C#N)c(C#N)cc4nc32)c2cccc21</chem>	2Cz2tCzBn	

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
11-AcBPdCN	<chem>CC1(C)c2cccc2N(c2ccc3nc4c5ccc(C#N)cc5c5cc(C#N)ccc5c4nc3c2)c2cccc21</chem>	2tCz2CzBn	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c(-n2c3cccc3c3cccc32)c(C#N)c1-n1c2cccc2c2cccc21</chem>
PID-3	<chem>c1ccc(-n2c(-c3ccc(-c4cccs4)s3)nc3c4cccc4c4cccc4c32)cc1</chem>	PmPmP	<chem>cccc50c5cccc54)cc3)cc(-c3cncnc3)n2)c1</chem>
3-ACBPmCN	<chem>CC1(C)c2cccc2N(c2ccc3c(c2)c2cc(C#N)ccc2c2nc4cccc4nc32)c2cccc21</chem>	PLQYs	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1cc2ccc3c(-n4c5ccc(C(C)(C)C)cc5c5cc(C(C)(C)C)ccc54)cc4ccc(c1)c2c34</chem>
3-DTPA-BBI	<chem>O=c1c2ccc(-c3ccc(N(c4cccc4)c4cccc4)cc3)c3c(-c4cc(N(c5cccc5)c5cccc5)cc4)ccc(c32)c2nc3cccc3n12</chem>	PXZ-ND	<chem>CC1(C)c2cccc2-c2cc3c(cc2cc2ccc(N3c4cccc40c4cccc43)nc2n1</chem>
IABTCN	<chem>CC1(C)c2cccc2-c2cc3c(cc21)N(c1cccc1)c1ccc(-c2ccc(C#N)c4nsnc24)cc1C3(c1ccc1)c1cccc1</chem>	CPPyC	<chem>c1cc(-c2cncc(-n3c4cccc4c4cccc43)c2)cc(-n2c3cccc3c3cccc32)c1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
BTZPP	<chem>CC(C)(C)c1ccc(-n2c(-c3ccc(-c4cccc5nsnc45)cc3)cc3c2cc(-c2ccc(-c4cccc5nsnc45)cc2)n3-c2ccc(C(C)(C)C)cc2)cc1</chem>	DMAC-ND	<chem>CC1(C)c2cccc2N(c2ccc3ccc(N4c5cccc5C(C)(C)c5cccc54)nc3n2)c2cccc21</chem>
DPimdPPA	<chem>N#C/C(=C\c1ccc(-c2nc(-c3cccc3)c(-c3cccc3)[nH]2)c c1)c1cccc1</chem>	PrDPhAc	<chem>c1ccc(C2(c3cccc3)c3cccc3N(c3cncc(N4c5cccc5C(c5cccc5)(c5cccc5)c5cccc54)n3)c3cccc32)cc1Cc1nc2cccc2n1-c1cc(S(=O)(=O)c2ccc(C(C)(C)C)cc2)c(-n2c(C)nc3cccc32)cc1S(=O)(=O)c1ccc(C(C)(C)C)cc1CC(C)(C)c1ccc(N2C(=O)c3ccc4c(N5c6cccc6C(C)(C)c6ccc7c(oc8cccc87)c65)ccc(c34)C2=O)cc1</chem>
BCzPh	<chem>c1ccc(-n2c3cccc3c3cc(-c4ccc5c(c4)c4cccc4n5-c4ccc(cc4)ccc32)cc1N#Cc1ccc(-c2ccc3c(c2)c2cc(-c4ccc(C#N)cc4)ccc2c2c3nc(-c3ccc(-c4ccc(N(c5cccc5)c5cccc5)cc4)cc3)n2-c2cccc2)cc1N#Cc1ccc(-c2ccc3c(c2)c2cc(-c4ccc(C#N)cc4)ccc2c2c3nc(-c3ccc(N(c4cccc4)c4ccc(cc4)cc3)n2-c2cccc2)cc1</chem>	2Mi2SB	
TBPM2CN		BFDMAc-NAI	
TPM2CN		HPIPXZ	

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Molecule Id	SMILES	Molecule Id	SMILES
PT-BZ-DMAC	<chem>CC1(C)c2ccccc2N(c2ccc(C(=O)c3ccc(Sc4ccccc4)cc3)cc2)c2ccccc21</chem>	3BPy-mDTC	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1cc(C(=O)c2cccn2)cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1</chem>
PS-BZ-DMAC	<chem>CC1(C)c2ccccc2N(c2ccc(C(=O)c3ccc(S(=O)(=O)c4ccccc4)cc3)cc2)c2ccccc21</chem>	2BPy-mDTC	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1cc(C(=O)c2cccn2)cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1</chem>
2PCzBN-tPh	<chem>CC(C)(C)c1ccc(-c2cc(-n3c4ccc(-c5ccccc5)cc4c4cc(-c5ccccc5)ccc43)c(C#N)c(-n3c4ccc(-c5ccccc5)cc4c4cc(-c5ccccc5)ccc43)c2)cc1N#Cc1c(-n2c3ccc(-c4ccccc4)cc3c3cc(-c4ccccc4)ccc32)cccc1-n1c2ccc(-c3ccccc3)c2c2cc(-c3ccccc3)ccc21O=C1c2ccccc2C(=O)c2c1ccc1nc(-c3ccc(N(c4ccccc4)c4ccccc4)cc3)c(-c3ccc(N(c4ccccc4)c4ccccc4)cc3)nc21</chem>	BP-mDTC	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1cc(C(=O)c2ccccc2)cc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1</chem>
2PCzBN		Cz2ICz	<chem>c1ccc2c(c1)c1ccccc1n2-c1cc2c(c1)c1cccc3c4cc(-n5c6ccccc6c6ccccc65)ccc4n2c13</chem>
TPA-2		Ac-BPCN	<chem>CC1(C)c2ccccc2N(c2ccc3nc4c5cc(C#N)ccc5c5ccc(C#N)cc5c4nc3c2)c2ccccc21</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
36DTPAFM	<chem>N#CC(C#N)=C1c2ccc(-c3ccc(N(c4cccc4)c4cccc4)cc3)c2-c2cc(-c3ccc(N(c4cccc4)c4cccc4)cc3)ccc21</chem>	5CzICz	<chem>c1ccc2c(c1)c1cccc1n2-c1ccc2c(c1)c1cccc3c4cccc4n2c31</chem>
AQTC	<chem>N#Cc1cc2nc3c(nc2c(C#N)c1C#N)-c1cccc2cccc-3c12</chem>	DCPDAPM	<chem>CC1(C)c2cccc2N(c2ccc(C(=O)c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)cc2)c2cccc21</chem>
3TPAFM	<chem>N#CC(C#N)=C1c2cccc2-c2cc(-c3ccc(N(c4cccc4)c4cccc4)cc3)ccc21</chem>	SPFS-PXZ	<chem>O=S1(=O)c2ccc(N3c4cccc40c4cccc43)cc2C2(c3cccc3-c3cccc32)c2cc(N3c4cccc40c4cccc43)ccc21</chem>
BBPA	<chem>CCCCOc1cccc1-c1c2cccc2c(-c2cccc20CCCC)c2cccc12</chem>	DPS-PXZ	<chem>O=S(=O)(c1ccc(N2c3cccc30c3cccc32)cc1)c1ccc(N2c3cccc30c3cccc32)cc1</chem>
DMPA	<chem>COc1ccc(-c2c3cccc3c(-c3ccc(OC)cc3OC)c3cccc23)c(OC)c1</chem>	TX-CzPh	<chem>O=C1c2cccc3c(N4c5cccc50c5cccc54)ccc(c23)C(=O)N1c1cccc1</chem>
PXZ-NAI	<chem>1cccc1</chem>	bpipH	<chem>Brc1cccc(-c2nc3c4cccnc4c4ncccc4c3[nH]2)n1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
PTZ-NAI	<chem>O=C1c2cccc3c(N4c5cccc5Sc5cccc54)ccc(c23)C(=O)N1c1cccc1</chem>	Tri-PXZ-PCN	<chem>N#Cc1c(-c2ccc(N3c4cccc40c4cccc43)cc2)nc(-c2ccc(N3c4cccc40c4cccc43)cc2)c(C#N)c1-c1ccc(N2c3cccc30c3cccc32)cc1</chem>
AI-4Cz	<chem>O=C1c2c(c(-n3c4cccc4c4cccc43)c(-n3c4cccc4c4cccc43)c(-n3c4cccc4c4cccc43)c2-n2c3cccc3c3cccc32)C(=O)N1c1cccc1</chem>	ACID-BPPZ	<chem>CC1(C)c2cccc2N(c2ccc3c(c2)cccc2c2nc4c5cccn5c5ncccc5c4nc32)c2cc3c4cccc4n(-c4cccc4)c3cc21</chem>
DACIPN	<chem>CC1(C)c2cccc2N(c2cc(N3c4cccc4C(C)(C)c4cccc43)c(C#N)cc2C#N)c2cccc21</chem>	C-BPyIA	<chem>c1ccc(-n2c(-c3ccc(-c4c5ccc5c(-c5ccc(-c6nc7c8cccc9ccc%10cccc(c%10c98)c7n6-c6cccc6)cc5)c5cccc45)cc3)nc3c4cccc5ccc6ccc(c6c54)c32)cc1</chem>
TCz1	<chem>CCCCC(CC)Cn1c2ccc(-n3c4cccc4c4cccc43)cc2c2cc(-n3c4cccc4c4cccc43)ccc21</chem>	N-BPyIA	<chem>c1ccc(-c2nc3c4cccc5ccc6cc(cc(c6c54)c3n2-c2ccc(-c3c4cccc4c(-c4ccc(-n5c(-c6ccc6)nc6c7cccc8ccc9cccc(c9c87)c65)cc4)c4cccc34)cc2)cc1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
MBF	<chem>F[B-]1(F)OC(c2ccccc2)=Cc2cc(-c3ccc(N4c5ccccc5O5ccccc54)cc3)nc[n+]</chem> 21	2-TA-AN	<chem>c1ccc2c(c1)Sc1ccc(-c3c4ccccc4cc4ccccc34)cc1S2</chem>
MPA	<chem>O=C(Cc1cc(-c2ccc(N3c4ccccc4O3c4ccccc43)cc2)ncn1)c1cccc1</chem>	ANOP-MeFAC	<chem>Cc1ccc2c(c1)C1(c3ccccc3-c3ccccc31)c1cc(C)ccc1N2c1cccc2c3c(cccc13)-c1nc3nonc3nc1-2</chem>
3,3-NHC	<chem>O=c1oc2ccccc2c(O)c1C(c1ccc2ccccc2c1)c1c(O)c2ccccc2oc1=O</chem>	ANOP-DMAC	<chem>CC1(C)c2ccccc2N(c2ccc3c4c(cccc24)-c2nc4nonc4nc2-3)c2ccccc21</chem>
3CP-DPS-DMAC	<chem>CC1(C)c2ccccc2N(c2ccc(S(=O)(=O)c3ccc4c(c3)c3ccccc3n4-c3ccccc3)cc2)c2ccccc21</chem>	4PyCNICz	<chem>N#Cc1c(-c2ccccc2)nc(-c2cccc2)nc1-c1ccc(-n2c3ccccc3c3ccc4c(c5ccccc5n4-c4cccc4)c32)cc1</chem>
3CP-DPS-PXZ	<chem>O=S(=O)(c1ccc(N2c3ccccc3O3c3ccccc32)cc1)c1ccc2c(c1)c1cccc1n2-c1cccc1</chem>	2PyCNBCz	<chem>N#Cc1c(-c2ccccc2)nc(-c2ccc(-n3c4ccccc4c4cc(-c5ccc6c(c5)c5ccccc5n6-c5ccccc5)ccc43)cc2)nc1-c1cccc1Cc1ccc(N(c2ccc(C)cc2)c2c3ccccc3c(-c3ccc(P(=O)(c4cccc4)c4ccc(-n5c6ccccc6c6cccc65)cc4)cc3)c3ccccc23)cc1</chem>
3CzAnN	<chem>c1ccc(-n2c3ccccc3c3cc(-c4c5ccccc5c(-c5ccc6ccccc6c5)c5ccccc45)ccc32)cc1</chem>	CzA-PPO	

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Molecule Id	SMILES	Molecule Id	SMILES
DPQ-DMAC	<chem>CC1(C)c2cccc2N(c2ccc3nc(-c4ccccc4)c(-c4ccccc4)nc3c2)c2cccc21</chem>	2PyCNICz	<chem>N#Cc1c(-c2cccc2)nc(-c2cc(-n3c4ccccc4c4ccc5c(c6ccccc6n5-c5ccccc5)c43)cc2)n</chem>
DPAA	<chem>c1ccc(-c2c3ccccc3c(-c3cccc3)c3ccccc23)cc1</chem>	TAA-PPO	<chem>c1-c1cccc1Cc1ccc(N(c2ccc(C)cc2)c2c3ccccc3c(-c3ccc(P(=O)(c4ccccc4)c4ccccc4)cc3)c3ccccc23)cc1</chem>
t3Cz-SO	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc2c(c1)c1cc(-n3c4ccc(C(C)(C)C)c4c4cc(C(C)(C)C)ccc43)ccc1n2-c1ccc(S(=O)(=O)c2cccc2)cc1</chem>	pSFIAc2	<chem>Cc1ccc2c(c1)c1c3c4c(c5c1n2-c1ccc(C(C)(C)C)cc1C51c2ccccc2-c2cccc21)c1cc(C)cc1n4-c1ccc(C(C)(C)C)cc1C31c2cccc2-c2cccc21</chem>
Cz-SO	<chem>O=S(=O)(c1cccc1)c1ccc(-n2c3ccccc3c3ccccc32)cc1</chem>	TBN-TPA	<chem>CC(C)(C)c1ccc(N2c3ccc(C(C)(C)C)cc3B3c4cc(C(C)(C)C)ccc4N(c4ccc(C(C)(C)C)cc4)c4cc(-n5c6ccc(C(C)(C)C)cc6c6cc(C(C)(C)C)ccc65)cc2c43)cc1</chem>
DPQ-PXZ	<chem>c1ccc(-c2nc3ccc(N4c5ccccc50c5ccccc54)cc3nc2-c2ccccn2)nc1</chem>	CzBPDCb	<chem>c1cc(-c2cccc(-n3c4ccccc4c4ncccc43)c2)cc(-n2c3ccccc3c3ccccc32)c1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
pCzAnN	<chem>c1ccc2cc(-c3c4ccccc4c(-c4ccc(-n5c6ccccc6c6ccccc65)cc4)c4ccccc34)ccc2c1</chem>	[2,1-b]IF	<chem>C1=CC2(C=C3C1=CC1=C3C3(C4=C5ccccc5=CC4=C1)c1ccccc1Sc1ccccc13)c1ccccc1Sc1cccc12</chem>
3-BHH	<chem>O=c1oc2ccccc2c(=O)c1C(c1cc(Br)s1)c1c(=O)c2ccccc2oc1=O</chem>	cyclohexanone	<chem>BrCCCCCCCC1(CCCCCCBr)c2cc(Br)ccc2-c2ccc(Br)cc21</chem>
DPP-PXZ	<chem>c1ccc(-c2nc3cc(N4c5ccccc5Oc5ccccc54)cnc3nc2-c2cccn2)nc1</chem>	thio-ether	<chem>C1=CC2(C=C3C1=CC1=C3C3(C4=C5ccccc5=CC4=C1)c1ccccc1Sc1ccccc13)c1ccccc1Sc1cccc12</chem>
APPT-PXZ	<chem>c1ccc2c(c1)Oc1ccccc1N2c1ccc2c3c(cccc13)-c1nc3c4cccn4c4ncccc4c3nc1-2</chem>	ICz-2	<chem>c1ccc(-c2cccnc2-c2cc(-c3c[nH]nn3)cc(-c3cccn3)n2)c1</chem>
HITPE	<chem>Oc1ccc(-c2ccc(C(=C(c3cccc3)c3ccccc3)c3ccccc3)cc2)cc1-c1nc(-c2ccccc2)c(-c2ccccc2)n1-c1ccccc1</chem>	DDCzTrz	<chem>c1ccc(-c2nc(-c3cc(-n4c5ccccc5c5ccccc54)cc(-n4c5cccc5c5ccccc54)c3)nc(-c3cc(-n4c5ccccc5c5ccccc54)cc(-n4c5ccccc5c5ccccc54)c3)n2)cc1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
DV-3CzCN	<chem>C=Cc1ccc(COc2cccc3c2c2ccc cc2n3-c2cc(-n3c4cccc4c4c (OCc5ccc(C=C)cc5)cccc43)c (C#N)c(-n3c4cccc4c4c(OCc 5ccc(C=C)cc5)cccc43)c2)cc</chem>	B-O-dpa	<chem>CC(C)(C)c1cc2c3c(c1)N(c1c cccc1)c1cccc1B3c1cccc1O 2</chem>
SF-DPSO	<chem>O=S(=O)(c1ccc(-c2ccc3c(c2)C2(c4cccc4-c4cccc42)c2 cccc2-3)cc1)c1ccc(-c2ccc 3c(c2)C2(c4cccc4-c4cccc 42)c2cccc2-3)cc1 CC(C)(C)c1ccc2c(c1)c1cc(C (C)(C)C)ccc1n2-c1ccc(-n2c 3ccc(C(C)(C)C)cc3c3cc(C(C) (C)C)ccc32)c2nc(-c3cccc 3)c(-c3cccc3)nc12 CC1(C)c2cccc2N(c2ccc(N3c 4cccc4C(C)(C)c4cccc43)c 3nc(-c4cccc4)c(-c4cccc4)nc23)c2cccc21</chem>	B-dpa-SpiroAC	<chem>CC(C)(C)c1cc2c3c(c1)N1c4c cccc4C4(c5cccc5-c5cccc5 4)c4cccc(c41)B3c1cccc1N2 c1cccc1</chem>
BDQ-tBuCz		DCz-OTP	<chem>c1cc(-c2cccc2-c2ccc(-n3 c4cccc4c4cccc43)c2)cc(- n2c3cccc3c3cccc32)c1</chem>
BDQDMAC		B-dpa-Cz	<chem>CC(C)(C)c1cc2c3c(c1)-n1c4 cccc4c4cccc(c41)B3c1cccc c1N2c1cccc1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
BDQPCz	<chem>c1ccc(-c2nc3c(-c4ccc(-n5c6cccc6c6cccc65)cc4)ccc(-c4ccc(-n5c6cccc6c6cccc65)cc4)c3nc2-c2cccc2)cc1</chem>	c-D2T	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(N2c3cccc3C3(c4cccc4-c4cc5(cc43)C3(c4cccc4-5)c4cccc4N(c4ccc(-c5nc(-c6cccc6)nc(C6CCCCC6)n5)cc4)c4cccc43)c3cccc32)cc1O=C(c1cccc1)c1ccc2nc3c4c</chem>
4-SCZ	<chem>c1ccc2c(c1)-c1cccc1C21c2cccc2-c2c1ccc1[nH]c3cccc3c21</chem>	DPXZ-DPPM	<chem>cc(N5c6cccc60c6cccc65)c4c4cc(N5c6cccc60c6cccc65)ccc4c3nc2c1</chem>
Py-TPA	<chem>CC(C)(C)c1ccc(N(c2ccc(-c3cc4nc5c6ccc(-c7ccncc7)nc6c6nc(-c7ccncc7)ccc6c5nc4cc3C#N)cc2)c2ccc(C(C)(C)C)cc2)cc1</chem>	CBPMDN	<chem>N#Cc1ccc(-n2c(-c3ccc(-n4c5cccc5c5cccc54)cc3)nc3c4cccc4c4cccc4c32)cc1</chem>
DPACPhTPI	<chem>CC(C)(C)c1ccc(-n2c(-c3ccc(-c4ccc(N5c6cccc6C(c6cccc6)(c6cccc65)c4)cc3)nc3c4cccc4c4cccc4c32)cc1</chem>	CPBMDN	<chem>N#Cc1ccc(-n2c(-c3ccc(-c4cc(-n5c6cccc6c6cccc65)c4)cc3)nc3c4cccc4c4cccc4c32)cc1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
oPTBC	<chem>N#Cc1ccc(-c2cc(C#N)cc(-c3ccc(C#N)cc3)c2N2c3cccc30c3cccc32)cc1</chem>	TPB-AC	<chem>N#Cc1ccc(-c2cc(-c3cccc3)c(-c3ccc(N(c4cccc4)c4ccc4)cc3)cc2-c2cccc2)cc1</chem>
mPTBC	<chem>N#Cc1ccc(-c2cc(N3c4cccc40c4cccc43)cc(-c3ccc(C#N)cc3)c2C#N)cc1</chem>	Me-DMAC	<chem>N#Cc1c(-n2c(-c3cccc3)nc3cccc32)cc(-n2c(-c3cccc3)nc3cccc32)cc1-n1c(-c2ccc2)nc2cccc21</chem>
P3C	<chem>CCCC(Cc1c2cc(C#N)ccc2c2ccc(C#Cc3ccc4ccc5cccc6cc3c4c56)cc21</chem>	iTPBI-CN	<chem>c1ccc2c(c1)0c1cccc1N2c1cc(-n2c3cccc3c3cccc4ccc5c6ccc6c1c1c5c4c3c210=C1c2cccc2C(=O)c2cc(N(c3ccc4c(c3)C(=O)c3cccc3C4=O)c3ccc4c(c3)C(=O)c3cccc3C4=O)ccc21</chem>
DBCP	<chem>0=C(c1ccc(N2c3cccc30c3cccc32)cc1)c1ccc2c(c1)c1ccc1n2-c1cccc(-n2c3cccc3c3cccc32)c1</chem>	TCP	<chem>3cc(C(=O)c4ccc(N5c6cccc60c6cccc65)cc4)ccc32)cc1Cc1cc(-n2c3cccc3c3cccc32)ccc1-c1ccc(-n2c3cccc3c3cccc32)ccc(C(=O)c4ccc(N5c6cccc6C(C)(C)c6cccc65)cc4)ccc32)cc1C</chem>
NAQ3	<chem>0=C(c1ccc(N2c3cccc30c3cccc32)cc1)c1ccc2c(c1)c1ccc1n2-c1cccc(-n2c3cccc3c3cccc32)c1</chem>	CDBP-BP-PXZ	
mCP-BP-DMAC		CDBP-BP-DMAC	

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Molecule Id	SMILES	Molecule Id	SMILES
BFCZPZ3	<chem>c1ccc2c(c1)oc1ccc3c4ccccc4n(-c4cnc(-n5c6cccc6c6cc7oc8cccc8c7c65)cn4)c3c1</chem>	3CzPhPM	<chem>c1ccc(-c2ccc3c(c2)c2ccccc2n3-c2ccc(-c3cc(-c4ccccc4)nc(-c4ccccc4)n3)cc2)cc1</chem>
BTCZPZ1	<chem>c1ccc2c(c1)sc1c2ccc2c3ccc2cc3n(-c3cnc(-n4c5cccc5c5ccc6c7cccc7sc6c54)cn3)c2</chem>	SFI23pPM	<chem>c1ccc(-c2cc(-c3ccc(-n4c5cccc5c5cc6c(cc54)C4(c5cccc5-c5cccc54)c4cccc4-6)cc3)nc(-c3cccc3)n2)cc1</chem>
BFCZPZ2	<chem>c1ccc2c(c1)oc1c2ccc2c1c1cccc1n2-c1cnc(-n2c3cccc3c3c4oc5cccc5c4ccc32)cn1</chem>	3PCzPFP	<chem>c1ccc(-c2cc(-c3ccc(-n4c5cccc5c5c6c(ccc54)-c4cccc4C64c5cccc5-c5cccc54)cc3)nc(-c3cccc3)n2)cc1</chem>
CZ9CZPZ	<chem>c1ccc2c(c1)c1cccc1n2-c1cnc(-n2c3cccc3c3cc(-n4c5cccc5c5cccc54)ccc32)cn1</chem>	SFI34pPM	<chem>c1ccc(-c2cc(-c3ccc(-n4c5cccc5c5c6c(ccc54)-c4cccc4C64c5cccc5-c5cccc54)cc3)nc(-c3cccc3)n2)cc1</chem>
BFCZPZ1	<chem>c1ccc2c(c1)oc1c2ccc2c3ccc2cc3n(-c3cnc(-n4c5cccc5c5ccc6c7cccc7oc6c54)cn3)c2</chem>	SBA-2DPS	<chem>c1ccc(-c2cc(-c3ccc(-n4c5cccc5c5c6c(ccc54)-c4cccc4C64c5cccc5-c5cccc54)cc3)nc(-c3cccc3)n2)cc1</chem>
TPBPPI-PY	<chem>c1ccc(-c2cc(-c3ccc(-c4nc5cccc6c6cccc6c5n4-c4cc(-c5cccn5)cc4)cc3)c(-c3cccc3)c(-c3cccc3)c2-c2cccc2)cc1</chem>	2DMAC-BP-F	<chem>c1ccc(-c2cc(-c3ccc(-n4c5cccc5C(C)(C)c5cccc54)ccc2c2nc4c(F)ccc4nc32)c2cccc21</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
TPBPPI-PBI	<chem>c1ccc(-c2cc(-c3ccc(-c4nc5c6cccc6c6cccc6c5n4-c4cc</chem> <chem>c(-c5ccc(-c6nc7cccc7n6-c6cccc6)cc5)cc4)cc3)c(-c3cccc3)c(-c3cccc3)c2-c2cccc2)cc1</chem>	DCA	<chem>N#Cc1cc(C#N)cc(-c2c3cccc3c(-c3cc(C#N)cc(C#N)c3)c3cccc23)c1</chem>
PTZ-2PTO	<chem>O=S1(=O)c2cccc2N(c2cccc2)c2cc(N3c4cccc4Sc4cccc43)ccc21</chem> <chem>Oc1cccc(N(c2cccc(C)c2)c2cc3c4c(c2)N(c2c(F)cccc2F)c2cc5c(cc2B4c2ccc(C)cc2N3c2cccc(C)c2)B2c3ccc(C)cc3N(c3cccc(C)c3)c3cc(N(c4cccc(C)c4)c4cccc(C)c4)cc(c32)N5c2c(F)cccc2F)c1</chem>	DPFPhe	<chem>CCCCCN1C2=CCC(=CC3c4cc(-c5cccc5)ccc4-c4ccc(-c5ccc5)cc43)C=C2Oc2cccc21</chem>
4F-m- ν -DABNA	<chem>c2ccccc(C)c2)B2c3ccc(C)cc3N(c3cccc(C)c3)c3cc(N(c4cccc(C)c4)c4cccc(C)c4)cc(c32)N5c2c(F)cccc2F)c1</chem>	DFPFPhe	<chem>CCCCCN1C2=CCC(=CC3c4cc(-c5ccc(F)cc5)ccc4-c4ccc(-c5ccc(F)cc5)cc43)C=C2Oc2cccc21</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
m- ν -DABNA	<chem>Cc1cccc(N(c2cccc(C)c2)c2c</chem>	3ai	<chem>CCCCc1cc2c(s1)-c1sc(CCCC)</chem>
	<chem>c3c4c(c2)N(c2ccccc2)c2cc5</chem>		
	<chem>c(cc2B4c2ccc(C)cc2N3c2ccc</chem>		
	<chem>c(C)c2)B2c3ccc(C)cc3N(c3c</chem>		
	<chem>ccc(C)c3)c3cc(N(c4cccc(C)</chem>		
TPA-PyI	<chem>c4)c4cccc(C)c4)cc(c32)N5c</chem>	DNFPh	<chem>CCCCCN1C2=CCC(=CC3c4cc(-</chem>
	<chem>2ccccc2)c1</chem>		
	<chem>Br c1cccc(-n2c(-c3ccc(-c4c</chem>		
	<chem>cc(N(c5ccccc5)c5ccccc5)cc</chem>		
	<chem>4)cc3)nc(-c3ccccc3)c2-c2c</chem>		
4F- ν -DABNA	<chem>cccc2)c1</chem>	pCzAnBzt	<chem>2Sc2ccccc21</chem>
	<chem>Fc1cccc(F)c1N1c2cc3c(cc2B</chem>		
	<chem>2c4ccccc4N(c4ccccc4)c4cc(</chem>		
	<chem>N(c5ccccc5)c5ccccc5)cc1c4</chem>		
	<chem>2)B1c2ccccc2N(c2ccccc2)c2</chem>		
CBPTZ	<chem>cc(N(c4ccccc4)c4ccccc4)cc</chem>	DPE-DPXZ	<chem>C(#Cc1ccc(N2c3ccccc30c3cc</chem>
	<chem>(c21)N3c1c(F)cccc1F</chem>		
	<chem>c1ccc(-c2nc(-c3ccccc3)nc(</chem>		
	<chem>-c3cccc(-c4cc(-n5c6ccccc6</chem>		
	<chem>c6ccccc65)ccc4-c4ccc(-n5c</chem>		
	<chem>6ccccc6c6ccccc65)cc4)c3)n</chem>		<chem>30c3ccccc32)cc1</chem>
	<chem>2)cc1</chem>		

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
CBPBM	<chem>c1ccc(-n2c(-c3ccc(-c4cc(-n5c6cccc6c6cccc65)ccc4-c4ccc(-n5c6cccc6c6cccc65)cc4)cc3)nc3cccc32)cc1</chem>	3CzAnBzt	<chem>N#Cc1ccc(-c2c3cccc3c(-c3ccc4c(c3)c3cccc3n4-c3cccc3)c3cccc23)cc1</chem>
CBPPY	<chem>c1cncc(-c2cc(-n3c4cccc4c4cccc43)ccc2-c2ccc(-n3c4cccc4c4cccc43)cc2)c1</chem>	TABD-COOH	<chem>O=C(O)c1ccc(/C(=C\C=C(\c2cccc2)c2ccc(C(=O)O)cc2)c2cccc2)cc1</chem>
N-B	<chem>Cc1cc(C)c(B(c2cccc2)c2c(C)cc(C)cc2C)c(C)c1</chem>	ADN	<chem>c1ccc2cc(-c3c4cccc4c(-c4ccc5cccc5c4)c4cccc34)cc2c1</chem>
DPACpBr	<chem>CC1(C)c2cccc2N(c2ccc(-c3ccc(-c4nc5c6cccc6c6cccc6c5n4-c4ccc(C#N)cc4)cc3)c2)c2cccc21</chem>	Et3SiH	<chem>FC1=C(F)C(c2c(F)c(F)c(F)c(F)c2F)(c2c(F)c(F)c(F)c(F)c2F)C(c2c(F)c(F)c(F)c(F)c2F)(c2c(F)c(F)c(F)c(F)c2F)B1c1c(F)c(F)c(F)c(F)c1F</chem>
DPCN	<chem>CC1(C)c2cccc2N(c2ccc(C#C)c3ccc(N4c5cccc5C(C)(C)c5cccc54)cc3)cc2)c2cccc21</chem>	DPE-DDMAc	<chem>CC1(C)c2cccc2N(c2ccc(C#C)c3ccc(N4c5cccc5C(C)(C)c5cccc54)cc3)cc2)c2cccc21</chem>
BZC-PXZ	<chem>O=C(CC(=O)c1cccc10)c1ccc(N2c3cccc30c3cccc32)cc1</chem>	mCz-TAn-CN	<chem>n4c5cccc5c5cccc54)c3)c3cc(C(C)(C)C)ccc3c(-c3ccc(C#N)cc3)c2c1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
CzBN	<chem>N#Cc1c(-n2c3cccc3c3cccc32)cccc1-n1c2cccc2c2cccc</chem>	indenoacridine	<chem>c1ccc2[nH]c3ccc4c5cccc5cc4c3cc2c1</chem>
2CzBN	<chem>c21CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1cccc(-n2c3ccc(C(C)(C)C)cc3c3cc(C(C)(C)C)ccc32)c1C#N</chem>	MPAc	<chem>Cc1ccc2c(c1)C(c1cccc1)(c1cccc1)c1cc(C)ccc1N2</chem>
TPA-DPPI	<chem>c1ccc(-c2nc3c4cccc4c4cccc4c3n2-c2ccc(-n3c(-c4ccc(-c5ccc(N(c6cccc6)c6cccc6)cc5)cc4)nc4c5cccc5c5cccc5c43)cc2)cc1</chem>	TBAc	<chem>CC(C)(C)C1=CC=CC(C(C)(C)C)(C2(C3(C(C)(C)C)C=CC=C(C(C)(C)C)C3)c3cccc3N(c3cccc3)c3cccc32)C1</chem>
ANQDC-PSTA	<chem>N#Cc1cc2nc3c(nc2cc1C#N)-c1ccc(N2c4ccc(-c5cccc5)cc4C(c5cccc5-c5cccc54)c4ccc5c(sc6cccc65)c42)c2cc</chem>	poCz-SO	<chem>O=P(c1cccc1)(c1cccc1)c1ccc2[nH]c3ccc(P(=O)(c4cccc4)c4cccc4)cc3c2c1</chem>
pDPAPQ-PXZ	<chem>cc-3c12c1ccc2c(c1)Oc1cccc1N2c1ccc2c3c(cccc13)-c1nc3cc(-c4ccncc4)c(-c4ccncc4)cc3nc1-2</chem>	H4TCPE	<chem>O=C(O)c1ccc(CC(c2ccc(C(=O)O)cc2)(c2ccc(C(=O)O)cc2)c2ccc(C(=O)O)cc2)cc1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
m-CZ-DPS-DMAC	<chem>CC1(C)c2cccc2N(c2cccc(S(=O)(=O)c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)c2)c2cccc21c1ccc2c(c1)Oc1cccc1N2c1c</chem>	CbBPCb	<chem>c1cc(-c2cccc(-n3c4cccc4c4ccnc43)c2)cc(-n2c3cccc3c3ccnc32)c1</chem>
pDPBPZ-PXZ	<chem>cc2c(c1)c1cccc1c1nc3cc(-c4ccncc4)c(-c4ccncc4)cc3n</chem>	TMOE	<chem>C0c1ccc(C(=C(c2ccc(OC)cc2)c2ccc(OC)cc2)c2ccc(OC)cc2)cc1</chem>
DDPhCz-DPPZ	<chem>c21c1ccc(-c2ccc3c(c2)c2cc(-c4cccc4)ccc2n3-c2cc3nc4c5ccnc5c5ncccc5c4nc3cc2-n2c3ccc(-c4cccc4)cc3c3cc(-c4cccc4)ccc32)cc1c1cnc2c(c1)c1nc3cc(-n4c5cccc5c5cccc54)c(-n4c5cccc5c5cccc54)cc3nc1c1ccn</chem>	FAPbCl3	<chem>c1ccc(-c2ccc(-c3cccc3)c(-c3cccc3)c2-c2cccc2)cc1</chem>
DCz-DPPZ	<chem>c12c1ccc2c(c1)c1cccc1n2-c1c</chem>	DBT-SADF	<chem>O=S1(=O)c2cccc2-c2cc(N3c4cccc4C4(c5cccc5-c5cccc54)c4cccc43)ccc21</chem>
Oxd-bCZ	<chem>cc(-c2ccc(-c3nnc(-c4ccc(-c5ccc(-n6c7cccc7c7cccc76)cc5)cc4)o3)cc2)cc1c1ccc2c(c1)Oc1cccc1N2c1c</chem>	mPXTMP	<chem>Cc1c(-c2ccc3c(c2)Oc2cccc2S3)cccc1-c1ccc2c(c1)Oc1cccc1S2</chem>
Fene	<chem>cc2cccc(N3c4cccc4Oc4cccc43)c2n1</chem>	APDC-tPh	<chem>N#Cc1nc2c(nc1C#N)-c1ccc(-c3cc(-c4cccc4)cc(-c4cccc4)c3)c3cccc-2c13</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
Fens	<chem>c1ccc2c(c1)Sc1cccc1N2c1c</chem> <chem>cc2cccc(N3c4cccc4Sc4cccc</chem> <chem>c43)c2n1</chem>	POZDBPHZ	<chem>c1ccc2c(c1)Oc1cccc1N2c1c</chem> <chem>cc2c(ccc3nc4ccc5cc(N6c7cc</chem> <chem>ccc7Oc7cccc76)ccc5c4nc32</chem> <chem>)c1</chem> <chem>CC(C)(C)c1ccc2c(c1)C(c1cc</chem> <chem>c(-n3c4cccc4c4cccc43)cc</chem> <chem>1)(c1ccc(-n3c4cccc4c4ccc</chem> <chem>cc43)cc1)c1cc(C(C)(C)C)cc</chem> <chem>c1-2</chem>
Yad	<chem>CC1(C)c2cccc2N(c2ccc3ccc</chem> <chem>c(N4c5cccc5C(C)(C)c5cccc</chem> <chem>c54)c3n2)c2cccc21</chem>	CPTBF	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C</chem> <chem>(C)(C)C)ccc1n2-c1ccc(-c2c</chem> <chem>cc3ccc4c(-c5ccc(-c6nnc(-c</chem> <chem>7cccc7)n6-c6cccc6)cc5)c</chem> <chem>cc5ccc2c3c54)cc1</chem> <chem>O=S1(=O)c2cccc2S(=O)(=O)</chem> <chem>c2cc(-c3ccc(N(c4cccc4)c4</chem> <chem>cccc4)cc3)ccc21</chem> <chem>c1ccc(-c2nc(-c3cc(-c4ccnc</chem> <chem>c4)cc(-c4ccncc4)c3)cc(-c3</chem> <chem>cc(-c4ccncc4)cc(-c4ccncc4</chem> <chem>)c3)n2)cc1</chem>
DTPCZPYTZ		trimesitylboron	<chem>Cc1cc(C)c(B(c2c(C)cc(C)cc</chem> <chem>2C)c2c(C)cc(C)cc2C)c(C)c1</chem>
TPADSO2		CMP	<chem>c1ccc(C(c2cccc2)(c2cccc</chem> <chem>2)c2cccc2)cc1</chem>
B4PyPPM		CH3OH-CHCl3	<chem>O=P(c1cccc1)(c1cccc1)c1</chem> <chem>cccc(-c2ccc3ccc4cccc5ccc2</chem> <chem>c3c45)c1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
mAP	<chem>CC1(C)c2cccc2N(c2cccc(N3c4cccc4C(C)(C)c4cccc43)c2)c2cccc21c1ccc(-c2cc3nc4c5ccc(N6c7cccc70c7cccc76)cc5c5cc(N6c7cccc70c7cccc76)ccc5c4nc3cc2-c2cccc2)cc1c1ccc2c(c1)0c1cccc1N2c1cc2c(c1)c1cc(N3c4cccc40c4cccc43)ccc1c1nc3cc(-c4cncnc4)c(-c4cncnc4)cc3nc210=C(c1ccc(N(c2cccc2)c2ccc2)cc1)c1cccc(C(=O)c2ccc(N(c3cccc3)c3cccc3)cc2)c1c1cncc(-c2cc3nc4c5ccc(N6c7cccc70c7cccc76)cc5c5cc(N6c7cccc70c7cccc76)ccc5c4nc3cc2-c2cccnc2)c1</chem>	s-triazine	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc([Si](c2cccc2)(c2cccc2)c2ccc(B(O)O)cc2)cc1</chem>
DBBPZ-DPXZ	<chem>CCCCCCC1(CCCCCC)c2cccc2-c2ccc(Br)cc21</chem>	[BMIM][PF6]	<chem>CCCCCCC1(CCCCCC)c2cccc2-c2ccc(Br)cc21</chem>
DPIBPZ-DPXZ	<chem>c1ccc2c(c1)-c1cccc1C21c2cccc2-c2ccc(-c3cccc4cccc34)cc21</chem>	1SBFN	<chem>c1ccc2c(c1)-c1cccc1C21c2cccc2-c2ccc(-c3cccc4cccc34)cc21</chem>
m-DTPACO	<chem>c1ccc2c(c1)-c1cccc1C21c2cccc2-c2ccc(-c3cccc4cccc4c3)cc21</chem>	2SBFN	<chem>c1ccc2c(c1)-c1cccc1C21c2cccc2-c2ccc(-c3cccc4cccc4c3)cc21</chem>
mDPBPZ-DPXZ	<chem>CC1(C)c2cccc2-c2ccc(N(c3cccc3)c3ccc4c(c3)C(C)(C)c3cccc3-4)cc21</chem>	F2PA	<chem>CC1(C)c2cccc2-c2ccc(N(c3cccc3)c3ccc4c(c3)C(C)(C)c3cccc3-4)cc21</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
p-DTPACO	<chem>O=C(c1ccc(C(=O)c2ccc(N(c3cccc3)c3cccc3)cc2)cc1)c1ccc(N(c2cccc2)c2cccc2)cc1</chem>	TSBFB	<chem>c1ccc2c(c1)-c1cccc1C21c2cccc2-c2ccc(-c3cc(-c4ccc5c(c4)C4(c6cccc6-c6cccc64)c4cccc4-5)cc(-c4ccc5c(c4)C4(c6cccc6-c6cccc64)c4cccc4-5)c3)cc21</chem>
2TDBA-SBA	<chem>CC(C)(C)c1ccc2c(c1)B1c3cc(C(C)(C)C)ccc3Oc3cc(N4c5cccc5C5(c6cccc64)c4cccc4N(c4cc6c7c(c4)Oc4ccc(C(C)(C)C)cc4B7c4cc(C(C)(C)C)ccc4O6)c4cccc45)cc(c31)O2</chem>	DPPP	<chem>c1ccc(-c2cc(-c3cccc3)cc(-c3cccc3-c3ccc4ccc5c(-c6cccc6-c6cc(-c7cccc7)cc(-c7cccc7)c6)ccc6ccc3c4c65)c2)cc1</chem>
TPA-AP _m	<chem>c1ccc(N(c2cccc2)c2ccc(-c3cnc4nc5c(nc4n3)-c3cccc4ccc-5c34)cc2)cc1</chem>	TPA-NZP	<chem>c1ccc(-c2c3cccc3c(-c3ccc(N(c4cccc4)c4cccc4)cc3)c3nsc23)cc1</chem>
TPA-AP	<chem>c1ccc(N(c2cccc2)c2ccc(-c3ccc4nc5c(nc4c3)-c3cccc4ccc-5c34)cc2)cc1</chem>	mTPA-PPI	<chem>c1ccc(N(c2cccc2)c2ccc(-c3cccc(-c4nc5c6cccc6c6cccc6c5n4-c4cccc4)c3)cc2)c1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
TDBA-SBA	<chem>CC(C)(C)c1ccc(N2c3cccc3C3(c4cccc42)c2cccc2N(c2cc4c5c(c2)0c2ccc(C(C)(C)C)cc2B5c2cc(C(C)(C)C)ccc2O4)c2cccc23)cc1</chem>	BSBFB	<chem>c1cc(-c2ccc3c(c2)C2(c4cccc4-c4cccc42)c2cccc2-3)cc(-c2ccc3c(c2)C2(c4cccc4-c4cccc42)c2cccc2-3)c1</chem>
TPA-APy	<chem>c1ccc(N(c2cccc2)c2ccc(-c3cnc4nc5c(nc4c3)-c3cccc4cccc-5c34)cc2)cc1</chem>	BN-DBC-Ph2	<chem>C1=CB2C(=C3C(=C4C=C(c5cccc5)C=CN4c4ccc(-c5cccc5)cc43)c3cccc32)C=C1c1ccc(-c2nc(-c3cccc3)nc(-c3cc(-n4c5cccc5c5ncccc54)c(-n4c5cccc5c5ncccc54)c3c(-n4c5cccc5c5ncccc54)cc(-n4c5cccc5c5ncccc54)c3)cc12</chem>
TPD4PA	<chem>c1ccc(N(c2cccc2)c2cc3c4c(c2)N(c2cccc2)c2cc5c6c(c2B4c2cccc2N3c2cccc2)0c2cccc2B6c2cccc2O5)cc1</chem>	TCbTrz	<chem>CC1(C)c2cccc2N(c2ccc(C(=O)c3cccc3C(=O)c3ccc(N4c5cccc5C(C)(C)c5cccc54)cc3)cc2)c2cccc21</chem>
o-Ac2BBP	<chem>Cc1ccc(N(c2ccc(C3(c4cccnc4)c4cc(C(C)(C)C)ccc4-c4cc(c(C(C)(C)C)cc43)cc2)c2ccc(C3(c4cccnc4)c4cc(C(C)(C)C)ccc4-c4ccc(C(C)(C)C)cc43)cc2)cc1</chem>	Xm-mCP	<chem>CC1(C)OB(c2cc(-n3c4cccc4c4cccc43)cc(-n3c4cccc4c4cccc43)c2)OC1(C)C</chem>
TPA-PyF2	<chem>CC1(C)c2cccc2N(c2ccc(C(=O)c3cccc3C(=O)c3ccc(N4c5cccc5C(C)(C)c5cccc54)cc3)cc2)c2cccc21</chem>	Xp-mCP	

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Molecule Id	SMILES	Molecule Id	SMILES
o-Px2BBP	<chem>O=C(c1ccc(N2c3cccc30c3ccc32)cc1)c1cccc1C(=O)c1ccc(N2c3cccc30c3cccc32)cc1</chem>	ethylacetate	<chem>O=S(=O)(c1ccc(N2c3cccc3C=Cc3cccc32)cc1)c1ccc(N2c3cccc3C=Cc3cccc32)cc1</chem>
D-BP-PXZ	<chem>O=C(c1ccc(N2c3cccc30c3ccc32)cc1)c1cccc1-c1cccc1C(=O)c1ccc(N2c3cccc30c3cccc32)cc1</chem>	3DMACIPN	<chem>CC1(C)c2cccc2N(c2c(F)c(N3c4cccc4C(C)(C)c4cccc43)c(C#N)c(N3c4cccc4C(C)(C)c4cccc43)c2C#N)c2cccc21</chem>
BTD-PXZ	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc2nsnc2c1Cc1cc(-n2c3ccc(C(C)(C)C)c3c3cc(C(C)(C)C)ccc32)cc(C)c1B1c2cccc2B(c2c(C)cc(-n3c4ccc(C(C)(C)C)cc4c4cc(C(C)(C)C)ccc43)cc2C)c2ccc21</chem>	ICZ	<chem>CCCCC1c2[nH]c3cccc3c2cc2[nH]c3cccc3c12</chem>
tBuCzDBA	<chem>CC(C)(C)N1C(=O)c2cc(-n3c4cccc4c4cccc43)ccc2S1(=O)=O</chem>	BDAVBi	<chem>C(=Cc1ccc(N(c2cccc2)c2ccc2)cc1)c1ccc(-c2ccc(C=Cc3ccc(N(c4cccc4)c4cccc4)cc3)cc2)cc1</chem>
5-Cz-Sac	<chem>CC(C)(C)N1C(=O)c2cc(-n3c4cccc4c4cccc43)ccc2S1(=O)=O</chem>	DPVBi	<chem>C(=C(c1cccc1)c1cccc1)c1ccc(-c2ccc(C=C(c3cccc3)c3cccc3)cc2)cc1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
6-Cz-Sac	<chem>CC(C)(C)N1C(=O)c2ccc(-n3c4cccc4c4cccc43)cc2S1(=O)=O</chem>	CBTD2	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-c2ccc(-c3ccc(-n4c5ccc(C(C)(C)C)cc5c5cc(C(C)(C)C)ccc54)cc3)c3nsnc23)cc1</chem>
DiCz-Sac	<chem>CC(C)(C)N1C(=O)c2cc(-n3c4cccc4c4cccc43)c(-n3c4cccc4c4cccc43)cc2S1(=O)=O</chem>	MeOCzPCN	<chem>C0c1ccc2c(c1)c1cc(OC)ccc1n2-c1ncnc(-n2c3ccc(OC)cc3c3cc(OC)ccc32)c1C#N</chem>
2CM	<chem>CCCCCCCCn1c2cccc2c2cc(Nc3nc(Nc4ccc5c(c4)c4cccc4n5CCCCCCCC)nc(OC)n3)ccc21</chem>	Na-DMAC	<chem>CC1(C)c2cccc2N(c2ccc(-c3c(C#N)c(-c4cccc5cccc45)n c(-c4cccc5cccc45)c3C#N)c c2)c2cccc21</chem>
CzoB	<chem>Cc1cc(C)c(B(c2cccc2-n2c3cccc3c3cccc32)c2c(C)cc(C)cc2C)c(C)c1</chem>	3Py-DMAC	<chem>CC1(C)c2cccc2N(c2ccc(-c3c(C#N)c(-c4cccnc4)nc(-c4ccnc4)c3C#N)cc2)c2cccc21</chem>
CzBroB	<chem>Cc1cc(C)c(B(c2ccc(Br)cc2-n2c3cccc3c3cccc32)c2c(C)cc(C)cc2C)c(C)c1</chem>	6tBPA	<chem>C0c1ccc2c(c1)c1cccc1n2-c1cc2c(cc1-n1c3cccc3c3ccc cc(C(C)(C)C)cc23)cc1</chem>
PxCN	<chem>N#Cc1c(-c2ccc(N3c4cccc40c4cccc43)cc2)cncc1-c1ccc(N2c3cccc30c3cccc32)cc1</chem>	C-O	<chem>C0c1ccc2c(c1)c1cccc1n2-c1cc2c(cc1-n1c3cccc3c3ccc cc31)Cc1cccc102</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
AcCN	<chem>CC1(C)c2cccc2N(c2ccc(-c3cncc(-c4ccc(N5c6cccc6C(C)(C)c6cccc65)cc4)c3C#N)c2)c2cccc21</chem>	IDFL-2DPA	<chem>CC1(C)c2cccc2-c2c1cc1c3cc(N(c4cccc4)c4cccc4)ccc3n3c4ccc(N(c5cccc5)c5cccc5)cc4c2c13</chem>
CzCNoB	<chem>Cc1cc(C)c(B(c2ccc(C#N)cc2)-n2c3cccc3c3cccc32)c2c(C)cc(C)cc2C)c(C)c1</chem>	ACRF	<chem>N#Cc1ccc2c(c1)C1(c3cc(C#N)ccc3-2)c2cccc2N(c2cccc2)c2cccc21</chem>
CNPP-TPA	<chem>N#Cc1nc2c3ccc(-c4ccc(N(c5cccc5)c5cccc5)cc4)nc3c3nc(-c4ccc(N(c5cccc5)c5cccc5)cc4)ccc3c2nc1C#N</chem>	TPSi-F	<chem>O=C1c2ccc(N3c4cccc4C4(c5cccc5-c5cccc54)c4cccc43)cc2-c2cc(N3c4cccc4C4(c5cccc5-c5cccc54)c4cccc43)ccc21</chem>
SPAC-FO	<chem>CC1(C)c2cccc2N(c2ccc3c(c2)-c2cc(N4c5cccc5C(C)(C)c5cccc54)ccc2C3=O)c2cccc21</chem>	PAB	<chem>CC(C)(C)c1ccc(N2c3ccc(C(C)(C)C)cc3B3c4cc(C(C)(C)C)ccc4N(c4ccc(C(C)(C)C)cc4)c4cc(N(c5cccc5)c5cccc5)cc2c43)cc1</chem>
DMAC-FO	<chem>CC1(C)c2cccc2N(c2ccc3c(c2)-c2cc(N4c5cccc5C(C)(C)c5cccc54)ccc2C3=O)c2cccc21</chem>	2tPAB	

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Molecule Id	SMILES	Molecule Id	SMILES
6AcBIQ	<chem>CC1(C)c2cccc2N(c2ccc3c4c(c4)C(=O)N(c2cccc2)C3=O)c2cccc21</chem>	pDTPA-DTPZ	<chem>c1ccc(N(c2cccc2)c2ccc(-c3ccc(-c4ccc(N(c5cccc5)c5cccc5)cc4)c4nc5c(nc34)c3sc3c3ccsc35)cc2)cc1c1ccc(N(c2cccc2)c2ccc(-c3cc4nc5c(nc4cc3-c3ccc(N(c4cccc4)c4cccc4)cc3)c3sc3c3ccsc35)cc2)cc1</chem>
PtB	<chem>CC(C)(C)c1ccc(-n2c(-c3ccc4ccc5cccc6ccc3c4c56)nc(-c3cccc3)c2-c2cccc2)cc1</chem>	oDTPA-DTPZ	<chem>3cc4nc5c(nc4cc3-c3ccc(N(c4cccc4)c4cccc4)cc3)c3sc3c3ccsc35)cc2)cc1</chem>
BCPBT	<chem>c1ccc2c(c1)c1cccc1n2-c1cc(-c2ccc(-c3cc(-n4c5cccc5c5cccc54)cc(-n4c5cccc5c5cccc54)c3)c3nsnc23)cc(-n2c3cccc3c3cccc32)c1</chem>	oTPAPO-DTPZ	<chem>0=P(c1cccc1)(c1cccc1)c1cc2nc3c(nc2cc1-c1ccc(N(c2cccc2)c2cccc2)cc1)c1sc3c1c1ccsc13</chem>
InDM	<chem>CN(C)c1ccc(-n2c(-c3cccc3)nc(-c3cccc3)c2-c2cccc2)cc1</chem>	BPZPP	<chem>c1ccc(-c2cc3nc4c5ccc(N6c7cccc70c7cccc76)nc5c5nc(N6c7cccc70c7cccc76)ccc5</chem>
PnB	<chem>CCCCc1ccc(-n2c(-c3ccc4ccc5cccc6ccc3c4c56)nc(-c3ccc3)c2-c2cccc2)cc1</chem>	BIPOCz	<chem>c4nc3cc2-c2cccc2)cc1c1ccc(-c2nc(-c3ccc(OC)cc4cccc5c4[nH]c4cccc45)cc3)n(-c3cccc3)c2-c2cccc2)cc1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
BANE	<chem>c1ccc(-c2ccc(-c3c4ccccc4c(-c4c5ccccc5c(-c5ccc(-c6ccccc6)cc5)c5ccccc45)c4ccc(cc34)cc2)cc1</chem>	tDPAC-BN	<chem>CC(C)(C)c1ccc(N2c3ccc(C(C)(C)C)cc3B3c4cc(C(C)(C)C)ccc4N(c4ccc(C(C)(C)C)cc4)c4cc(N5c6ccccc6C(c6ccccc6)(c6ccccc6)c6ccccc65)cc2c43)cc1</chem>
CBM	<chem>CCCCn1c2ccc(C=C(C#N)C#N)c2c2cc(C=C(C#N)C#N)ccc21</chem>	tDMAC-BN	<chem>CC(C)(C)c1ccc(N(c2ccc(C(C)(C)C)cc2)c2cc3c4c(c2)N2c5ccccc5C(C)(C)c5cccc(c52)B4c2cc(C(C)(C)C)ccc2N3c2ccc(C(C)(C)C)cc2)cc1</chem>
DCzBNPh-1	<chem>N#Cc1ccc(-c2cccc(-c3ccc(C#N)c(-n4c5ccccc5c5ccccc54)c3)c2)cc1-n1c2ccccc2c2ccccc21</chem>	2ICz-OXD	<chem>c1ccc(-n2c3ccccc3c3c2ccc2c4ccccc4n(-c4ccc(-c5nnc(-c6ccc(-n7c8ccccc8c8ccc9c(c%10ccccc%10n9-c9ccccc9)c87)cc6)o5)cc4)c23)cc1</chem>
MCZ-P2-DTM	<chem>C0c1ccc(-c2ccc3c(c2)c2cc(-c4ccc(OC)cc4)ccc2n3-c2ccc(C(=O)c3ccc4sc5ccccc5c4c3)cn2)cc1</chem>	SIR-B	<chem>Cc1ccc2c(c1)C1(c3ccc(-c4cccc4)cc3C(=O)c3ccc(-c4cccc4)cc31)c1cc(C)ccc1N2c1ccc(C(C)(C)C)cc1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
DBTO-PTZ	<chem>O=S1(=O)c2ccc(N3c4cccc4O c4cccc43)cc2-c2cc(N3c4cc ccc4Oc4cccc43)ccc21</chem>	SIR-C	<chem>Cc1ccc2c(c1)C1(c3ccc(-c4c cccc4-c4cccc4)cc3C(=O)c3 ccc(-c4cccc4-c4cccc4)cc 31)c1cc(C)ccc1N2c1ccc(C(C) (C)C)cc1</chem>
α -DMAC-DBP	<chem>CC1(C)c2cccc2N(c2cccc3nc 4c5cccc5c5cccc5c4nc23)c 2cccc21</chem>	SIR-A	<chem>Cc1ccc2c(c1)C1(c3cccc3C(=O)c3cccc31)c1cc(C)ccc1N 2c1ccc(C(C)(C)C)cc1 c1ccc(N(c2cccc2)c2ccc(-c 3nc(-c4ccc(N(c5cccc5)c5c cccc5)cc4)nc(-c4ccc(N(c5c cccc5)c5cccc5)cc4)n3)cc2)cc1</chem>
α -2DMAC-DBP	<chem>CC1(C)c2cccc2N(c2ccc(N3c 4cccc4C(C)(C)c4cccc43)c 3nc4c5cccc5c5cccc5c4nc2 3)c2cccc21</chem>	TRZ3DPA	<chem>Nc1ccc2c(c1)Oc1cc(N)cc3c1 B2c1ccc(Nc2cccc4cc5ccc6cc 7cc8cccc8cc7cc6c5cc24)cc 103 Cc1ccc(C)c(N(c2ccc3ccc4c(N(c5cc(C)ccc5C)c5cccc6c5o c5cccc56)ccc5ccc2c3c54)c 2cccc3c2oc2cccc23)c1</chem>
β -DI-DBP	<chem>c1ccc(-n2c3cccc3c3c2c2c4 cccc4n(-c4ccc5nc6c7cccc 7c7cccc7c6nc5c4)c2c2c4cc ccc4n(-c4cccc4)c32)cc1</chem>	BO3N	
DIC-TRZ	<chem>c1ccc(-c2nc(-c3cccc3)nc(-n3c4cccc4c4ccc5c6cccc6 n(-c6cccc6)c5c43)n2)cc1</chem>	BPPyA	

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Molecule Id	SMILES	Molecule Id	SMILES
PO-TPA	<chem>CC(C)(C)c1ccc(N(c2ccc(-c3cc4c(cc3C#N)=NC3c5ccc(P(=O)(c6cccc6)c6cccc6)nc5-c5nc(P(=O)(c6cccc6)c6cccc6)ccc5C3N=4)cc2)c2ccc(C(C)(C)C)cc2)cc1</chem>	SiCN2Cz	<chem>N#Cc1cc(-c2nc(-n3c4cccc4c4cccc43)nc(-n3c4cccc4c4cccc43)n2)cc([Si](c2cccc2)(c2cccc2)c2cccc2)c1</chem>
DMAC-Cz-TTR	<chem>CC1(C)c2cccc2N(c2ccc3c(c2)c2cc(N4c5cccc5C(C)(C)c5cccc54)ccc2n3-c2ccc3c(c2)S(=O)(=O)c2cccc2S3(=O)=O)c2cccc21</chem>	PhPC	<chem>Cc1ccc2c(c1)c1cc(C)ccc1n2</chem>
DCzPO	<chem>Cc1ccc2c(c1)c1cc(C)ccc1n2</chem>	1MOBI	<chem>C0c1cc(/C=C/c2ccc(-n3cnc4cccc43)cc2)c(OC)cc1/C=C/c1ccc(-n2cnc3cccc32)cc1</chem>
IQ-oTPA	<chem>0=C1c2cccc2-c2nc3cc(-c4ccc(N(c5cccc5)c5cccc5)cc4)c(-c4ccc(N(c5cccc5)c5cccc5)cc4)cc3nc21</chem>	3BrBI	<chem>Brc1cc(/C=C/c2ccc(-n3cnc4cccc43)cc2)c(Br)cc1/C=C/c1ccc(-n2cnc3cccc32)cc1</chem>
2C	<chem>N#CC(C#N)=Cc1ccc(-c2ccc3ccc4cccc5ccc2c3c45)cc1F</chem>	DMBFTX	<chem>CC1(C)c2cccc2-c2cccc(-c3ccc4sc5ccc(-c6cccc7c6C(C)(C)c6cccc6-7)cc5c(=O)c4c3)c21</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
C-H	<chem>N#CC(C#N)=Cc1c(F)cc(-c2ccc3ccc4cccc5ccc2c3c45)cc1F</chem>	SBF-AnCN	<chem>N#Cc1ccc(-c2c3cccc3c(-c3ccc4c(c3)C3(c5cccc5-c5ccc3)cc1CC1(C)c2cccc2-c2ccc(-c3c4cccc4c(-c4ccc(C#N)cc4)c4cccc34)cc21</chem>
PSPP	<chem>O=S(=O)(c1cccnc1)c1cccc(N2c3cccc30c3cccc32)c1</chem>	DMF-AnCN	<chem>N#Cc1ccc(-c2c3cccc3c(-c3ccc4c(c3)C(c3cccc3)(c3ccc3)c3cccc3-4)c3cccc23)cc1</chem>
2DN	<chem>CC(C)c1cccc(C(C)C)c1N1C(=O)c2cc(N3c4cccc4C(C)(C)c4cccc43)cc3cc(N4c5cccc5C(C)(C)c5cccc54)cc(c23)C1=O</chem>	DPF-AnCN	<chem>c1ccc(-c2c3cccc3c(-c3ccc3)c3cccc23)cc1</chem>
PSPBP	<chem>O=S(=O)(c1cccnc1)c1cc(N2c3cccc30c3cccc32)cc(N2c3cccc30c3cccc32)c1</chem>	DPA	<chem>c1ccc(N(c2cccc2)c2ccc(-c3c4cccc4c(-c4ccc5ccc6c(-c7c8cccc8c(-c8ccc(N(c9ccc9)c9cccc9)cc8)c8cccc78)ccc7ccc4c5c76)c4cccc34)cc2)cc1</chem>
2PN	<chem>CC(C)c1cccc(C(C)C)c1N1C(=O)c2cc(N3c4cccc40c4cccc43)cc3cc(N4c5cccc50c5cccc54)cc(c23)C1=O</chem>	TPA-DAP-TPA	

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Molecule Id	SMILES	Molecule Id	SMILES
pCbz-PYR	<chem>c1ccc(-c2ccc3c(c2)c2cc(-c4ccccc4)ccc2n3-c2cc(-n3c4ccc(-c5ccccc5)cc4c4cc(-c5ccccc5)ccc43)ncn2)cc1c1ccc(-c2nc(-c3ccccc3)n(-c3ccc(-c4c5ccccc5c(-c5ccc(-c6nc7c8ccccc8c8ccccc8c7n6-c6ccccc6)cc5)c5ccccc45)cc3)c2-c2ccccc2)cc1Fc1cc(N2c3ccccc3Sc3ccccc32)c(F)cc1N1c2ccccc2Sc2ccc</chem>	AcDbp	<chem>CC1(C)c2ccccc2N(c2ccc3c4c5cc(N5c6ccccc6C(C)(C)c6cccc65)cc4c4nc5ccccc5nc4c3c2)c2ccccc21</chem>
PPI-An-NPIM	<chem>c3ccc(-c4c5ccccc5c(-c5ccc(-c6nc7c8ccccc8c8ccccc8c7n6-c6ccccc6)cc5)c5ccccc45)cc3)c2-c2ccccc2)cc1Fc1cc(N2c3ccccc3Sc3ccccc32)c(F)cc1N1c2ccccc2Sc2ccc</chem>	4-Ac	<chem>CC(C)(C)c1ccc2c(c1)C(C)(C)c1cc(C(C)(C)C)ccc1N2c1ccc(C(F)(F)F)cc1</chem>
2PS-2FPh	<chem>c21C(=Cc1ccc(-n2c3ccccc3c3ccc32)cc1)c1ccc(C(=C(c2ccccc2)c2ccccc2)c2ccccc2)cc</chem>	czpzpy	<chem>c1cc(-c2cccc3c2[nH]c2ccccc23)nc(-n2cccn2)c1c1ccc2c(c1)Oc1ccccc1N2c1ccc(-c2ccc3nc(-c4ccc(N5c6ccccc6Oc6ccccc65)cc4)ccc3n2)cc1Cc1nc(N2c3ccccc3C(c3ccccc3)(c3ccccc3)c3ccccc32)cc(N2c3ccccc3C(c3ccccc3)(c3ccccc3)c3ccccc32)n1CCc1ccccc1P(c1ccccc1CC)c1ccccc1P(c1ccccc1CC)c1ccccc1CC</chem>
9CzTPE	<chem>c21Cc1nc(N2c3ccccc3C(c3ccccc3)(c3ccccc3)c3ccccc32)cc(N2c3ccccc3C(c3ccccc3)(c3ccccc3)c3ccccc32)n1CCc1ccccc1P(c1ccccc1CC)c1ccccc1P(c1ccccc1CC)c1ccccc1CC</chem>	NyDPO	
2CzTPE	<chem>c21Cc1nc(N2c3ccccc3C(c3ccccc3)(c3ccccc3)c3ccccc32)cc(N2c3ccccc3C(c3ccccc3)(c3ccccc3)c3ccccc32)n1CCc1ccccc1P(c1ccccc1CC)c1ccccc1P(c1ccccc1CC)c1ccccc1CC</chem>	1MPA	
FTAT-FBO	<chem>c21Cc1nc(N2c3ccccc3C(c3ccccc3)(c3ccccc3)c3ccccc32)cc(N2c3ccccc3C(c3ccccc3)(c3ccccc3)c3ccccc32)n1CCc1ccccc1P(c1ccccc1CC)c1ccccc1P(c1ccccc1CC)c1ccccc1CC</chem>	LEt	

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Molecule Id	SMILES	Molecule Id	SMILES
DmPy-PPI	<chem>c1ccc(-n2c(-c3ccc(-c4cc(-c5cccc(-c6cccnc6)c5)cc(-c5cccc(-c6cccnc6)c5)c4)cc3)nc3c4cccc4c4cccc4c32)c</chem>	DC-TC	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(C(=O)C(=O)c2ccc(-n3c4ccc(C(C)(C)C)cc4c4cc(C(C)(C)C)ccc43)cc2)cc1</chem>
mCP-BmPy	<chem>c1ccc(C(C)(C)c1ccc2c(c1)C(c1cc(-c3cccnc3)cc(-c3cccnc3)c1)(c1ccc3c(c1)c1cccc1n3-c1cccc(-n3c4cccc4c4cccc43)c1)c1cc(C(C)(C)C)ccc1-2</chem>	DC-ACR	<chem>CC1(C)c2cccc2N(c2ccc(C(=O)C(=O)c3ccc(N4c5cccc5C(C)(C)c5cccc54)cc3)cc2)c2cccc21</chem>
mCP-Py	<chem>c1ccc(C(C)(C)c1ccc2c(c1)C(c1ccnnc1)(c1ccc3c(c1)c1cccc1n3-c1cccc(-n3c4cccc4c4cccc43)c1)c1cc(C(C)(C)C)cc</chem>	1G-F1	<chem>c1ccc(P(c2cccc2)c2cscc2P(c2cccc2)c2cccc2)cc1</chem>
mCP-Ph	<chem>c1-2CC(C)(C)c1ccc2c(c1)C(c1cccc1)(c1ccc3c(c1)c1cccc1n3-c1cccc(-n3c4cccc4c4cccc43)c1)c1cc(C(C)(C)C)cc</chem>	P-Cu-C	<chem>c1ccc(P(c2cccc2)c2cccc2-c2cccc2P(c2cccc2)c2cccc2)cc1</chem>
	<chem>c1-2</chem>		

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Molecule Id	SMILES	Molecule Id	SMILES
DMTAC	<chem>Cc1ccc(-c2c(-c3ccccc3)c(-c3ccccc3)c3c(N)cc4c5ccccc5c(N)cc4c3c2-c2ccc(C)c(C)c2)cc1C</chem>	BPCP-2CPC	<chem>N#Cc1cc(-n2c3ccccc3c3cc(-c4ccc(C5(c6ccc(-n7c8ccccc8c8ccccc87)cc6)CCCCC5)cc4)ccc32)c(-n2c3ccccc3c3cc(-c4ccc(C5(c6ccc(-n7c8ccccc8c8ccccc87)cc6)CCCCC5)cc4)ccc32)cc1C#N</chem>
IPTAC	<chem>Cc1ccc(-c2c(-c3ccc(C(C)C)cc3)c(-c3ccc(C(C)C)cc3)c3c(c(N)cc4c5ccccc5c(N)cc43)c2-c2ccc(C)cc2)cc1</chem>	2H-Qz	<chem>c1ccc(-c2cc(-c3ccccn3)cc(-c3nc(-c4ccc(N5c6ccccc60c6ccccc65)cc4)c4cccc4n3)c2)nc1</chem>
TBTAC	<chem>Cc1ccccc1-c1c(-c2ccc(C(C)C)cc2)c(-c2ccc(C(C)C)cc2)c2c(c(N)cc3c4ccccc4c(N)cc32)c1-c1ccccc1C</chem>	PPINCN-Cz	<chem>N#Cc1ccc(-n2c(-c3ccc4ccc5cccc6ccc3c4c56)nc3c4ccc(-c5ccc(-n6c7ccccc7c7ccccc76)cc5)cc4c4cc(-c5ccc(-n6c7ccccc7c7ccccc76)cc5)ccc4c32)c2ccccc12</chem>
p2Cz2Trz	<chem>c1ccc(-c2nc(-c3ccccc3)nc(-c3cc(-n4c5ccccc5c5ccccc54)c(-c4nc(-c5ccccc5)nc(-c5ccccc5)n4)cc3-n3c4ccccc4c4ccccc43)n2)cc1</chem>	PPINCN	<chem>N#Cc1ccc(-n2c(-c3ccc4ccc5cccc6ccc3c4c56)nc3c4ccccc4c4ccccc4c32)c2ccccc12</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
m2Cz2Trz	<chem>c1ccc(-c2nc(-c3ccccc3)nc(-c3cc(-c4nc(-c5ccccc5)nc(-c5ccccc5)n4)c(-n4c5ccccc5c5ccccc54)cc3-n3c4ccccc4c4ccccc43)n2)cc1</chem>	BDAPM	<chem>O=C(c1ccc(N2c3ccccc3[Si](c3ccccc3)(c3ccccc3)c3cccc32)cc1)c1ccc(N2c3ccccc3[Si](c3ccccc3)(c3ccccc3)c3ccccc32)cc1</chem>
2,4-CzPyCl3	<chem>Clc1nc(-n2c3ccccc3c3ccccc32)c(Cl)c(-n2c3ccccc3c3ccccc32)c1Cl</chem>	12AcCz-PM	<chem>CC1(C)c2ccccc2N(c2ccc(-c3cc(-c4ccccc4)nc(-c4ccccc4)n3)cc2)c2c1ccc1c2c2ccccc2n1-c1ccccc1</chem>
DMAC-PCN	<chem>CC1(C)c2ccccc2N(c2ccc(-c3c(C#N)c(-c4ccccc4)nc(-c4ccccc4)c3C#N)cc2)c2ccccc21</chem>	23AcCz-PM	<chem>CC1(C)c2ccccc2N(c2ccc(-c3cc(-c4ccccc4)nc(-c4ccccc4)n3)cc2)c2cc3c(cc21)c1ccc1n3-c1ccccc1</chem>
2PXZP	<chem>Cn1cnc2c(-c3ccc(N4c5ccccc50c5ccccc54)cc3(N4c5ccccc50c5ccccc54)cc3)nc21</chem>	CDDPI	<chem>c1ccc2c(-n3c4ccccc4c4ccccc43)ccc(-c3nc4c5ccccc5c5ccccc5c4n3-c3ccc4c(c3)OCCO4)c2c1</chem>
2,4,6-CzPyCl2	<chem>Clc1c(-n2c3ccccc3c3ccccc32)nc(-n2c3ccccc3c3ccccc32)c(Cl)c1-n1c2ccccc2c2ccccc21</chem>	DDBP	<chem>(N(c5ccccc5)c5ccccc5)cc4)c4ccccc34)n(-c3ccc4c(c3)OCCO4)c2-c2ccccc2)cc1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
1PXZP	<chem>Cn1cnc2c(-c3ccc(N4c5ccccc5Oc5ccccc54)cc3)ncnc21</chem>	CCDD	<chem>O=C(c1cccc1)c1ccc(-n2c3ccc(N(c4ccccc4)c4ccccc4)cc3c3cc(N(c4ccccc4)c4ccccc4)ccc32)cc1</chem>
3,6-T2C	<chem>N#Cc1ccc(-c2nc3c4ccc(-c5ccc(N(c6ccccc6)c6ccccc6)cc5)cc4c4cc(-c5ccc(N(c6ccccc6)c6ccccc6)cc5)ccc4c3n2-c2ccc(C#N)cc2)cc1</chem>	DPNC	<chem>N#Cc1ccc(-c2nc3c4ccc(-c5ccc(N(c6ccccc6)c6ccccc6)cc5)ccc4c3n2-c2ccc(C#N)cc2)cc1</chem>
3,6-TC	<chem>N#Cc1ccc(-c2nc3c4ccc(-c5ccc(N(c6ccccc6)c6ccccc6)cc5)ccc4c3n2-c2ccccc2)cc1</chem>	LiPr	<chem>CC(C)c1cccc1P(c1cccc1C(C)C)c1cccc1P(c1cccc1C(C)C)c1cccc1C(C)C</chem>
Ph-CF3	<chem>c1ccc2c(c1)c1ccccc1n2-c1ccc(Nc2ccc(-n3c4ccccc4c4ccccc43)cc2)cc1</chem>	Spiro-CN	<chem>Cc1ccc(N(c2ccc(C)cc2)c2ccc3c(c2)C2(c4cc(C#N)ccc4-c4ccc(C#N)cc42)c2cc(N(c4ccc(C)cc4)c4ccc(C)cc4)ccc2-3)cc1</chem>
arylene	<chem>c1ccc2c(c1)c1ccccc1n2-c1ccc(Nc2ccc(-n3c4ccccc4c4ccccc43)cc2)cc1</chem>	TMDBP	<chem>CC1(C)c2ccccc2Nc2c1ccc1c2C(C)(C)c2ccccc2N1</chem>

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Molecule Id	SMILES	Molecule Id	SMILES
TS-1	<chem>O=S(=O)(c1ccccc1)c1ccc(-n2c3ccccc3c3ccccc32)cc1</chem>	ACRSA	<chem>O=C1c2ccccc2C2(c3ccccc31)c1ccccc1N(c1ccccc1)c1cccc12</chem>
3,6-MeCzOXD	<chem>Cc1ccc2c(c1)c1cc(C)ccc1n2-c1ccccc1-c1nnc(-c2ccccc2-n2c3ccc(C)cc3c3cc(C)ccc32</chem>	CAPI	<chem>cccc6c(-n6c7ccccc7c7cccc76)c6ccccc56)cc4)nc4c5ccc5c5ccccc5c43)cccc2c1</chem>
p-PO15NCzDPA	<chem>O=P(c1ccccc1)(c1ccccc1)c1ccc(-c2c3ccccc3c(-c3ccc(-n4c5cccn5c5cccn54)cc3)c3ccccc23)cc1</chem>	PhCzSPOTz	<chem>O=P(c1ccccc1)(c1ccccc1)c1nc(-c2ccccc2)nc(-n2c3ccccc3c3ccccc32)n1</chem>
2,7-MeCzOXD	<chem>Cc1ccc2c3ccc(C)cc3n(-c3ccc3-c3nnc(-c4ccccc4-n4c5cc(C)ccc5c5ccc(C)cc54)o3)c2c1</chem>	CCAPI	<chem>N#Cc1ccc(-n2c(-c3ccc(-c4c5ccccc5c(-n5c6ccccc6c6cccc65)c5ccccc45)cc3)nc3c4cccc4c4ccccc4c32)c2ccccc12</chem>
BPCN-Cz2Ph	<chem>N#Cc1ccccc1-c1ccc(-n2c3ccc(-c4ccccc4)cc3c3cc(-c4ccc4)ccc32)cc1</chem>	2CzCbPy	<chem>c1ccc(-n2c3ccccc3c3ccccc32)nc(-n2c3ccc(-n4c5ccccc5c5ccccc54)cc3c3ncccc32)c1</chem>
o-CzOXD	<chem>c1ccc(-n2c3ccccc3c3ccccc32)c(-c2nnc(-c3ccccc3-n3c4ccccc4c4ccccc43)o2)c1</chem>	CzCbPy	<chem>c1ccc(-n2c3ccc(-n4c5ccccc5c5ccccc54)cc3c3ncccc32)n</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
BPCN-2Cz	<chem>N#Cc1cccc1-c1ccc(-n2c3ccc3cc3cc(-c4ccc5c(c4)c4ccc4n5-c4cccc4)ccc32)cc1</chem>	DCE	<chem>CC(C)CCC[C@@H](C)[C@H]1CC[C@H]2[C@@H]3CC=C4C[C@@H](O)CC[C@]4(C)[C@H]3CC[C@]12C</chem>
Cbz-MI	<chem>CCCCCN1C(=O)C(c2ccc(-c3ccc(-n4c5cccc5c5cccc54)c3)cc2)=C(c2ccc(-c3ccc(-n4c5cccc5c5cccc54)cc3)cc2)C1=O</chem>	PICFOCz	<chem>FC(F)(F)c1cccc(-n2c(-c3ccc(OCc1n4c5cccc5c5cccc54)cc3)nc3c4cccc4c4cccc4c32)c1</chem>
DBFDPO	<chem>O=P(c1cccc1)(c1cccc1)c1cccc2c1oc1c(P(=O)(c3cccc3)c3cccc3)cccc12</chem>	dpmt	<chem>Cc1sc(C)c(P(c2cccc2)c2ccc2)c1P(c1cccc1)c1cccc1</chem>
XCBP	<chem>Cc1cc(-n2c3cccc3c3cccc32)c(C)cc1-c1ccc(C(=O)c2ccc2)cc1</chem>	BICFOCz	<chem>1FC(F)(F)c1cccc(-n2c(-c3ccc(OCc1n4c5cccc5c5cccc54)cc3)nc(-c3cccc3)c2-c2cccc2)c1</chem>
Cbz-PYR	<chem>c1ccc2c(c1)c1cccc1n2-c1cc(-n2c3cccc3c3cccc32)nc1</chem>	19a	<chem>c1ccc2cc(-c3ccc4cc5cc(-c6ccc7cccc7c6)ccc5cc4c3)cc2c1</chem>
tCbz-mPYRs	<chem>n1CSc1nc(-n2c3ccc(C(C)(C)C)cc3c3c(C(C)(C)C)cccc32)c(C)cc(-n2c3ccc(C(C)(C)C)cc3c3c(C(C)(C)C)cccc32)n1</chem>	BTPBFCz	<chem>c1ccc(-c2nc(-c3cccc3-n3c4cccc4c4c5oc6cccc6c5ccc43)c3sc4cccc4c3n2)cc1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
PCBP	<chem>O=C(c1cccc1)c1ccc(-c2ccc(-n3c4cccc4c4cccc43)cc2)cc1</chem>	Ph-OBNA	<chem>c1ccc(-n2c3cccc3c3c4c5c(cc32)0c2cccc2B5c2cccc204)cc1</chem>
MeCZ-TRZ	<chem>Cc1ccc2c(c1)c1cccc1n2-c1ccc(-c2nc(-c3ccc(-n4c5ccc5c5cc(C)ccc54)c(C)c3)nc(-c3ccc(-n4c5cccc5c5cc(C)ccc54)c(C)c3)n2)cc1C</chem>	HPVAnt	<chem>CCCCCc1ccc(C=Cc2ccc3cc4cccc4cc3c2)cc1</chem>
3ca	<chem>C0c1ccc(-c2cc(-c3ccc(C(F)(F)F)cc3)c(-c3nc4cccc4[nH]3)s2)cc1</chem>	BTPBFCz-D3	<chem>c1ccc(-c2nc(-c3ccc(-n4c5cccc5c5ccc6c(c7cccc7n6-c6cccc6)c54)cc3-n3c4cccc4c4c5oc6cccc6c5ccc43)c3sc4cccc4c3n2)cc1</chem>
2MeCZ-TRZ	<chem>Cc1ccc2c(c1)c1cc(C)ccc1n2-c1ccc(-c2nc(-c3ccc(-n4c5ccc(C)cc5c5cc(C)ccc54)c(C)c3)nc(-c3ccc(-n4c5ccc(C)cc5c5cc(C)ccc54)c(C)c3)n2)cc1C</chem>	ICBNTrz4	<chem>CC1(C)c2cccc2-c2cc3c4cccc4n(-c4ccc(-c5nc(-c6cccc6)nc(-c6cccc6)n5)cc4C#N)c3cc21</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
CZ-TRZ	<chem>Cc1cc(-c2nc(-c3ccc(-n4c5c cccc5c5cccc54)c(C)c3)nc(-c3ccc(-n4c5cccc5c5cccc 54)c(C)c3)n2)ccc1-n1c2ccc cc2c2cccc21</chem>	1PCTRz	<chem>c1ccc(-c2nc(-c3cccc3)nc(-c3ccc(-n4c5cccc5c5cc(-n 6c7cccc7c7cccc76)ccc54) c(-c4cccc4)c3)n2)cc1</chem>
DMAC-AQ	<chem>CC1(C)c2cccc2N(c2ccc3c(c 2)C(=O)c2cccc2C3=O)c2ccc cc21</chem>	DMACN-B	<chem>Cc1cc(C)c(N(c2ccc3c(c2)C(c2cccc2)(c2cccc2)c2cccc 4c2N3c2cccc2C4(C)C)c2c(C)cc(C)cc2C)c(C)c1</chem>
DTPTCzDP-CN	<chem>CC(C)(C)c1nc(-c2ccc(-n3c4 ccc(N(c5cccc5)c5cccc5)c c4c4cc(N(c5cccc5)c5cccc 5)ccc43)c(C#N)c2)nc(C(C)(C)C)n1</chem>	CzSO-1CO	<chem>CC1(OCOC(=O)c2cccc2)CC=C c2c1c1cccc1n2-c1ccc(S(=O) (=O)c2cccc2)cc1</chem>
mP2MPC	<chem>Cc1cc(-c2cccc(-n3c4cccc4 c4cccc43)c2)c(C)cc1-c1cc cc(-c2nc3c4cccc4c4cccc4 c3n2-c2cccc2)c1</chem>	PPO21	<chem>CC(C)(C)c1ccc(-c2nnc(-c3c ccc(-c4nnc(-c5ccc(C(C)(C) C)cc5)o4)c3)o2)cc1</chem>
XPhos	<chem>CC(C)(C)C(P(C2CCCCC2)C 2CCCCC2)C1</chem>	OXD-7	

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
TPXZ-as-TAZ	<chem>c1ccc2c(c1)Oc1ccccc1N2c1c</chem>	TCZ1	<chem>CCCCC(CC)Cn1c2ccc(-n3c4cc</chem>
	<chem>cc(-c2nnc(-c3ccc(N4c5cccc</chem>		<chem>ccc4c4ccccc43)cc2c2cc(-n3</chem>
	<chem>c50c5ccccc54)cc3)c(-c3ccc</chem>		<chem>c4ccccc4c4ccccc43)ccc21</chem>
2Trz6CNDBT	<chem>(N4c5ccccc50c5ccccc54)cc3</chem>	APPT	<chem>c1cc2c3c(cccc3c1)-c1nc3c4</chem>
	<chem>)n2)cc1</chem>		
	<chem>N#Cc1cccc2c1sc1ccc(-c3nc(</chem>		
TPAPyPI	<chem>-c4ccccc4)nc(-c4ccccc4)n3)</chem>	ANQDC	<chem>N#Cc1cc2nc3c(nc2cc1C#N)-c</chem>
	<chem>cc12</chem>		
	<chem>c1ccc(N(c2ccccc2)c2ccc(-c</chem>		
TPACzPI	<chem>3ccc(-c4nc5c6ccccc6c6cccc</chem>	ECDPTT	<chem>CCn1c2ccccc2c2cc(C(=C(c3c</chem>
	<chem>c6c5n4-c4cccc(-c5ccc6ccc7</chem>		
	<chem>cccc8ccc5c6c78)c4)cc3)cc2</chem>		
NA-TNA	<chem>)cc1</chem>	1CM	<chem>CCCCCCCCn1c2ccccc2c2cc(Nc</chem>
	<chem>c1ccc(N(c2ccccc2)c2ccc(-c</chem>		
	<chem>3ccc(-c4nc5c6ccccc6c6cccc</chem>		
TPACzPI	<chem>c6c5n4-c4cccc(-c5ccc6c(c5</chem>	ECDPTT	<chem>ccccc3)c3ccccc3)c3ccc(-c4c</chem>
	<chem>)c5ccccc5n6-c5ccccc5)c4)c</chem>		
	<chem>c3)cc2)cc1</chem>		
NA-TNA	<chem>CC(C)(C)c1ccc(N2C(=O)c3cc</chem>	1CM	<chem>3nc(OC)nc(OC)n3)ccc21</chem>
	<chem>cc4c(/C=C/c5ccc6cc(N(c7cc</chem>		
	<chem>ccc7)c7ccccc7)ccc6c5)ccc(</chem>		
	<chem>c34)C2=O)cc1</chem>		

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
CzPrSBI	<chem>O=C1c2ccccc2S(=O)(=O)N1CC</chem>	NPE-AcDPS	<chem>CC(c1ccccc1)N1c2cc(N3c4ccc</chem>
	<chem>Cn1c2ccccc2c2ccccc21</chem>		<chem>ccc4C(C)(C)c4ccccc43)ccc2</chem>
	<chem>Cc1cc(-c2cccc(-n3c4ccccc4</chem>		<chem>S(=O)(=O)c2ccc(N3c4ccccc4</chem>
m-CzCNDMT	<chem>c4cc(C#N)ccc43)c2)c(C)cc1</chem>	DMAC-Cz	<chem>C(C)(C)c4ccccc43)cc21</chem>
	<chem>-c1cccc(-n2c3ccccc3c3cc(C</chem>		<chem>CC1(C)c2ccccc2N(c2ccc3[nH</chem>
	<chem>#N)ccc32)c1</chem>		<chem>]c4ccc(N5c6ccccc6C(C)(C)c</chem>
	<chem>Cc1cc(N(c2ccccc2)c2cccc3c</chem>		<chem>6ccccc65)cc4c3c2)c2ccccc2</chem>
DTXO-TPA2	<chem>cccc23)ccc1-c1ccc(N(c2ccc</chem>	VBC-PTZ	<chem>1</chem>
	<chem>cc2)c2cccc3ccccc23)cc1C</chem>		<chem>CCN(CC)c1ccc2c(c1)oc(=O)c</chem>
	<chem>c1ccc2c(-c3ccc(-n4c5ccccc</chem>		<chem>1c(=O)oc3cc(N4c5ccccc5Sc5</chem>
24PQ-Cz	<chem>5c5ccccc54)cc3)nc(-c3ccc(</chem>	t-BuPCz	<chem>cccc54)ccc3c12</chem>
	<chem>-n4c5ccccc5c5ccccc54)cc3)</chem>		<chem>CC(C)(C)c1ccc(-c2ccc3[nH]</chem>
	<chem>nc2c1</chem>		<chem>c4ccc(-c5ccc(C(C)(C)C)cc5</chem>
	<chem>O=S(=O)(c1ccc(N2c3ccccc3S</chem>		<chem>)cc4c3c2)cc1</chem>
SFPC	<chem>c3ccccc32)cc1)c1ccc(-n2c3</chem>	4BPy-mDTC	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C</chem>
	<chem>ccccc3c3ccccc32)cc1</chem>		<chem>(C)(C)C)ccc1n2-c1cc(C(=O)</chem>
	<chem>C0c1cc(=O)oc2cc3c(cc12)c1</chem>		<chem>c2ccncc2)cc(-n2c3ccc(C(C)</chem>
2CzMC	<chem>ccccc1n3-c1ccccc1</chem>	DC-TC	<chem>(C)C)cc3c3cc(C(C)(C)C)ccc</chem>
			<chem>32)c1</chem>
			<chem>Cc1ccc(N(c2ccc(C)cc2)c2cc</chem>
			<chem>c(C3(c4ccc(N(c5ccc(C)cc5)</chem>
			<chem>c5ccc(C)cc5)cc4)CCCC3)cc</chem>
			<chem>2)cc1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
PPI-PIM-2	<chem>FC(F)(F)c1ccc(-n2c(-c3ccc(-c4ccc(-n5c(-c6cccc6)nc(-c6cccc6)c5-c5cccc5)cc4)cc3)nc3c4cccc4c4cccc4c32)cc1</chem>	DFDBQPXZ	<chem>Fc1cc2nc(-c3ccc(N4c5cccc50c5cccc54)cc3)c(-c3ccc(N4c5cccc50c5cccc54)cc3)nc2cc1F</chem>
4CzMC	<chem>C0c1cc(=O)oc2c1ccc1c2c2ccccc2n1-c1cccc1</chem>	DPE	<chem>C(#Cc1ccc(N2c3cccc30c3cccc32)cc1)c1ccc(N2c3cccc30c3cccc32)cc1</chem>
PPI-PIM-1	<chem>FC(F)(F)c1ccc(-n2c(-c3ccc(-c4ccc(-c5nc(-c6cccc6)c(-c6cccc6)n5-c5cccc5)cc4)cc3)nc3c4cccc4c4cccc4c32)cc1</chem>	CBZ	<chem>CCCCC0c1ccc2c3ccc(OC(C)cc3n(-c3ccnc(-c4ccc(F)c(C#N)c4F)c3)c2c1</chem>
CzTPA	<chem>c1ccc(N(c2cccc2)c2cccc2-n2c3cccc3c3cccc32)cc1</chem>	tBuCz2	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc2[nH]c3ccc(-n4c5ccc(C(C)(C)C)c5c5cc(C(C)(C)C)ccc54)cc3c2c1</chem>
SFX-bPy	<chem>c1ccc2c(c1)0c1ccc(-c3ccc4cccc4n3)cc1C21c2cccc2-c2cccc21</chem>	26DCzPPy	<chem>c1cc(-c2cccc(-c3cccc(-n4c5cccc5c5cccc54)c3)n2)cc(-n2c3cccc3c3cccc32)c1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
t-BuCz2BP	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(C(=O)c2ccc(-n3c4ccc(C(C)(C)C)cc4c4cc(C(C)(C)C)ccc43)cc2)cc1</chem>	mCBP	<chem>c1cc(-c2cccc(-n3c4cccc4c4cccc43)c2)cc(-n2c3cccc3c3cccc32)c1</chem>
SAF-2CzB	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-c2ccc3c(c2)C2(c4cc(-c5ccc(-n6c7ccc(C(C)(C)C)cc7c7cc(C(C)(C)C)ccc76)cc5)ccc4-3)c3cccc3N(c3cccc3)c3cccc32)cc1</chem>	benzoxazole	<chem>Cc1sc2cccc2c1C1=C(c2c(C)sc3cccc23)C(F)(F)C(F)(F)C1(F)F</chem>
F-2CzB	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-c2ccc3c(c2)C(C)(C)c2cc(-c4cc(-n5c6ccc(C(C)(C)C)cc6c6cc(C(C)(C)C)ccc65)cc4)ccc2-3)cc1</chem>	dbG2TAZ	<chem>c1ccc2c(c1)ccc1[nH]c3cccc4cccc4c3c12</chem>
SAF-CzB	<chem>CC(C)(C)c1ccc2c(c1)c1cc(C(C)(C)C)ccc1n2-c1ccc(-c2ccc3c(c2)C2(c4cccc4-3)c3cccc3N(c3cccc3)c3cccc32)cc1</chem>	V-TPA	<chem>C=Cc1ccc(C0c2cccc3c2c2cccc2n3-c2ccc(N(c3ccc(-n4c5cccc5c5cccc54)cc3)c3ccc(-n4c5cccc5c5cccc54)cc3)cc2)cc1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
BPy-TP2	<chem>c1ccc(-c2ccc(-c3ccc4c5ccc(-c6ccc(-c7cccn7)nc6)cc5c5ccccc5c4c3)cn2)nc1</chem>	CCDC	<chem>O=C(c1cccc1)c1ccc(-n2c3ccc(-n4c5ccccc5c5ccccc54)cc3c3cc(-n4c5ccccc5c5ccccc54)ccc32)cc1</chem>
Cz-AQ	<chem>O=C1c2cccc2C(=O)c2cc(-n3c4ccccc4c4ccccc43)ccc21</chem>	PXZPDO	<chem>O=C(C=C(O)c1ccc(N2c3ccccc30c3ccccc32)cc1)c1ccc(N2c3ccccc30c3ccccc32)cc1</chem>
-ph-pCzAnBzt	<chem>Cc1cc(-c2c3ccccc3c(-c3ccc(C#N)cc3)c3ccccc23)c(C)cc1-c1ccc(-n2c3ccccc3c3ccccc32)cc1</chem>	DMAC-BP	<chem>CC1(C)c2ccccc2N(c2ccc(C(=O)c3ccc(N4c5ccccc5C(C)(C)c5ccccc54)cc3)cc2)c2ccccc21</chem>
PyI	<chem>c1cc2ccc3cccc4c5[nH]cnc5c(c1)c2c34</chem>	SBTF	<chem>c(ccc3cc(-c4ccc5ccccc5c4)ccc23)-c2ccc(-c3cc4ccccc4c4ccccc34)c3cccc1c23</chem>
DVCz-2CzCN	<chem>C=Cc1ccc(COc2cccc3c2c2cccc2n3CCCCCOc2cccc3c2c2cccc2n3-c2cccc(-n3c4ccccc4c4c(OCc7ccc(C=C)cc7)cccc65)cc43)c2C#N)cc1</chem>	3-DPM	<chem>N#CC(C#N)=C(c1cccc(N(c2cccc2)c2ccccc2)c1)c1cccc(N(c2ccccc2)c2ccccc2)c1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
ATDBT	<chem>c1ccc(-c2c3ccccc3c(-c3ccc(-n4c5ccccc5c5c6sc7ccccc7c6ccc54)cc3)c3ccccc23)cc1</chem>	4-DPM	<chem>N#CC(C#N)=C(c1ccc(N(c2cccc2)c2cccc2)cc1)c1ccc(N(c2cccc2)c2cccc2)cc1</chem>
DPhBCz	<chem>c1ccc(-n2c3ccccc3c3cc(-c4ccc5c(c4)c4ccccc4n5-c4ccc4)ccc32)cc1</chem>	2CzPy	<chem>4ccccc43)cc2-n2c3ccccc3c3ccccc32)nc1</chem>
AcHPM	<chem>CC1(C)c2ccccc2N(c2ccc(-c3cc(-c4ccc(N5c6ccccc6C(C)(C)c6ccccc65)cc4)ncn3)cc2)c2ccccc21</chem>	2PBCzPy	<chem>c1ccc(-n2c3ccccc3c3cc(-c4ccc5c(c4)c4ccccc4n5-c4ccc(-c5ccccc5)c(-n5c6ccccc6c6cc(-c7ccc8c(c7)c7ccccc7n8-c7ccccc7)ccc65)c4)ccc32)cc1</chem>
PhCO-DMAC	<chem>CC1(C)c2ccccc2N(c2ccc(C(=O)c3ccccc3)cc2)c2ccccc21</chem>	2BFCzPy	<chem>4c5oc6ccccc6c5ccc43)cc2-n2c3ccccc3c3c4oc5ccccc5c4cc32)nc1</chem>
Diphenylsulfane	<chem>c1ccc(-c2cccc(-c3nc(-c4ccc(-c5ccccc5)c4)nc(-c4ccc(-c5ccccc5)c4)n3)c2)cc1</chem>	CNTPA-CZ	<chem>N#Cc1ccc(N(c2ccccc2)c2cc(-n3c4ccccc4c4ccccc43)cc(-n3c4ccccc4c4ccccc43)c2)cc1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
2BrCPT	<chem>BrC1CCC2C(C1)C1CC(Br)CCC1</chem>	CNTPA-PXZ	<chem>N#Cc1ccc(N(C2CCCCC2)C2CC(N3C4CCCCC4O4CCCCC43)CC(N3C4CCCCC4O4CCCCC43)C2)CC1</chem>
DCbz-Ph-Bz	<chem>C1CCC(-N2C(-C3CCC(-C4CC(-N5C6CCCCC6C6CCCCC65)CC(-N5C6CCCCC6C6CCCCC65)C4)CC3)NC3CCCCC32)CC1</chem>	BMZ-TZ	<chem>C1CC(-C2NC3CCCCC3[NH]2)CC(-C2NNNC(-C3CCCC(-C4NC5CCCCC5[NH]4)C3)C2-C2CCCC(-C3NC4CCCCC4[NH]3)C2)C1</chem>
DDPA-Ph-Bz	<chem>C1CCC(N(C2CCCCC2)C2CC(-C3CCC(-C4NC5CCCCC5N4-C4CCCCC4)CC3)CC(N(C3CCCCC3)C3CCCCC3)C2)CC1</chem>	TPAP _m	<chem>N#Cc1c(-C2CCC(N(C3CCCCC3)C3CCCCC3)CC2)NCNC1-C1CCC(N(C2CCCCC2)C2CCCCC2)CC1</chem>
IDAC-MCO	<chem>Cc1cc2cc(-n3c4CCCCC4C4CC5C(Cc43)C(C)(C)C3CCCCC3N5C3CCCCC3)CCC2C(=O)O1</chem>	CzP _m	<chem>N#Cc1c(-C2CCC(-N3C4CCCCC4C4CCCCC43)CC2)NCNC1-C1CCC(-N2C3CCCCC3C3CCCCC32)CC1</chem>
diarylbiindole	<chem>C1CCC(-C2C(-C3CCCCC3)C3CC4CCCC5CC(-C6CCCCC6)C(C2-C2CCCCC2)C3C45)CC1</chem>	TCPZ	<chem>C1=C(C2CCCC(-N3C4CCCCC4C4CCCCC43)C2)NN(C2CCCC(-N3C4CCCCC4C4CCCCC43)C2)N=C1C1CCCC(-N2C3CCCCC3C3CCCCC32)C1</chem>

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Table S1 – Continued from previous page

Molecule Id	SMILES	Molecule Id	SMILES
CNDSB	<chem>N#CC(=Cc1cccc1)c1ccc(C(C#N)=Cc2cccc2)cc1</chem>	TrisCz-Trz	<chem>c1ccc(-n2c3cccc3c3cccc32)c(-c2nc(-c3cccc3-n3c4cccc4c4cccc43)nc(-c3cccc3-n3c4cccc4c4cccc43)n2)c1</chem>
TMAB	<chem>Cc1cc(C)c(CC#Cc2ccc(N(C)C)cc2)c(B(c2c(C)c(C)cc(C)c2CC#Cc2ccc(N(C)C)cc2)c1C</chem>	PTZ-TAZ	<chem>c1ccc2c(c1)Oc1cccc1N2c1ccc(-c2nnc(-c3ccc(N4c5cccc5O4)cc3)o2)cc1</chem>
MOF	<chem>c1cncc(CC(c2nc3cccc3[nH]2)N(C(Cc2ccnc2)c2nc3cccc3[nH]2)C(Cc2ccnc2)c2nc3cccc3[nH]2)c1</chem>	DDMA-TXO2	<chem>CC1(C)c2cccc2N(c2ccc3c(c2)C(C)(C)c2cc(N4c5cccc5C(C)(C)c5cccc54)ccc2S3(=O)=O)c2cccc21</chem>
DICzTrz	<chem>N#Cc1cc(-c2nc(-c3cccc3)nc(-c3cccc3)n2)ccc1-n1c2cccc2c2cc3c4cccc4n4c5cccc5c(c21)c34</chem>	UGH3	<chem>O=S1(=O)c2cccc2N(c2cccc2)c2cc([Si](c2cccc2)(c2ccccc2)c2cccc([Si](c3cccc3)(c3cccc3)c3cccc3)c2)cc1</chem>
PXZ2PTO	<chem>2)c2cc(N3c4cccc4O4cccc43)ccc21</chem>		

Table S2: Complete validation dataset for emission wavelength predictions. Predictions from sTDA and sTD-DFT methods in gas and toluene phases are compared against experimental/computational literature values. References are provided in the last column.

Molecule	sTDA (Gas)	sTD-DFT (Gas)	sTDA (Tol)	sTD-DFT (Tol)	Ref. (nm)	Type	Citation
-ph-pCzAnBzt	501.4	512.6	467.6	477.7	450.0	experimental	S1
1SBFN	420.2	421.1	399.3	400.0	385.0	experimental	S2
2,7-MeCzOXD	524.7	524.2	507.0	506.3	520.0	experimental	S3
25CzBPym	631.6	632.7	645.1	646.8	487.3	experimental	S4
25tCzBPym	643.1	644.4	682.5	684.2	506.0	experimental	S4
2BPy-mDTC	691.4	691.6	686.1	686.0	491.0	experimental	S5
2C	560.0	556.6	595.2	591.0	534.0	experimental	S6
2CM	384.9	391.7	366.5	372.7	374.0	experimental	S7
2Cz-DMAC-BTB	474.2	471.3	535.1	531.4	529.0	experimental	S8
2Cz-DMAC-TTR	469.5	472.9	517.6	522.9	601.0	experimental	S8
2Cz-DMAC-TXO	528.6	528.0	561.8	559.3	566.0	experimental	S8
2Cz2tCzBn	483.6	486.9	484.3	488.3	488.0	experimental	S9
2CzPN	474.8	475.8	490.5	492.1	480.4	experimental	S10
2DN	775.6	773.4	736.4	751.2	659.4	experimental	S11
2F	776.6	788.7	700.4	711.6	493.3	computational	S12
2PN	823.0	825.1	1059.4	1062.0	673.3	experimental	S11
2SBFN	412.2	409.1	394.4	391.7	382.0	experimental	S2
2TDBA-SBA	447.6	454.2	448.2	454.8	448.0	experimental	S13
2tCz2CzBn	477.6	481.1	482.4	486.6	472.0	experimental	S9
3,3-NHC	369.4	370.2	426.0	428.7	412.7	experimental	S14
3,6-MeCzOXD	513.1	512.5	500.5	499.9	520.0	experimental	S3

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Table S2 – continued from previous page

Molecule	sTDAs	TD-DFTs	sTDAs	TD-DFT	Ref.	Type	Citation
	(Gas)	(Gas)	(Tol)	(Tol)	(nm)		
3,6-TC	531.4	532.0	551.0	552.0	488.0	experimental	S15
3-BHH	358.9	362.0	339.8	342.9	478.5	experimental	S16
3-DTPA-BBI	631.2	632.6	675.2	676.6	604.0	experimental	S17
35CzBPym	610.7	614.0	615.1	618.5	478.0	experimental	S4
35tCzBPym	622.7	625.9	647.0	650.6	496.0	experimental	S4
3BPy-mDTC	581.2	584.0	580.3	583.2	478.0	experimental	S5
3CzBN	438.3	437.9	437.5	437.4	431.0	experimental	S18
3CzFTFP	402.8	407.6	430.0	435.6	484.0	experimental	S10
3DPTIA	496.4	503.0	474.3	480.7	452.3	experimental	S19
3TCPM	496.5	496.7	524.8	521.8	482.5	experimental	S20
4CzIPN	528.4	531.6	537.7	540.9	502.7	experimental	S21
4NTAZ-PPI	486.9	484.6	468.1	465.7	456.7	experimental	S22
5CzICz	418.8	423.4	409.3	413.9	410.0	experimental	S23
ACRXTN	456.8	459.1	448.7	451.4	454.7	experimental	S24
APDC-DTPA	793.9	798.5	831.4	825.1	767.5	experimental	S25
AcCN	562.0	557.5	571.8	567.2	512.0	experimental	S26
BACF	442.0	448.7	422.0	428.7	427.0	experimental	S27
BACH	433.6	439.5	409.9	415.7	427.7	experimental	S27
BACN	459.7	465.4	448.1	454.2	443.2	experimental	S28
BAPCN	475.5	480.5	478.1	483.4	436.5	experimental	S28
BCN	474.9	478.7	467.0	471.1	444.0	experimental	S29
BDPCC	437.3	440.3	425.7	428.4	483.5	experimental	S30
BDQ-tBuCz	838.1	839.6	745.9	747.6	557.0	experimental	S31

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Table S2 – continued from previous page

Molecule	sTDAs	TD-DFTs	sTDAs	TD-DFT	Ref.	Type	Citation
	(Gas)	(Gas)	(Tol)	(Tol)	(nm)		
BDQDMAC	962.9	962.8	868.4	868.3	601.7	experimental	S31
BDQPCz	820.6	814.9	721.4	715.1	502.0	experimental	S31
BP-mDTC	522.0	524.3	527.8	529.1	461.5	experimental	S5
BSBFB	398.7	398.1	376.9	376.6	388.0	experimental	S2
BTZPP	664.8	660.4	712.3	707.8	594.0	experimental	S32
BuCzCF3oB	550.1	550.8	544.8	545.6	519.0	experimental	S33
C-H	540.9	538.1	589.8	585.5	526.0	experimental	S6
C3F7	498.1	498.2	472.2	471.2	558.0	experimental	S33
CBM	505.0	504.9	501.4	501.4	458.0	experimental	S34
CDBP-BP-DMAC	511.7	508.8	501.7	498.6	500.7	experimental	S35
CDBP-BP-PXZ	501.8	502.0	503.8	504.1	525.8	experimental	S35
CN-QP	859.4	854.1	945.1	936.2	602.0	experimental	S36
CPzPC	581.1	584.6	524.4	527.7	468.0	experimental	S37
CPzPyC	607.3	609.0	589.9	592.2	457.5	experimental	S37
CPzPzC	754.3	757.6	690.7	693.9	477.0	experimental	S37
CTM	473.7	477.1	485.6	483.0	502.7	experimental	S38
CZ-DPS-DMAC	415.3	418.7	437.2	436.7	458.7	experimental	S39
Cz-SO	368.3	372.9	367.3	371.7	424.0	experimental	S40
Cz2ICz	446.3	450.8	436.7	441.1	420.0	experimental	S23
CzB-FMPIM	492.0	496.3	479.7	484.5	420.0	experimental	S41
CzB-FMPPI	502.9	504.4	491.9	492.9	432.0	experimental	S41
CzPh-PPO	458.3	459.8	450.2	451.5	390.0	experimental	S42
CzPrSBI	473.5	479.4	444.5	450.9	477.3	experimental	S43

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Table S2 – continued from previous page

Molecule	sTDAs	TD-DFTs	sTDAs	TD-DFT	Ref.	Type	Citation
	(Gas)	(Gas)	(Tol)	(Tol)	(nm)		
CzTPA-m-Trz	552.3	555.5	549.2	553.2	849.7	experimental	S44
CzoB	497.9	498.0	472.3	471.4	517.0	experimental	S33
Cz-AQ	727.9	732.7	710.7	715.7	586.5	experimental	S45
DBQ-3PXZ	851.7	854.2	868.0	870.1	618.0	experimental	S46
DBT-BZ-DMAC	636.7	634.8	665.7	664.3	535.5	experimental	S47
DCbz-Ph-Bz	456.3	460.3	437.7	441.7	404.0	experimental	S48
DCzBNPh-1	472.4	475.7	442.1	445.7	472.5	experimental	S49
DCzIPN	454.2	457.1	457.6	460.4	453.0	experimental	S50
DDPA-Ph-Bz	469.1	468.3	465.0	464.7	435.0	experimental	S48
DDPhCz-DCPP	755.3	757.7	739.8	742.3	571.3	experimental	S51
DFPFPh	503.0	506.8	475.0	478.4	537.0	experimental	S52
DHID-DBS	461.5	467.0	488.2	492.2	478.0	experimental	S53
DHID-DPS	433.3	437.4	477.1	481.1	514.0	experimental	S53
DMAC-11-DPPZ	1031.3	1030.1	891.9	891.3	577.5	experimental	S54
DMAC-BPP	641.0	636.9	599.3	594.8	505.0	experimental	S55
DMAC-Cz-TTR	500.1	503.8	581.4	584.4	449.0	experimental	S56
DMAC-DPS	729.4	729.1	628.2	628.1	470.0	experimental	S57
DMAC-TRZ	619.5	616.8	602.9	602.4	500.6	theoretical study	S58
DMACMNPTO	348.6	350.2	346.3	347.9	382.0	experimental	S59
DMeCzIPN	479.2	481.5	486.4	488.8	474.0	experimental	S50
DNFPh	540.6	544.7	468.9	470.1	521.5	experimental	S52
DPACPhTPI	495.3	496.3	483.9	485.0	436.0	experimental	S60
DPACpBr	367.6	367.9	365.0	365.9	448.7	experimental	S61

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Table S2 – continued from previous page

Molecule	sTDAs	TD-DFTs	sTDAs	TD-DFT	Ref.	Type	Citation
	(Gas)	(Gas)	(Tol)	(Tol)	(nm)		
DPCN	511.7	511.6	508.0	508.3	424.0	experimental	S62
DPFPhe	494.3	494.7	461.9	464.2	532.5	experimental	S52
DPPP	532.0	535.3	486.6	489.5	390.7	experimental	S2
DPPSPF	405.3	406.4	393.8	395.0	385.0	experimental	S63
DPS-PXZ	485.1	485.2	464.8	465.0	537.0	experimental	S64
DPXZ-DPPM	1090.2	1093.3	1048.7	1051.3	636.0	experimental	S65
DPhBCz	395.1	399.7	381.8	386.2	365.0	experimental	S66
DQP	1207.5	1183.6	1057.9	1036.8	562.0	experimental	S47
DTPCZTZ	468.9	473.7	455.8	460.7	474.6	experimental	S67
DV-3CzCN	470.6	472.0	442.0	442.6	473.6	experimental	S68
DVCz-2CzCN	434.5	438.3	453.4	454.9	461.5	experimental	S69
Diphenylsulfane	541.6	535.9	511.4	507.5	454.0	experimental	S70
F2AcBO	491.3	489.9	515.6	516.6	476.7	experimental	S71
F2PA	435.9	435.7	426.3	426.2	403.3	experimental	S2
FTAT-FBO	399.3	398.4	375.0	374.2	459.0	experimental	S72
FlCz	553.3	548.0	525.0	520.0	412.0	experimental	S73
IABTCN	659.1	658.8	694.6	696.9	650.0	experimental	S74
ImPy-1	463.0	469.9	437.0	443.5	479.0	experimental	S75
ImPy-2	465.7	472.2	441.1	447.4	471.0	experimental	S75
ImPy-3	557.2	556.0	533.0	531.4	480.0	experimental	S75
InDM	416.2	422.6	399.3	405.0	490.0	experimental	S76
MBF	605.6	604.9	662.7	665.5	614.0	experimental	S77
MPA	553.7	554.2	560.5	561.2	574.7	experimental	S77

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Table S2 – continued from previous page

Molecule	sTDAs	TD-DFTs	sTDAs	TD-DFT	Ref.	Type	Citation
	(Gas)	(Gas)	(Tol)	(Tol)	(nm)		
MPPA	805.6	809.3	941.6	947.7	679.1	experimental	S78
Me-DMAC	527.8	524.1	543.5	539.4	517.0	experimental	S79
Me2ACBO	498.1	499.9	509.1	508.8	474.0	experimental	S71
NA-TNA	648.4	645.2	683.6	680.0	619.2	experimental	S80
NAI-BiFA	558.5	562.8	573.2	578.7	596.1	experimental	S81
NAI-PhBiFA	569.8	566.6	647.7	645.0	556.8	experimental	S81
NAP-1,5-DPA	501.4	502.3	477.3	478.1	507.2	experimental	S82
NAQ3	688.2	692.4	685.2	689.2	530.0	experimental	S83
OCzBN	449.2	453.1	434.7	436.7	393.0	experimental	S84
P4CzCN-BCz	387.1	383.9	334.2	331.4	426.5	experimental	S85
PBTPA	736.2	732.1	753.7	750.2	642.5	experimental	S86
PCTXO	561.6	557.6	568.3	564.3	616.4	experimental	S87
PDA-DP	477.9	477.5	472.4	472.2	480.0	experimental	S88
PIAnTPh	498.9	506.6	462.9	470.2	440.0	experimental	S89
PO-ABN	509.7	519.7	472.9	482.2	434.0	experimental	S90
POBBPE	362.1	364.1	341.9	344.0	320.0	experimental	S91
PS-BZ-DMAC	474.2	478.9	477.1	482.3	574.0	experimental	S92
PSPSF	406.3	407.2	394.8	395.6	386.0	experimental	S63
PT-BZ-DMAC	573.1	568.5	583.9	580.7	531.8	experimental	S92
PTZ-2PTO	459.1	460.2	434.2	434.2	452.2	experimental	S93
PTZ-DBTO2	473.4	479.6	471.4	477.9	520.0	experimental	S55
PX-TRZ	541.9	543.5	554.0	554.6	531.0	experimental	S94
PXZ-10-DPPZ	1099.7	1102.8	958.6	960.6	614.0	experimental	S54

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Table S2 – continued from previous page

Molecule	sTDAs	TD-DFTs	sTDAs	TD-DFT	Ref.	Type	Citation
	(Gas)	(Gas)	(Tol)	(Tol)	(nm)		
PXZ-11-DPPZ	987.5	991.7	863.8	867.0	628.5	experimental	S54
PXZ2PTO	430.0	430.4	420.1	420.5	496.7	experimental	S95
PmPmP	653.0	653.8	643.5	644.3	563.0	experimental	S96
PnB	518.0	514.4	488.4	484.9	465.0	experimental	S76
Pra-2DMAC	370.2	371.9	359.0	360.5	513.0	experimental	S97
PtB	527.9	525.4	487.3	483.6	446.3	experimental	S76
PxCN	558.1	558.2	576.1	576.2	527.5	experimental	S26
PyDCN	423.8	422.9	403.4	402.5	472.0	experimental	S98
PyDCN-DMAC	508.7	504.4	496.9	493.2	494.2	experimental	S98
PyIANtPh	509.5	519.5	470.9	480.2	572.5	experimental	S89
SBA	351.0	352.5	342.3	343.8	350.0	experimental	S99
SBDBQ-PXZ	843.2	845.7	751.8	753.8	594.0	experimental	S46
SF-DPSO	431.1	431.1	384.5	384.4	399.5	experimental	S100
SFAC-BP-SFAC	592.9	592.6	597.8	597.2	506.0	experimental	S101
SFCz	556.1	553.2	529.4	526.6	413.0	experimental	S73
STAC-BP-STAC	582.5	582.1	575.3	575.0	506.5	experimental	S101
SXAC-BP-SXAC	585.8	585.4	584.9	584.4	500.0	experimental	S101
SiCz1Py3	405.9	410.7	387.3	392.0	479.0	experimental	S102
SiCz2Py2	399.8	404.0	389.3	393.7	478.0	experimental	S102
SiCz3Py1	409.9	414.1	386.3	390.5	476.0	experimental	S102
SpiroAC-TRZ	610.1	610.1	591.5	591.0	484.5	experimental	S47
TATP-BP	558.1	557.8	568.1	566.4	527.7	experimental	S103
TAZ-PPI	499.8	500.6	487.4	488.5	456.7	experimental	S22

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Table S2 – continued from previous page

Molecule	sTDAs	TD-DFTs	sTDAs	TD-DFT	Ref.	Type	Citation
	(Gas)	(Gas)	(Tol)	(Tol)	(nm)		
TBN-TPA	436.7	443.9	423.1	430.0	453.0	experimental	S104
TBPe	542.6	553.7	500.3	510.7	471.0	experimental	S105
TBRb	677.3	696.5	610.9	628.2	538.0	experimental	S105
TCz-3PA-TCz	580.2	578.6	534.0	532.6	452.0	experimental	S106
TDBA-SBA	422.5	428.8	422.4	428.7	464.0	experimental	S13
TFM-QP	857.2	857.2	929.3	923.2	566.5	experimental	S36
TMAB	434.0	432.4	407.6	413.6	452.0	experimental	S107
TPA-DCPP	765.0	762.7	781.4	779.2	669.3	experimental	S47
TPA-L-BN	451.4	452.9	391.4	392.8	394.0	experimental	S108
TPA-PPO	493.0	490.9	500.0	497.8	472.0	experimental	S42
TPA-PyF2	390.2	390.8	379.4	381.1	376.0	experimental	S109
TPADSO2	450.5	452.0	480.0	482.0	533.3	experimental	S110
TPB-AC	469.2	469.4	475.9	476.0	370.0	experimental	S111
TPBPPI-PBI	472.9	473.4	461.4	462.6	429.0	experimental	S112
TPBPPI-PY	472.2	472.0	459.4	459.9	420.0	experimental	S112
TPXZ-as-TAZ	888.7	887.8	889.5	891.5	529.0	computational	S113
TRZ-3SO2	577.4	572.9	540.6	539.9	706.0	experimental	S114
TSBFB	404.5	403.6	385.1	384.4	395.0	experimental	S2
TXO-TPA	570.6	566.0	598.6	595.7	603.0	experimental	S87
VBNO	432.7	433.5	419.5	421.1	482.3	experimental	S115
XT-T	360.0	360.0	349.2	350.5	478.0	experimental	S116
[2,1-b]IF	1031.9	586.3	770.2	571.9	347.0	experimental	S117
dTolmp	435.6	434.6	397.2	396.9	548.0	experimental	S118

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Table S2 – continued from previous page

Molecule	sTDAs	TD-DFTs	sTDAs	TD-DFT	Ref.	Type	Citation
	(Gas)	(Gas)	(Tol)	(Tol)	(nm)		
iTPBI-CN	441.4	440.7	432.7	432.1	443.5	experimental	S119
m-CF3PIBI	492.2	492.6	485.5	486.1	435.3	experimental	S120
m-CZ-DPS-DMAC	425.1	428.5	423.9	427.9	490.8	experimental	S121
m-DTPACO	484.6	485.3	501.6	502.8	375.0	experimental	S122
m-PCzTPD	522.4	522.9	522.1	519.4	506.0	experimental	S123
m-PO-ABN	509.5	519.6	472.0	481.3	450.0	experimental	S90
mCP-BP-DMAC	503.7	504.0	504.6	505.0	505.5	experimental	S124
mCP-BmPy	395.9	399.3	384.9	389.0	410.0	experimental	S125
mCP-Ph	390.6	394.3	371.4	375.4	350.0	experimental	S125
mCP-Py	385.2	388.5	372.2	376.2	350.0	experimental	S125
mCz-TAn-CN	519.1	529.1	483.8	492.6	438.0	experimental	S126
mP2MPC	450.6	452.1	430.5	432.5	399.0	experimental	S127
mTPA-PPI	457.5	457.9	443.8	445.0	404.0	experimental	S2
o-CzOXD	500.3	500.5	477.5	477.5	492.7	experimental	S3
o-PCzTPD	558.4	555.3	518.5	519.1	485.0	experimental	S123
oCBP	388.3	392.6	377.0	381.2	389.0	experimental	S128
p-DTPACO	508.7	510.4	531.9	533.9	540.0	experimental	S122
p-PCzTPD	555.3	553.3	563.4	561.2	476.0	experimental	S123
p-PO-ABN	514.8	521.5	488.0	494.6	456.0	experimental	S90
p-PO15NCzDPA	508.1	518.1	467.2	477.0	427.0	experimental	S129
pSFIAC2	470.9	479.8	461.4	469.7	451.5	experimental	S104
pzpy	510.1	519.9	466.8	476.8	458.0	experimental	S130
t-DABNA	431.1	438.5	418.4	425.5	464.0	experimental	S131

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Table S2 – continued from previous page

Molecule	sTDA (Gas)	sTD-DFT (Gas)	sTDA (Tol)	sTD-DFT (Tol)	Ref. (nm)	Type	Citation
t3Cz-SO	434.5	440.1	442.9	448.2	446.0	experimental	S40
tBuCzDBA	757.8	751.2	684.8	678.9	558.5	experimental	S132
tCTM	483.8	487.0	483.3	483.5	459.0	experimental	S38
thio-ether	1032.4	586.4	770.0	572.2	366.0	experimental	S117
t-BuCz2BP	488.8	490.6	490.8	490.3	440.0	experimental	S133
α -2DMAC-DBP	1177.9	1176.8	1039.5	1038.2	637.5	experimental	S134
α -DMAC-DBP	1092.4	1092.0	952.2	951.8	622.0	experimental	S134
β -DI-DBP	961.4	952.3	832.6	825.2	582.5	experimental	S134

Table S3: Complete validation dataset for singlet-triplet gap predictions. Predictions from sTDA and sTD-DFT methods in gas and toluene phases are compared against experimental/computational literature values. References are provided in the last column.

Molecule	sTDA (Gas)	sTD-DFT (Gas)	sTDA (Tol)	sTD-DFT (Tol)	Ref. (eV)	Type	Citation
11-AcBPdCN	0.3009	0.2809	0.3667	0.3437	0.0650	experimental	S135
12AcCz-PM	0.2661	0.2631	0.3416	0.3407	0.3900	experimental	S136
12BTCzTPN	0.1955	0.1766	0.1720	0.1610	0.0829	experimental	S137
1CM	0.5074	0.4471	0.4934	0.4316	0.2900	experimental	S7
1CzSO	0.3195	0.2961	0.2972	0.2779	0.2150	experimental	S138
1PXZP	0.1653	0.1764	0.1692	0.1646	0.2600	computational	S139
23AcCz-PM	0.2003	0.2120	0.2544	0.2647	0.0600	experimental	S136
25CzBPym	0.0833	0.0797	0.0929	0.0879	0.0604	experimental	S4
25DAcBPy	0.0576	0.0575	0.3301	0.3383	0.0235	computational	S140

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Table S3 – continued from previous page

Molecule	sTDA (Gas)	sTD-DFT (Gas)	sTDA (Tol)	sTD-DFT (Tol)	Ref. (eV)	Type	Citation
25tCzBPym	0.0848	0.0811	0.0917	0.0872	0.0700	experimental	S4
26DAcBPy	0.1948	0.1907	0.1799	0.1764	0.0070	computational	S140
26DPXZBPy	0.2012	0.1987	0.1685	0.1661	0.0520	computational	S140
2BPy-mDTC	0.0798	0.0793	0.0847	0.0849	0.0650	experimental	S5
2BrCPT	0.2696	0.2431	0.3015	0.2718	0.0400	experimental	S141
2CM	0.4735	0.4172	0.4762	0.4198	0.3100	experimental	S7
2Cz-DMAC-BTB	0.2665	0.2823	0.1276	0.1438	0.0200	experimental	S8
2Cz-DMAC-TTR	0.2933	0.2743	0.3442	0.3199	0.0470	experimental	S8
2Cz-DMAC-TXO	0.3679	0.3706	0.3220	0.3319	0.0360	experimental	S8
2Cz2tCzBn	0.1908	0.1730	0.2168	0.1959	0.1300	experimental	S9
2CzPN	0.2427	0.2373	0.2677	0.2593	0.2525	experimental	S30
2CzTPEPCz	0.3096	0.3171	0.3311	0.3352	0.1300	experimental	S142
2DMAC-BP-F	0.2632	0.2475	0.3177	0.2988	0.1367	experimental	S143
2DN	0.0920	0.0965	0.4000	0.3669	0.0700	experimental	S11
2F	0.2152	0.1908	0.2374	0.2097	0.2300	computational	S12
2H-Qz	0.2138	0.2118	0.2326	0.2307	0.1100	experimental	S144
2Mi2SB	0.3464	0.3356	0.3865	0.3721	0.0300	computational	S145
2PCzBN	0.3041	0.3010	0.3170	0.3108	0.2800	experimental	S146
2PCzBN-tPh	0.2207	0.2191	0.2414	0.2386	0.2600	experimental	S146
2PN	0.1324	0.1285	0.1074	0.1046	0.0100	experimental	S11
2PS-2FPh	0.2729	0.2650	0.3064	0.2983	0.2050	computational	S147
2PXZ-OXD	0.2031	0.2024	0.1787	0.1783	0.2300	experimental	S148
2PXZP	0.2176	0.2301	0.2422	0.2410	0.1340	computational	S139

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Table S3 – continued from previous page

Molecule	sTDA (Gas)	sTD-DFT (Gas)	sTDA (Tol)	sTD-DFT (Tol)	Ref. (eV)	Type	Citation
2SPBAC-BP	0.4008	0.4004	0.4024	0.3933	0.2200	experimental	S149
2tCz2CzBn	0.1995	0.1807	0.2266	0.2046	0.1300	experimental	S9
3-ACBPmCN	0.0470	0.0582	0.0519	0.0651	0.1800	computational	S135
3-AcBPdCN	0.2112	0.2004	0.2667	0.2540	0.1800	experimental	S135
3-AcBPtCN	0.0637	0.0643	0.2413	0.2319	0.1300	experimental	S135
3-DPM	0.1689	0.1581	0.1558	0.1492	0.0987	experimental	S150
35CzBPym	0.1491	0.1382	0.1503	0.1393	0.1100	experimental	S4
35tCzBPym	0.1484	0.1383	0.1537	0.1431	0.1100	experimental	S4
36DTPAFM	0.2073	0.2235	0.2241	0.2391	0.1200	experimental	S151
3BPy-mDTC	0.1555	0.1449	0.1562	0.1456	0.1200	experimental	S5
3CzBN	0.3294	0.3321	0.3069	0.3081	0.3100	experimental	S18
3CzPhpPM	0.2409	0.2185	0.2659	0.2406	0.3300	experimental	S152
3DMAC-BP-Br	0.0203	0.0210	0.2215	0.2151	0.0400	experimental	S153
3DMAC-BP-CN	0.1852	0.1962	0.2209	0.2142	0.0200	experimental	S153
3DPA3CN	0.2230	0.2266	0.2186	0.2212	0.1030	experimental	S154
3Py-DMAC	0.0653	0.0776	0.3069	0.2963	0.0400	experimental	S155
3TCPM	0.2621	0.2614	0.1934	0.2072	0.2600	experimental	S20
3ai	0.8060	0.7332	0.8521	0.7687	0.1800	experimental	S156
4-Ac	0.5593	0.5465	0.6006	0.5815	0.0800	experimental	S157
4-DPM	0.4478	0.4605	0.4672	0.4791	0.1427	experimental	S150
4-SCZ	0.4456	0.4094	0.4687	0.4303	0.2400	experimental	S158
4BPy-mDTC	0.1541	0.1431	0.1477	0.1375	0.1300	experimental	S5
4CzIPN	0.1507	0.1367	0.1624	0.1487	0.1103	experimental	S159

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Table S3 – continued from previous page

Molecule	sTDA (Gas)	sTD-DFT (Gas)	sTDA (Tol)	sTD-DFT (Tol)	Ref. (eV)	Type	Citation
4CzTPN	0.2100	0.1874	0.2339	0.2079	0.0800	experimental	S160
4F-m- ν -DABNA	0.2638	0.2488	0.2657	0.2515	0.0700	experimental	S161
4F- ν -DABNA	0.2709	0.2560	0.2770	0.2625	0.0500	experimental	S161
5,6PXZ-PIDO	0.2119	0.2186	0.1857	0.1851	0.4500	computational	S162
5-Cz-Sac	0.3410	0.3079	0.3315	0.3064	0.2800	experimental	S163
6,7-DCNQx-DICz	0.0088	0.0143	0.0207	0.0279	0.0276	computational	S164
6AcBIQ	0.0859	0.0897	0.1004	0.1012	0.1000	experimental	S165
9CzFDBFSPO	0.3962	0.3615	0.3951	0.3550	0.1200	experimental	S166
ACID-BPPZ	0.2041	0.2014	0.1581	0.1669	0.0150	computational	S167
ACRFLCN	0.0002	0.0047	0.0001	0.0002	0.1000	computational	S168
ACRSA	0.0025	0.0217	-0.0001	0.0000	0.0350	experimental	S169
ACRXTN	0.4781	0.4642	0.5345	0.5175	0.1730	experimental	S24
AI-4Cz	0.2055	0.1909	0.2287	0.2140	0.0200	experimental	S170
ANQDC	0.2365	0.2634	0.2815	0.3132	0.0200	computational	S171
ANQDC-PSTA	0.0965	0.0948	0.1344	0.1296	0.0800	experimental	S171
APDC-DTPA	0.2802	0.2713	0.2891	0.3004	0.1400	experimental	S172
APPT	0.1857	0.1995	0.2290	0.2449	0.0700	experimental	S173
APPT-PXZ	0.1829	0.1755	0.2201	0.2100	0.0100	computational	S173
Ac-BPCN	0.3133	0.2940	0.3797	0.3575	0.1300	experimental	S174
AcCN	0.1610	0.1785	0.1956	0.2129	0.0700	experimental	S26
AcCYP	0.0725	0.0850	0.0824	0.0981	0.2000	experimental	S175
B-dpa-Cz	0.4397	0.3954	0.4512	0.4060	0.1500	experimental	S176
B-dpa-SpiroAC	0.4062	0.3635	0.4135	0.3709	0.1400	experimental	S176

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Table S3 – continued from previous page

Molecule	sTDA (Gas)	sTD-DFT (Gas)	sTDA (Tol)	sTD-DFT (Tol)	Ref. (eV)	Type	Citation
BCZ-DPS-AD	0.3282	0.2926	0.3383	0.3004	0.0967	experimental	S177
BDAPM	0.3393	0.3383	0.3593	0.3599	0.0600	experimental	S178
BDPCC	0.3333	0.3139	0.3425	0.3241	0.1100	experimental	S30
BDQ-tBuCz	0.1592	0.1565	0.1872	0.1835	0.1800	experimental	S31
BDQC-2	0.1647	0.1628	0.1856	0.1829	0.1200	experimental	S179
BDQDMAC	0.0645	0.0647	0.0735	0.0738	0.0750	experimental	S31
BDQPCz	0.1517	0.1623	0.1728	0.1880	0.4950	experimental	S31
BFCZPZ1	0.2679	0.2584	0.2793	0.2685	0.3267	experimental	S180
BFCZPZ2	0.2685	0.2615	0.2917	0.2840	0.3100	experimental	S180
BMIM	0.2801	0.2712	0.2898	0.3011	0.1400	experimental	S172
BP-mDTC	0.1928	0.1825	0.1832	0.1774	0.0700	experimental	S5
BPCN-2Cz	0.3558	0.3243	0.3590	0.3265	0.2200	experimental	S181
BPCN-Cz2Ph	0.3042	0.2773	0.3250	0.2965	0.3100	experimental	S181
BPCP-2CPC	0.1738	0.1724	0.2227	0.2182	0.1660	experimental	S182
BPZPP	0.1329	0.1344	0.1529	0.1531	0.2000	experimental	S183
BTCZPZ1	0.2457	0.2342	0.2592	0.2483	0.2400	experimental	S180
BTD-PXZ	0.1863	0.1829	0.1718	0.1684	0.1300	experimental	S184
BTITrz	0.2219	0.2224	0.2289	0.2290	0.1025	computational	S185
BTPBFCz	0.2662	0.2440	0.2687	0.2490	0.0900	experimental	S186
BTPBFCz-D3	0.2264	0.2157	0.2175	0.2055	0.1200	experimental	S186
BTPO	0.4356	0.4211	0.4061	0.3864	0.4300	experimental	S187
BTZPP	0.3066	0.3191	0.2829	0.2942	0.3000	experimental	S32
BTZ-DMAC	0.1274	0.1340	0.1126	0.1194	0.0200	computational	S188

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Table S3 – continued from previous page

Molecule	sTDA (Gas)	sTD-DFT (Gas)	sTDA (Tol)	sTD-DFT (Tol)	Ref. (eV)	Type	Citation
BZC-PXZ	0.2407	0.2284	0.2292	0.2175	0.0200	experimental	S189
BuPCzBCO	0.3702	0.3307	0.3938	0.3517	0.1850	computational	S190
BzITz	0.2513	0.2477	0.2575	0.2534	0.1850	computational	S191
CCDC	0.3066	0.2759	0.1276	0.1415	0.2220	computational	S192
CDBP-BP-DMAC	0.1829	0.1967	0.2389	0.2546	0.0393	experimental	S35
CDBP-BP-PXZ	0.2877	0.2868	0.2877	0.2862	0.0605	experimental	S35
CN-QP	0.1557	0.1648	0.1216	0.1341	0.0300	experimental	S36
CNPP-TPA	0.2225	0.2259	0.2549	0.2570	0.1450	experimental	S193
CNTPA-CZ	0.5282	0.5166	0.5729	0.5571	0.3500	experimental	S194
CNTPA-PXZ	0.3266	0.3257	0.3370	0.3364	0.1300	experimental	S194
CZ-DPS-DMAC	0.3076	0.2840	0.2851	0.2885	0.0700	experimental	S39
CZ-TRZ	0.2454	0.2302	0.2816	0.2641	0.1500	experimental	S195
CZ9CZPZ	0.2859	0.2751	0.3162	0.3040	0.3150	experimental	S180
Cz-SO	0.3954	0.3540	0.3935	0.3540	0.4100	experimental	S40
CzDCbTrz	0.1759	0.1760	0.2349	0.2348	0.1200	experimental	S196
CzPm	0.2873	0.2766	0.3039	0.2960	0.1800	experimental	S197
CzmPPC	0.1031	0.1273	0.0742	0.0936	0.2800	experimental	S198
CzoB	0.0786	0.0780	0.0975	0.1027	0.0900	computational	S199
DAcIPN	0.3427	0.3327	0.3332	0.3215	0.0350	experimental	S200
DBCP	0.0634	0.0645	0.0814	0.1017	0.0813	experimental	S201
DBQ-3PXZ	0.1057	0.1015	0.0955	0.0920	0.0300	experimental	S46
DBT-BZ-DMAC	0.3142	0.3200	0.3062	0.3102	0.0800	experimental	S47
DC-ACR	0.0033	0.0030	0.0366	0.0414	0.0100	experimental	S202

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Table S3 – continued from previous page

Molecule	sTDA	sTD-DFT	sTDA	sTD-DFT	Ref.	Type	Citation
	(Gas)	(Gas)	(Tol)	(Tol)	(eV)		
DC-TC	0.0052	0.0048	0.0056	0.0054	0.1400	experimental	S202
DCPDAPM	0.0701	0.0701	0.5171	0.4998	0.1300	computational	S203
DCz-DPPZ	0.1560	0.1490	0.2024	0.1931	0.1900	experimental	S204
DCzBNPh-1	0.2563	0.2378	0.3209	0.2983	0.5000	experimental	S49
DCzIPN	0.3588	0.3410	0.3593	0.3429	0.1433	experimental	S50
DDMA-TXO2	0.5414	0.5304	0.4180	0.3968	0.2200	experimental	S205
DDPhCz-DPPZ	0.1229	0.1229	0.1514	0.1470	0.1600	experimental	S204
DFDBQPXZ	0.1692	0.1691	0.1596	0.1589	0.0000	experimental	S206
DHID-DBS	0.4448	0.4129	0.4697	0.4490	0.3300	experimental	S53
DHID-DPS	0.5477	0.5208	0.4687	0.4470	0.1800	experimental	S53
DHPZ-2BI	0.2115	0.2147	0.2053	0.2075	0.1900	experimental	S207
DMAC-11-DPPZ	0.0560	0.0574	0.0714	0.0723	0.0803	experimental	S54
DMAC-AQ	0.4361	0.4090	0.4923	0.4663	0.0633	experimental	S208
DMAC-BP	0.0644	0.0658	0.5157	0.4999	0.0700	experimental	S209
DMAC-Cz	0.4201	0.3912	0.4611	0.4309	0.0007	experimental	S56
DMAC-Cz-TTR	0.2709	0.2527	0.2312	0.2200	0.0660	experimental	S56
DMAC-DPS	0.2222	0.2229	0.2340	0.2343	0.0100	experimental	S57
DMAC-FO	0.1247	0.1195	0.3839	0.3734	0.1400	experimental	S210
DMAC-PCN	0.1341	0.1504	0.1133	0.1292	0.1300	experimental	S125
DMAC-TRZ	0.0485	0.0571	0.0494	0.0510	0.0500	experimental	S211
DMACMNPTO	0.5377	0.5222	0.5471	0.5302	0.2050	experimental	S59
DMBFTX	0.5437	0.4736	0.5723	0.5002	0.2800	experimental	S212
DMIC-TRZ	0.5193	0.6832	0.7550	1.3766	0.0560	experimental	S213

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Table S3 – continued from previous page

Molecule	sTDA (Gas)	sTD-DFT (Gas)	sTDA (Tol)	sTD-DFT (Tol)	Ref. (eV)	Type	Citation
DMeCzIPN	0.2790	0.2669	0.2781	0.2657	0.0910	experimental	S50
DPACPhTPI	0.2551	0.2503	0.2498	0.2437	0.7200	experimental	S60
DPACpBr	0.3348	0.3322	0.4044	0.3961	0.0020	experimental	S61
DPE	0.2845	0.2840	0.2987	0.2980	0.1800	computational	S214
DPE-DDMAc	0.1751	0.1961	0.2781	0.3000	0.1800	computational	S214
DPE-DPXZ	0.2852	0.2849	0.2996	0.2988	0.2000	experimental	S214
DPIBPZ-DPXZ	0.1101	0.1057	0.1073	0.1034	0.0500	experimental	S215
DPS-PXZ	0.2879	0.2873	0.3129	0.3120	0.0500	experimental	S64
DPXZ-DPPM	0.1047	0.1014	0.1043	0.1013	0.0470	experimental	S65
DQP	0.0189	0.0396	0.0164	0.0402	0.0555	experimental	S47
DTPTCzDP-CN	0.2642	0.2402	0.2493	0.2321	0.0300	experimental	S216
DV-3CzCN	0.2477	0.2398	0.3404	0.3364	0.2300	experimental	S68
DVCz-2CzCN	0.2868	0.2623	0.2421	0.2332	0.1600	experimental	S69
DiCz-Sac	0.2404	0.2322	0.2562	0.2469	0.2000	experimental	S163
ECDPTT	0.3447	0.3433	0.3385	0.3331	0.0800	experimental	S217
F2AcBO	0.2847	0.2918	0.2588	0.2545	0.0750	experimental	S71
FAC	0.5171	0.4920	0.5458	0.5144	0.0900	computational	S218
FDQPXZ	0.1276	0.1217	0.1281	0.1225	0.0450	experimental	S219
FSF4A	0.2862	0.2912	0.2898	0.2923	0.0220	experimental	S220
Fene	0.1297	0.1378	0.1967	0.2089	0.0250	computational	S221
Fens	0.2404	0.2561	0.3481	0.3589	0.0200	computational	S221
FlCz	0.1952	0.2170	0.2129	0.2360	0.5100	experimental	S73
IABTCN	0.3551	0.3561	0.3843	0.3786	0.3998	experimental	S74

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Table S3 – continued from previous page

Molecule	sTDA (Gas)	sTD-DFT (Gas)	sTDA (Tol)	sTD-DFT (Tol)	Ref. (eV)	Type	Citation
ICBNTrz4	0.2041	0.1993	0.2097	0.2048	0.1000	experimental	S222
IDAC-MCO	0.4472	0.3996	0.4200	0.3755	0.0900	experimental	S223
IDFL-2DPA	0.2454	0.2221	0.2493	0.2259	0.4500	experimental	S224
IQ-Se	0.3316	0.3254	0.4327	0.4125	1.1800	computational	S225
IpCm-PhBzAc	0.5082	0.4902	0.5207	0.5015	0.3500	experimental	S226
MBF	0.2198	0.2225	0.2260	0.2181	0.0100	experimental	S77
MCZ-P2-DTM	0.2697	0.2587	0.2716	0.2621	0.2300	experimental	S227
MPA	0.2598	0.2577	0.2572	0.2543	0.1400	experimental	S77
MPPA	0.1960	0.1890	0.1952	0.1867	0.1743	experimental	S78
Me-DMAC	0.1117	0.1283	0.1387	0.1562	0.0100	experimental	S79
Me-DOC	0.4294	0.3786	0.4302	0.3763	0.2400	computational	S228
Me-MOC	0.3803	0.3386	0.3800	0.3370	0.2500	computational	S228
Me2ACBO	0.3142	0.3054	0.2883	0.2898	0.0500	experimental	S71
MeOCzPCN	0.4683	0.4467	0.4954	0.4673	0.0100	experimental	S229
NA-TNA	0.4044	0.4138	0.4040	0.4137	0.2400	experimental	S80
NAI-PhBiFA	0.3697	0.3820	0.2973	0.3055	0.1200	experimental	S81
NIDPA-1	0.3876	0.3588	0.4076	0.3773	0.6000	experimental	S230
NPE-AcDPS	0.4903	0.4850	0.3559	0.3563	0.0300	experimental	S231
NyDPAc	0.1657	0.1819	0.1808	0.1987	0.1520	experimental	S232
NyDPO	0.1494	0.1611	0.1945	0.2037	0.0333	computational	S233
OCzBN	0.3704	0.3466	0.4579	0.4450	0.1400	experimental	S84
OPDPO	0.1450	0.1626	0.0857	0.0813	0.1300	experimental	S142
P4CzCN-BCz	0.6006	0.6278	0.6999	0.7319	0.1700	experimental	S85

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Table S3 – continued from previous page

Molecule	sTDA (Gas)	sTD-DFT (Gas)	sTDA (Tol)	sTD-DFT (Tol)	Ref. (eV)	Type	Citation
PBICT	0.2132	0.1980	0.3424	0.3194	0.0280	experimental	S164
PFBP-2b	0.1515	0.1519	0.1663	0.1583	0.0600	experimental	S234
PIAnTPh	0.7770	0.7391	0.8291	0.7873	1.1400	experimental	S89
PO-TPA	0.1934	0.1883	0.3355	0.3190	0.2500	experimental	S235
PPO21	0.3775	0.3353	0.3825	0.3406	0.0253	computational	S236
PPZPPI	0.2203	0.2236	0.2151	0.2175	0.1100	experimental	S237
PPZTPI	0.2232	0.2276	0.2160	0.2192	0.1100	experimental	S237
PS-BZ-DMAC	0.5560	0.5307	0.5994	0.5716	0.0045	experimental	S92
PSPBP	0.2941	0.2927	0.2819	0.2814	0.1100	experimental	S238
PSPP	0.3194	0.3148	0.3162	0.3119	0.1100	experimental	S238
PT-BZ-DMAC	0.0678	0.0852	0.0520	0.0640	0.0084	experimental	S92
PTPC	0.2440	0.2387	0.2293	0.2245	0.6355	experimental	S239
PTZ-2PTO	0.2932	0.2868	0.2926	0.2929	0.2700	experimental	S93
PTZ-DCPP	0.2776	0.2665	0.3429	0.3333	0.1867	experimental	S240
PTZ-NAI	0.0003	0.0003	0.1172	0.1133	0.0867	experimental	S241
PX-TRZ	0.2424	0.2355	0.2529	0.2507	0.0130	experimental	S94
PXZ-10-DPPZ	0.0924	0.0891	0.0913	0.0886	0.0365	experimental	S54
PXZ-11-DPPZ	0.1207	0.1154	0.1309	0.1255	0.0507	experimental	S54
PXZ-BOO	0.2410	0.2297	0.2536	0.2410	0.0600	experimental	S242
PXZ-NAI	0.2654	0.2530	0.3109	0.2946	0.0600	experimental	S241
PXZ-PPO	0.5104	0.4859	0.5382	0.5229	0.6000	experimental	S242
PXZ-QL	0.3702	0.3444	0.3846	0.3576	0.1000	computational	S243
PXZ2PTO	0.3385	0.3358	0.3417	0.3392	0.0233	experimental	S95

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Table S3 – continued from previous page

Molecule	sTDA (Gas)	sTD-DFT (Gas)	sTDA (Tol)	sTD-DFT (Tol)	Ref. (eV)	Type	Citation
PhQLPXZ	0.1369	0.1437	0.1971	0.2118	0.0900	experimental	S219
PrDPhAc	0.4722	0.4514	0.5308	0.5046	0.0750	computational	S244
Pra-2DMAC	0.5074	0.4925	0.5244	0.5098	0.4400	experimental	S97
PxCN	0.2607	0.2601	0.2562	0.2557	0.0367	experimental	S26
PxCYP	0.2325	0.2438	0.2431	0.2422	0.2000	experimental	S175
PxPYM	0.1691	0.1801	0.2358	0.2454	0.0100	computational	S245
Py-TPA	0.1253	0.1357	0.1337	0.1335	0.1400	computational	S246
PyDCN-DMAC	0.2456	0.2665	0.3176	0.3359	0.0100	experimental	S98
PyIA _n TPh	0.7218	0.6752	0.7861	0.7350	1.1400	experimental	S89
SAF-2NP	0.0816	0.0812	0.0892	0.0874	0.0400	computational	S247
SBA-2DPS	0.4557	0.4531	0.3631	0.3818	0.0900	experimental	S248
SBDBQ-PXZ	0.1089	0.1046	0.1198	0.1154	0.0700	experimental	S46
SBF-PXZ	0.3045	0.3048	0.3249	0.3241	0.2600	experimental	S249
SBTF	0.5022	0.4954	0.5297	0.5222	0.2000	experimental	S250
SFAC-BP-SFAC	0.0774	0.0786	0.0898	0.0919	0.0360	experimental	S101
SFCz	0.2013	0.2131	0.2129	0.2257	0.4800	experimental	S73
SFI23pPM	0.2328	0.2146	0.2600	0.2406	0.3300	experimental	S152
SIR-A	0.0001	0.0002	0.0000	0.0001	0.0160	experimental	S251
SIR-B	0.0070	0.0083	0.0015	0.0017	0.0140	experimental	S251
SIR-C	0.0082	0.0089	-0.0005	-0.0002	0.0150	experimental	S251
SPAC-FO	0.1398	0.1339	0.1384	0.1326	0.1300	experimental	S210
SPFS-PXZ	0.3291	0.3293	0.3238	0.3229	0.0153	experimental	S252
STAC-BP-STAC	0.0858	0.0874	0.1261	0.1271	0.0520	experimental	S101

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Table S3 – continued from previous page

Molecule	sTDA (Gas)	sTD-DFT (Gas)	sTDA (Tol)	sTD-DFT (Tol)	Ref. (eV)	Type	Citation
SXAC-BP-SXAC	0.0835	0.0851	0.0929	0.0948	0.0610	experimental	S101
SiTCNCz	0.2497	0.2300	0.2626	0.2416	0.1300	experimental	S253
Spiro-CN	0.2889	0.2968	0.2696	0.2768	5.5000	experimental	S254
SpiroAC-TRZ	0.0677	0.0679	0.0597	0.0616	0.0720	experimental	S47
TATP-BP	0.1452	0.1464	0.1292	0.1354	0.1290	experimental	S103
TBPe	0.7123	0.6662	0.7738	0.7233	0.1200	computational	S255
TCPZ	0.5340	0.5069	0.5735	0.5469	0.1400	experimental	S256
TCZ1	0.4017	0.3602	0.4205	0.3767	0.0160	experimental	S200
TCz1	0.4013	0.3599	0.4206	0.3771	0.1000	experimental	S257
TCzTrz	0.1798	0.1783	0.2079	0.2050	0.1400	experimental	S196
TMDBP	0.5203	0.5192	0.5528	0.5493	0.1000	experimental	S258
TNPZ	0.0818	0.0930	0.0960	0.1087	0.1700	computational	S259
TPA-2	0.0416	0.0478	0.1131	0.1203	0.1450	experimental	S260
TPA-DCPP	0.1746	0.1796	0.1812	0.1855	0.1480	experimental	S47
TPADSO2	0.4055	0.3964	0.3925	0.3818	0.3800	experimental	S110
TPAPm	0.4500	0.4557	0.4763	0.4820	0.3200	experimental	S261
TPD4PA	0.3244	0.2950	0.3337	0.3056	0.0500	experimental	S262
TPSi-F	0.6295	0.6174	0.6494	0.6375	0.1000	experimental	S263
TPXZ-as-TAZ	0.0953	0.0968	0.1079	0.1048	0.2200	experimental	S113
TPXZ-as-TAZ	0.0983	0.0984	0.0271	0.0357	0.0300	experimental	S264
TRZ-pIC	0.4296	0.3745	0.4447	0.3877	0.2900	experimental	S265
TX-CzPh	0.5513	0.4893	0.5517	0.4916	0.6700	computational	S266
TXO-P-Si	-0.0053	-0.0056	-0.0027	-0.0027	0.5222	experimental	S236

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Table S3 – continued from previous page

Molecule	sTDA (Gas)	sTD-DFT (Gas)	sTDA (Tol)	sTD-DFT (Tol)	Ref. (eV)	Type	Citation
Tri-PXZ-PCN	0.1726	0.1682	0.1583	0.1532	0.0500	experimental	S267
TrisCz-Trz	0.1356	0.1368	0.2203	0.2251	0.0700	experimental	S268
UGH3	-0.0012	0.0083	0.0170	0.0280	0.2200	experimental	S205
VBC-PTZ	0.4336	0.4376	0.4823	0.4872	0.5300	experimental	S269
VBNO	0.3552	0.3499	0.4127	0.4015	0.1100	experimental	S115
V-TPA	0.2797	0.2756	0.2992	0.2941	0.6210	experimental	S270
Yad	0.3868	0.3562	0.4022	0.3704	0.0200	computational	S221
czpzpy	0.4202	0.3853	0.4351	0.3990	0.1300	experimental	S271
dbG2TAZ	0.4466	0.4517	0.4754	0.4820	0.5300	experimental	S272
m-CZ-DPS-DMAC	0.3204	0.2975	0.3722	0.3450	0.2100	experimental	S121
m-DTPACO	0.3172	0.3132	0.3495	0.3436	0.2100	experimental	S122
m- ν -DABNA	0.2892	0.2722	0.2900	0.2738	0.0700	experimental	S161
m2Cz2Trz	0.1515	0.1484	0.1759	0.1742	0.0900	experimental	S273
mCP-BP-DMAC	0.2872	0.2856	0.2875	0.2859	0.0160	experimental	S124
mDPBPZ-DPXZ	0.1066	0.1030	0.1053	0.1025	0.0200	computational	S215
mP2MPC	0.2837	0.2741	0.2978	0.2844	0.6500	experimental	S127
mPTBC	0.2343	0.2243	0.2175	0.2081	0.0060	experimental	S274
mTPA-PPI	0.2582	0.2558	0.2701	0.2628	0.7600	experimental	S275
oPTBC	0.1336	0.1284	0.1281	0.1232	0.0070	experimental	S274
p-DTPACO	0.4180	0.4096	0.4450	0.4365	0.0500	experimental	S122
pDPAPQ-PXZ	0.1777	0.1708	0.1956	0.1877	0.0600	experimental	S276
pDPBPZ-PXZ	0.0906	0.0982	0.0988	0.1070	0.0600	experimental	S276
pDTCz-2DPyS	0.2269	0.2264	0.2872	0.2793	0.2100	experimental	S277

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Table S3 – continued from previous page

Molecule	sTDA (Gas)	sTD-DFT (Gas)	sTDA (Tol)	sTD-DFT (Tol)	Ref. (eV)	Type	Citation
pSFIAc2	0.3667	0.3178	0.3779	0.3305	0.3100	experimental	S104
t-BuPCz	0.3330	0.2985	0.3504	0.3140	0.0800	experimental	S278
t-DABNA	0.4476	0.3993	0.4608	0.4111	0.1600	experimental	S279
t3Cz-SO	0.3555	0.3196	0.3436	0.3105	0.2600	experimental	S40
tCbz-mPYRs	0.2990	0.2843	0.3273	0.3107	0.4150	computational	S280
t-BuCz2BP	0.3370	0.3279	0.3345	0.3371	0.2150	experimental	S133
α -2CbPN	0.2931	0.2856	0.3139	0.3036	0.2600	experimental	S196
α -2DMAC-DBP	0.0552	0.0562	0.0580	0.0595	0.0250	experimental	S134
α -DMAC-DBP	0.0521	0.0525	0.0511	0.0517	0.0925	experimental	S134
β -DI-DBP	0.1070	0.1192	0.1275	0.1408	0.1600	experimental	S134
δ -2CbPN	0.2610	0.2545	0.2760	0.2673	0.1300	experimental	S196

Table S4: Outlier analysis for emission wavelength and singlet-triplet gap predictions. Outliers were identified using the Median Absolute Deviation (MAD) method with a threshold of $3 \times \text{MAD}$ from the median residual. For emission wavelengths, $\text{MAD} = 43.3 \text{ nm}$ (threshold: 130.0 nm); for ΔE_{ST} , $\text{MAD} = 0.11 \text{ eV}$ (threshold: 0.32 eV). All predictions are from sTDA-xTB calculations in toluene solvent.

Molecule	Property	Exp./Ref.	Predicted	Residual	Architecture	Citation
DQP	λ_{PL} [nm]	562.0	1 057.9	495.9	Polycyclic acceptor	S47
[2,1-b]IF	λ_{PL} [nm]	347.0	770.2	423.2	Other	S117
DPXZ-DPPM	λ_{PL} [nm]	636.0	1 048.7	412.7	Acridine/Phenoxazine	S65
thio-ether	λ_{PL} [nm]	366.0	770.0	404.0	Other	S117
α -2DMAC-DBP	λ_{PL} [nm]	637.5	1 039.5	402.0	Acridine/Phenoxazine	S134
2PN	λ_{PL} [nm]	673.3	1 059.4	386.1	Polycyclic acceptor	S11
TFM-QP	λ_{PL} [nm]	566.5	929.3	362.8	Polycyclic acceptor	S36

Table S4 – continued from previous page

Molecule	Property	Exp./Ref.	Predicted	Residual	Architecture	Citation
TPXZ-as-TAZ	λ_{PL} [nm]	529.0	889.5	360.5	Acridine/Phenoxazine	S113
PXZ-10-DPPZ	λ_{PL} [nm]	614.0	958.6	344.6	Acridine/Phenoxazine	S54
CN-QP	λ_{PL} [nm]	602.0	945.1	343.1	Polycyclic acceptor	S36
α -DMAC-DBP	λ_{PL} [nm]	622.0	952.2	330.2	Acridine/Phenoxazine	S134
DMAC-11-DPPZ	λ_{PL} [nm]	577.5	891.9	314.4	Acridine/Phenoxazine	S54
IQ-Se	ΔE_{ST} [eV]	1.180 0	0.432 7	−0.747 3	Heteroatom-containing	S225
DMIC-TRZ	ΔE_{ST} [eV]	0.056 0	0.755 0	0.699 0	Triazine acceptor	S213
3ai	ΔE_{ST} [eV]	0.180 0	0.852 1	0.672 1	Other	S156
TBPe	ΔE_{ST} [eV]	0.120 0	0.773 8	0.653 8	Other	S255
PS-BZ-DMAC	ΔE_{ST} [eV]	0.004 5	0.599 4	0.594 9	Acridine/Phenoxazine	S92
TPSi-F	ΔE_{ST} [eV]	0.100 0	0.649 4	0.549 4	Heteroatom-containing	S263
P4CzCN-BCz	ΔE_{ST} [eV]	0.170 0	0.699 9	0.529 9	Carbazole-based	S85
TXO-P-Si	ΔE_{ST} [eV]	0.522 2	−0.002 7	−0.525 0	Heteroatom-containing	S24

Table S5: Benchmarking of xTB-based methods against TD-DFT calculations. Vertical excitation energies and singlet-triplet gaps were calculated at the ω B97X-D/def2-TZVP level of theory using ORCA 6.1.0, with CPCM solvation in toluene. The benchmark set comprises 8 representative TADF emitters spanning different donor-acceptor architectures. All structures were optimized at the GFN2-xTB level in toluene solvent. Both sTDA-xTB and sTD-DFT-xTB systematically underestimate ΔE_{ST} by approximately 0.77–0.79 eV compared to TD-DFT, but preserve the relative ranking of molecules.

Molecule	TD-DFT S_1 [eV]	TD-DFT T_1 [eV]	TD-DFT ΔE_{ST} [eV]	sTDA-xTB ΔE_{ST} [eV]	sTD-DFT-xTB ΔE_{ST} [eV]	Deviation sTDA [eV]	Deviation sTD-DFT [eV]
2TCz-DPS	4.251	3.159	1.092	0.321	0.296	0.771	0.796
4CzIPN	3.685	2.801	0.884	0.234	0.208	0.650	0.676
CzS2	4.455	3.196	1.259	0.381	0.352	0.878	0.907
DMAC-DPS	4.168	3.329	0.839	0.234	0.234	0.605	0.605
DMAC-TRZ	4.095	3.071	1.024	0.049	0.051	0.975	0.973
PSPCz	4.481	3.197	1.284	0.404	0.363	0.880	0.921
Px2BP	3.827	2.991	0.836	0.226	0.214	0.610	0.622
TDBA-DI	4.052	2.889	1.163	0.380	0.344	0.783	0.819
Mean	4.127	3.079	1.048	0.279	0.258	0.769	0.790
MAE		—			0.769		0.790
RMSE		—			0.780		0.801

Notes:

- **Computational details:** TD-DFT calculations performed with ORCA 6.1.0 using ω B97X-D functional with def2-TZVP basis set and CPCM solvation model (toluene, $\epsilon = 2.374$). Vertical excitations calculated on GFN2-xTB/GBSA(toluene) optimized geometries.
- **Systematic underestimation:** Both sTDA-xTB and sTD-DFT-xTB consistently underestimate ΔE_{ST} compared to TD-DFT (mean deviation ~ 0.77 eV– 0.79 eV). This systematic bias does not affect relative ranking for high-throughput screening applications.
- **Correlation:** Despite the systematic offset, xTB methods correctly identify molecules with smaller vs. larger ΔE_{ST} values (e.g., Px2BP and 4CzIPN have the smallest gaps in both TD-DFT and xTB calculations).
- **Computational cost:** TD-DFT calculations require $\sim 2 - 6$ hours per molecule on 8

CPU cores, while xTB methods complete in $\sim 2 - 5$ minutes, enabling the screening of 747 molecules presented in this work.

- **Recommendation:** Use xTB methods for initial high-throughput screening to identify promising candidates, followed by TD-DFT validation for quantitative accuracy.

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